
FRC Shrinkage and Creep Characteristics

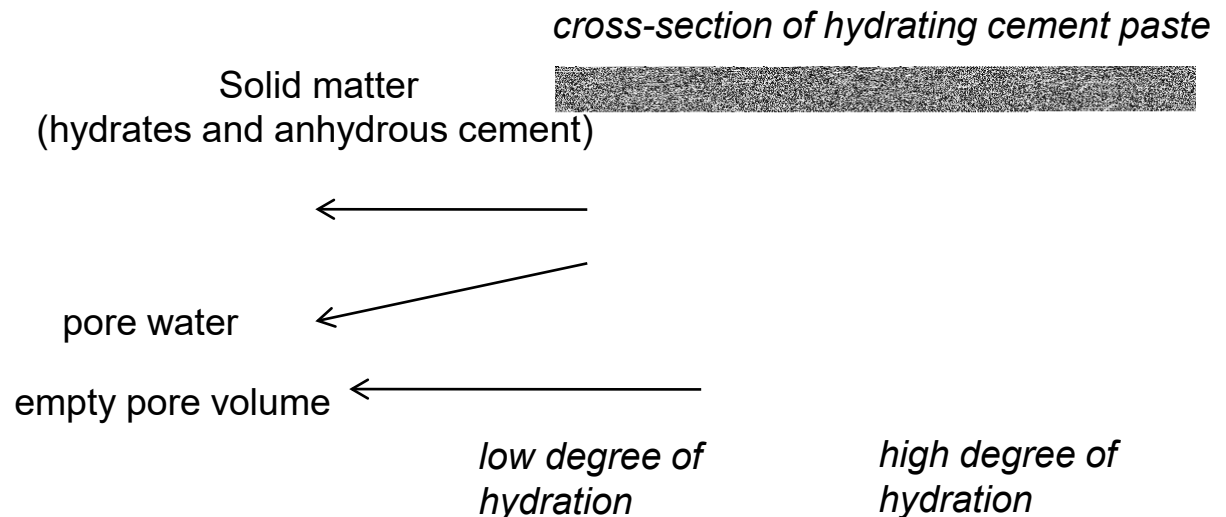
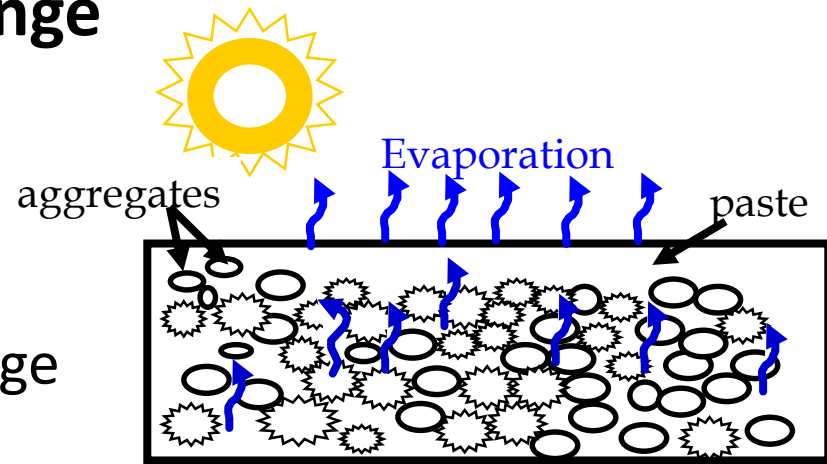
How Fiber Reinforced Concrete can be integrated as an economical, sustainable, and durable aspect of our infrastructure

PhD Summer School, Politecnico di Milano
July 2nd-7th 2021

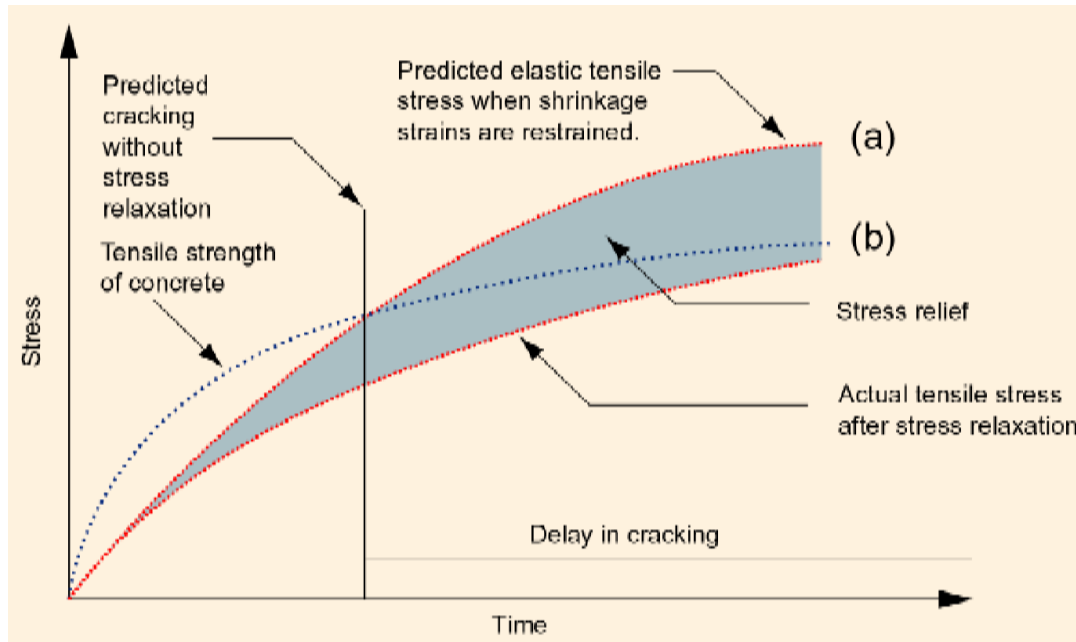
Barzin Mobasher
Arizona State University

Shrinkage as a Volumetric Change

- Drying and plastic shrinkage
(due to water loss by evaporation)
- Autogenous/self-desiccation shrinkage
(due to cement hydration)
 - Thermal shrinkage (due to decrease in temperature)
- Carbonation shrinkage
(reaction of hydrated cement w/carbon dioxide)



Significance

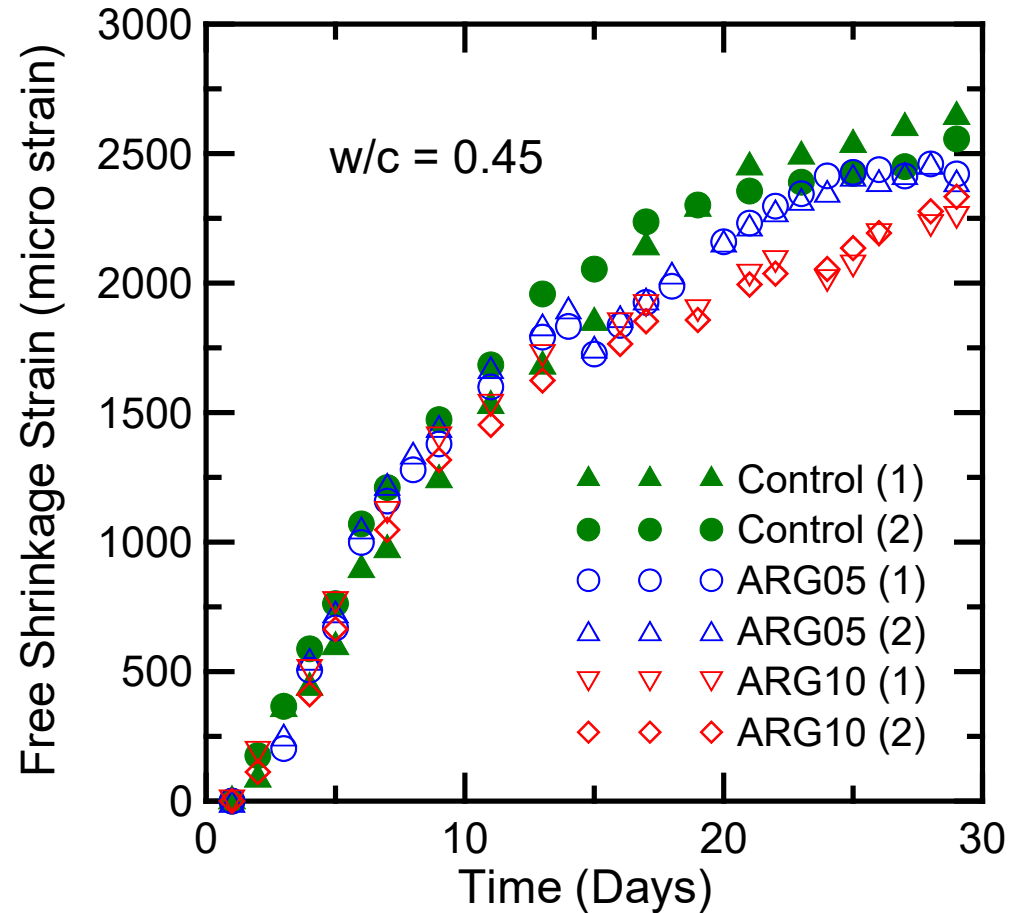


Cracking due to Low Tensile Strength of Cementitious Materials

Free shrinkage test for hardened concrete



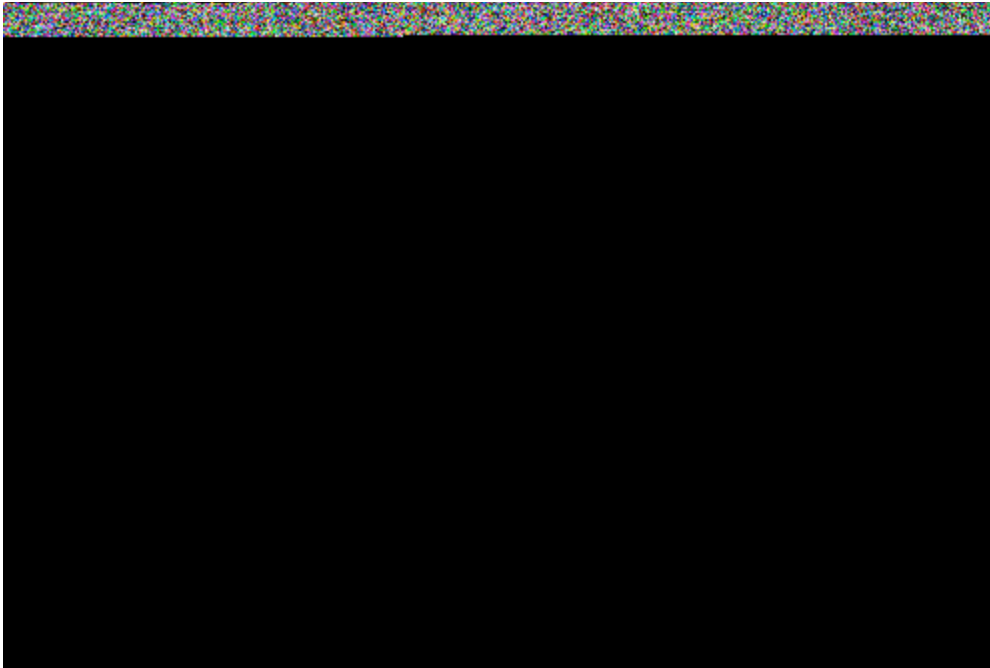
- Tests are performed according to ASTM C157



ACI nomograph for estimating rate of evaporation

- MENZEL'S FORMULA**

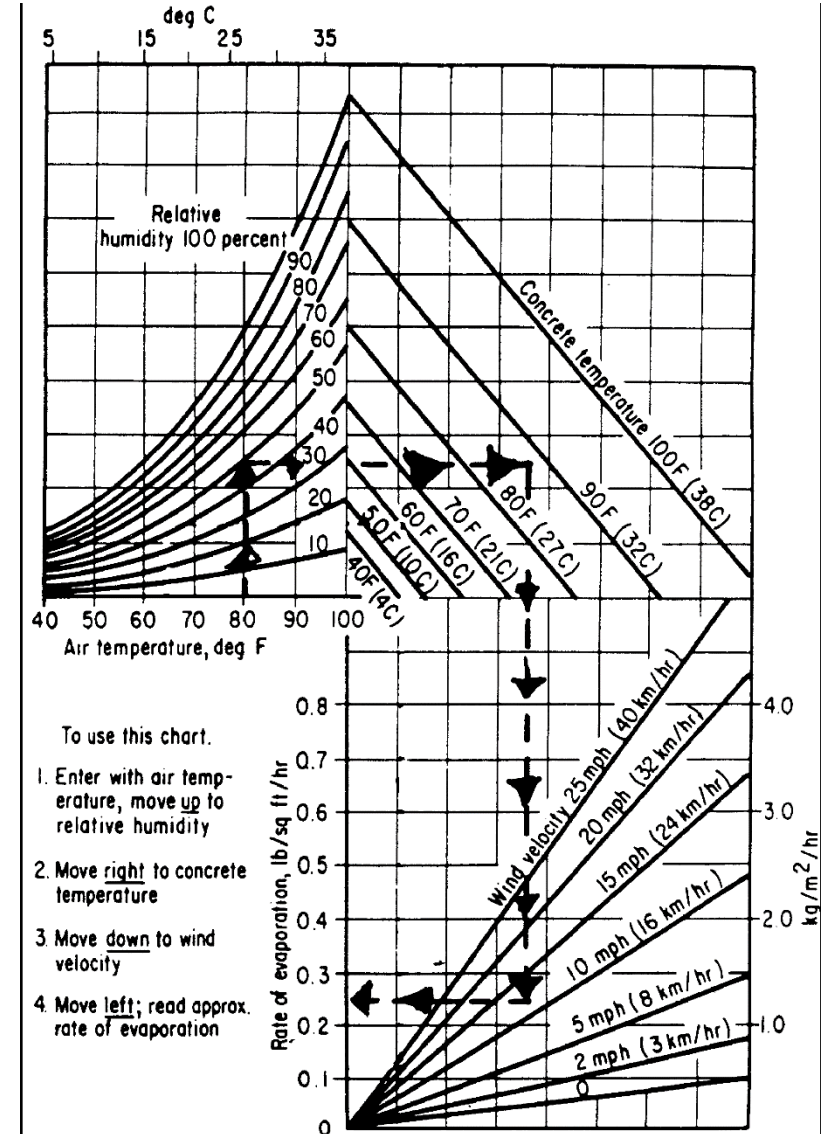
$$E = 0.44 (e_o - e_a)(0.253 + 0.096V)$$



- Water Evaporation from lake Hefner**

To Prevent Plastic Shrinkage Cracks:

- ACI:** Rate of Evaporation (RE) < 0.2 lb/ft²/hr (1.0 kg/m²/hr)



Prediction of free shrinkage using ACI 209-R 92 Model

The shrinkage strain $\epsilon_{sh}(t, t_c)$ at age of concrete t (days), measured from the start of drying at t_c (days), is calculated by

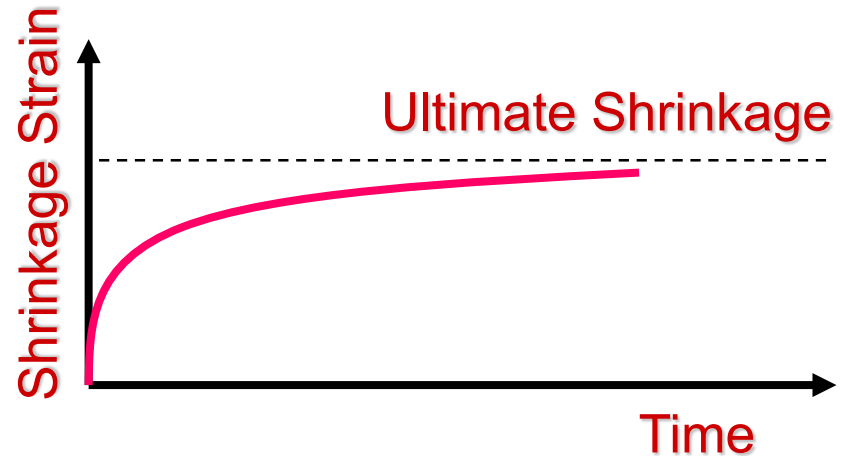
$$\epsilon_{sh}(t, t_c) = \frac{(t - t_c)^\alpha}{f + (t - t_c)^\alpha} \cdot \epsilon_{shu}$$

For standard condition, $f = 35$ for 7-day moist-cured specimens and $f = 55$ for 1- to 3-day steam cured specimens and $\alpha = 1.0$.

The shape and size effect can be considered by replacing $\alpha = 1.0$ which is an average value for a flatter hyperbolic curve and f as

$$f = 26.0e^{[0.36 (V/S)]}$$

where V/S is the volume-surface ratio.



Prediction of Ultimate Shrinkage

$$\varepsilon_{shu} = 780 \gamma_{sh,tc} \gamma_{sh,RH} \gamma_{sh,vs} \gamma_{sh,s} \gamma_{sh,\psi} \gamma_{sh,c} \gamma_{sh,\alpha} \times 10^{-6} \text{ in./in.}$$

$\gamma_{sh,tc}$: Initial moist curing coefficient for curing different from 7 days.

$\gamma_{sh,RH}$: Ambient relative humidity coefficient different than 30%

$\gamma_{sh,vs}$: Adjustment factor for the size of the member as volume-surface ratio

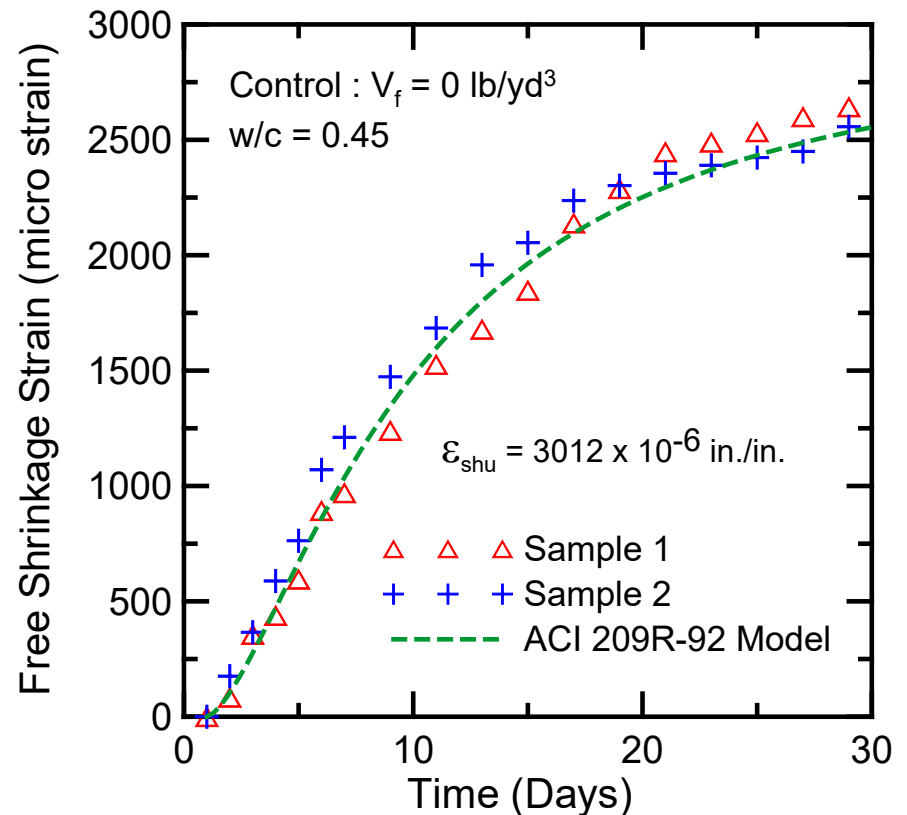
$\gamma_{sh,s}$: Slump factor

$\gamma_{sh,\psi}$: Fine aggregate factor

$\gamma_{sh,c}$: Cement content factor

$\gamma_{sh,\alpha}$: Air content factor

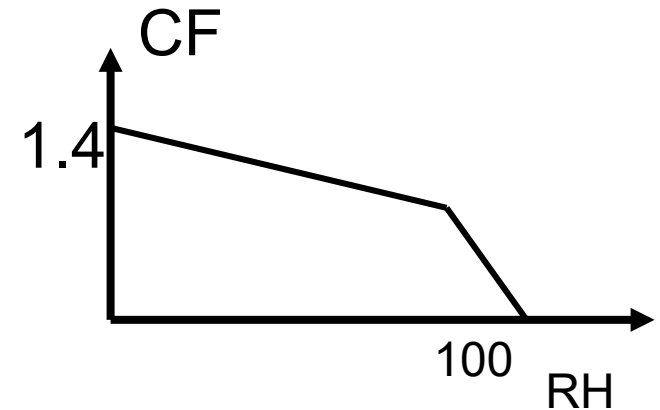
$$\varepsilon_{sh}(t, t_c) = \frac{(t - t_c)^\alpha}{f + (t - t_c)^\alpha} \cdot \varepsilon_{shu}$$



Simple ACI approach for Shrinkage Prediction

$$(\epsilon_{sh})_T = \frac{t}{35+t} (\epsilon_{sh})_{ult} \quad (\epsilon_{sh})_{ult} = 730 \times 10^{-6} * CF$$

- RH=Relative Humidity (%)
- RH>80% CF=3.0 – 0.03 RH
- RH<80% CF=1.4-0.01 RH
- CF is correction factor applied to the ultimate shrinkage



Free Shrinkage Strain as function of Time-and humidity (Bazant-Panula Model)

$$z = 0.381 \sqrt{f'_c} * \left[1.25 \sqrt{\frac{a}{c}} + 0.5 \left(\frac{g}{s} \right)^2 \right] \sqrt[3]{\frac{1+s/c}{w/c}} - 12.$$

$$y = (390 z^{-4} + 1)^{-1}$$

$$\epsilon_{sh\infty} = 1330 - 970y$$

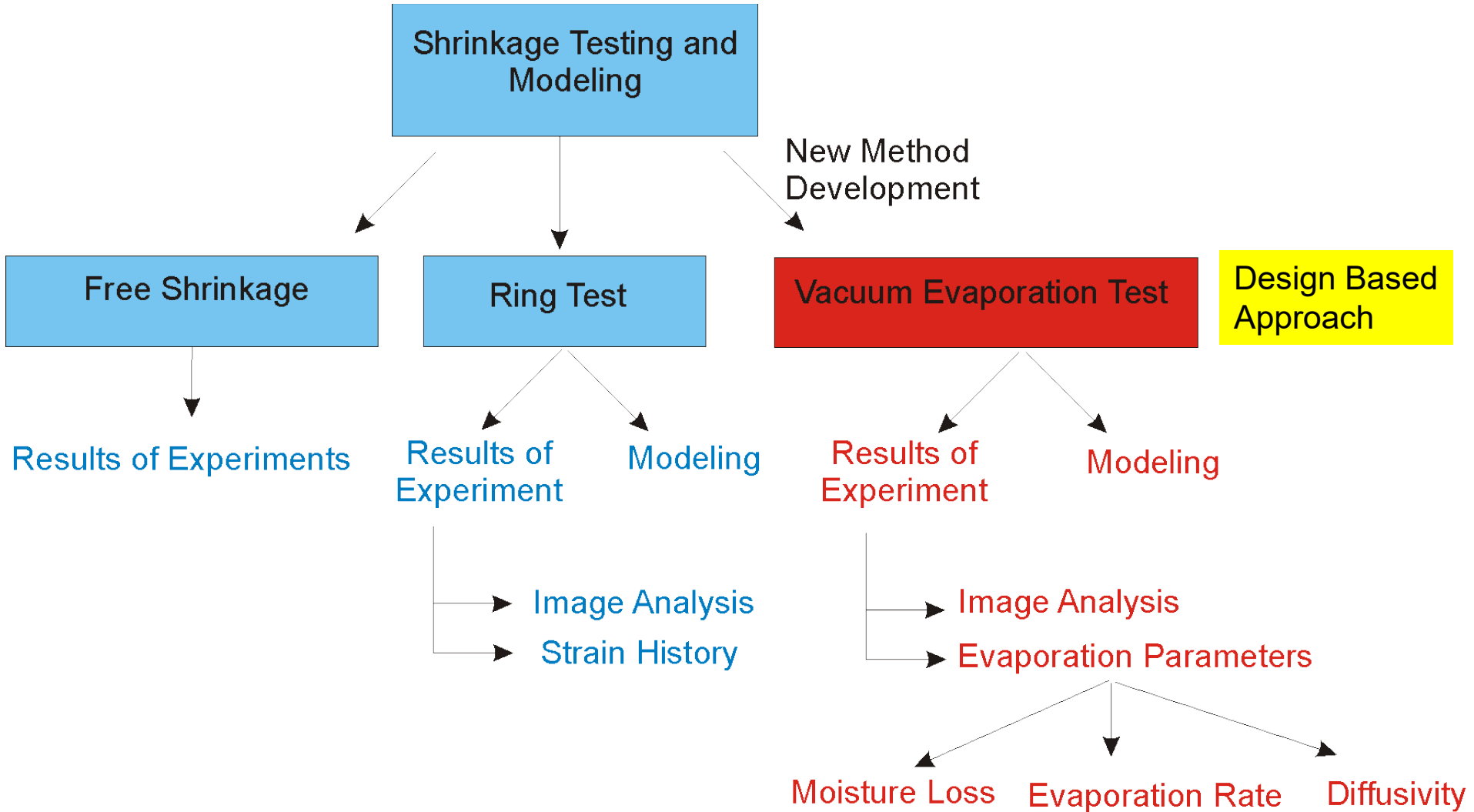
$$\epsilon_{sh} = \epsilon_{sh\infty} (1-h)^3$$

- a/c= aggregate to cement ratio
- s/c= sand to cement ratio
- g/s = coarse aggregate to fine aggregate ratio
- w/c= water cement ratio

Reinforcement Design for Shrinkage and Temperature control

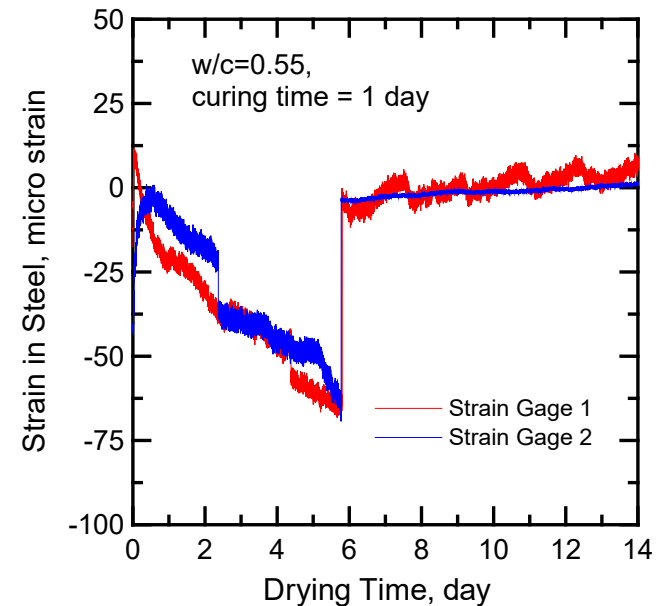
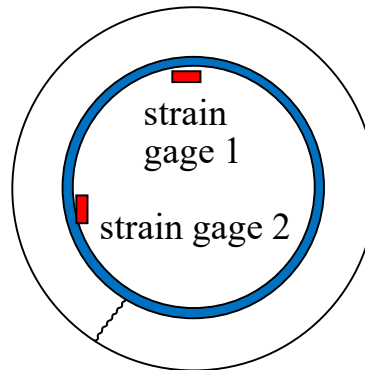
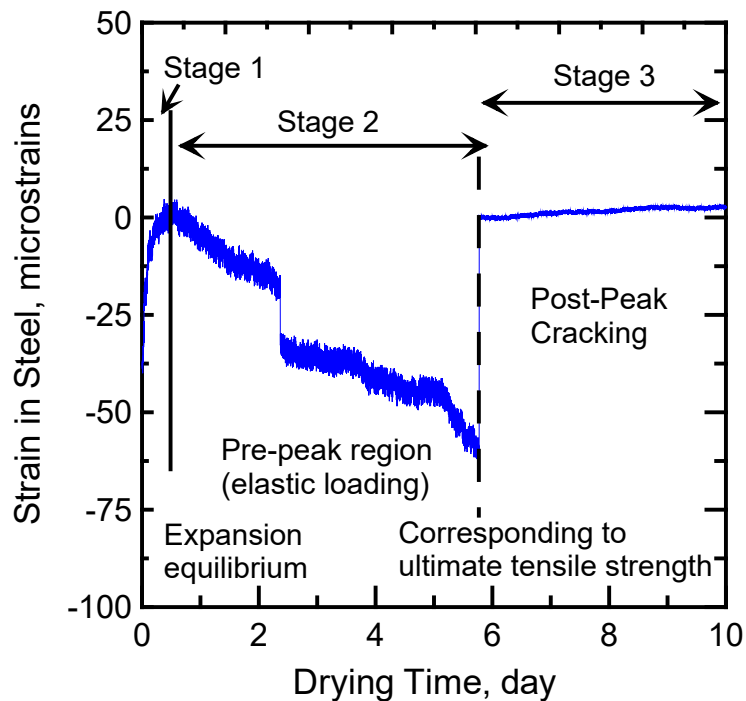
- empirical guidelines
 - ACI 318-05 specifies the reinforcement for control of temp and shrinkage cracking as 0.18% of the gross area for Grade 6 reinforcement.
 - For the restrained shrinkage case, ACI 318-05 states that the effects of shrinkage shall be considered, no further guidance.
 - ACI 318-05 requires that reinforcement not be spaced more than the lesser of three times the thickness of the deck or 18 in (450mm).
 - ACI 224-01: A reinforcement ratio of 0.18% will not control crack widths at acceptable levels, and a ratio of 0.60% should be used instead.
 - No serious treatment of fibers as the standard approach to address the deficiency.
 - Polymeric fibers and steel fibers perform quite well
 - primarily guided by manufacturer's recommendations of dosage

Multi-Scale testing and Modeling



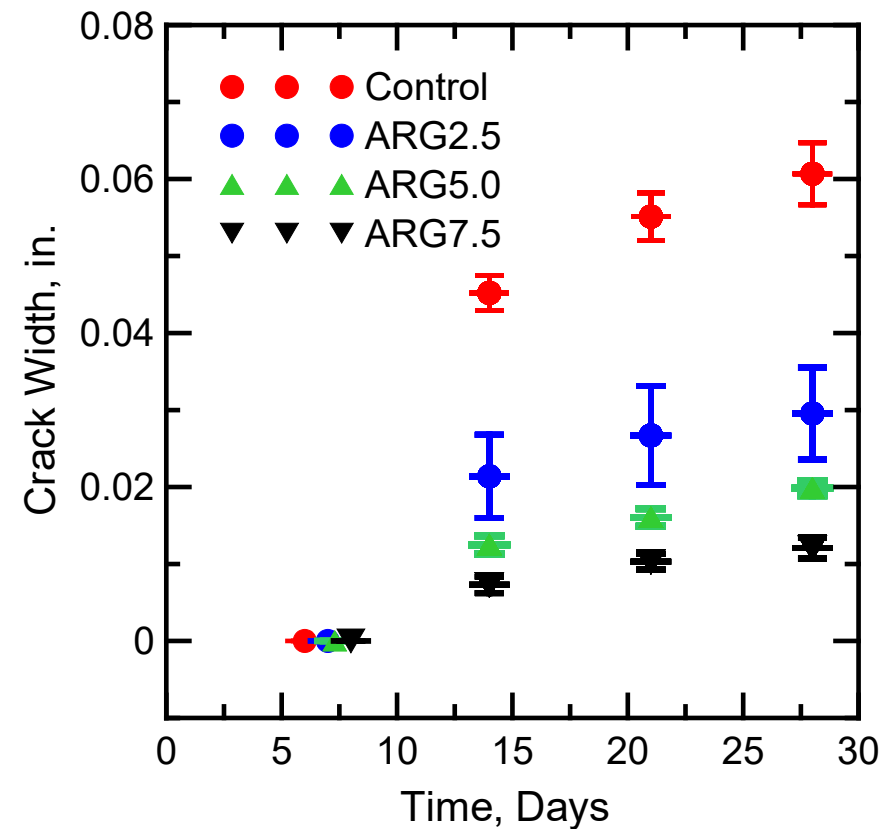
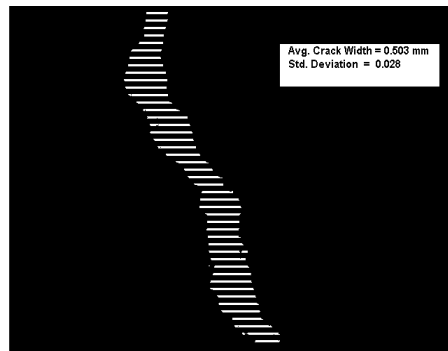
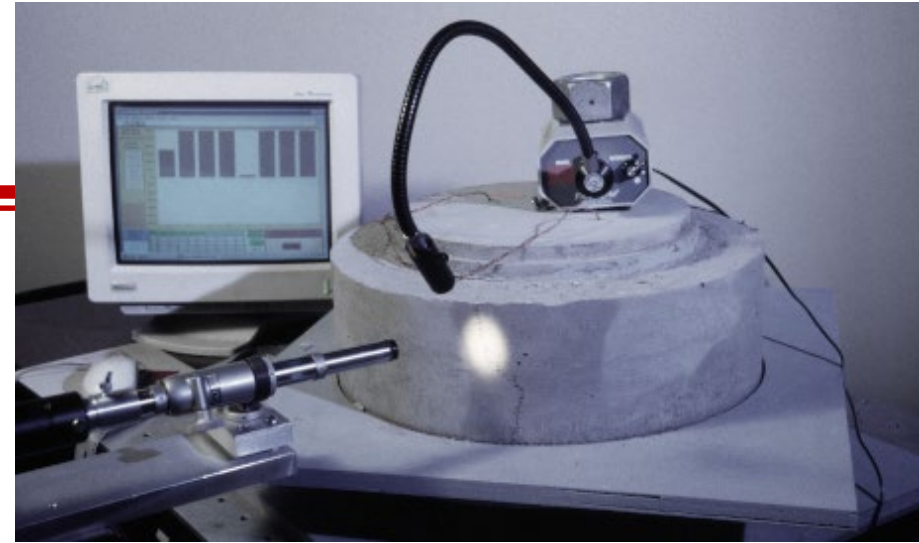
1D Shrinkage Cracking Ring Test AASHTO PP34-99

- Curing: 24 h under plastic sheets
- Relative humidity: $30\% \pm 1\%$
- Temperature: $40 \pm 1^\circ\text{C}$

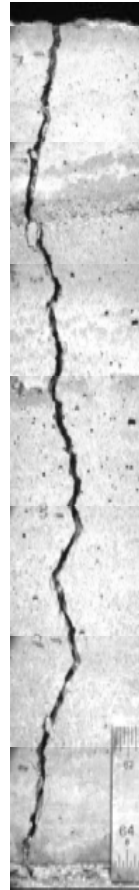
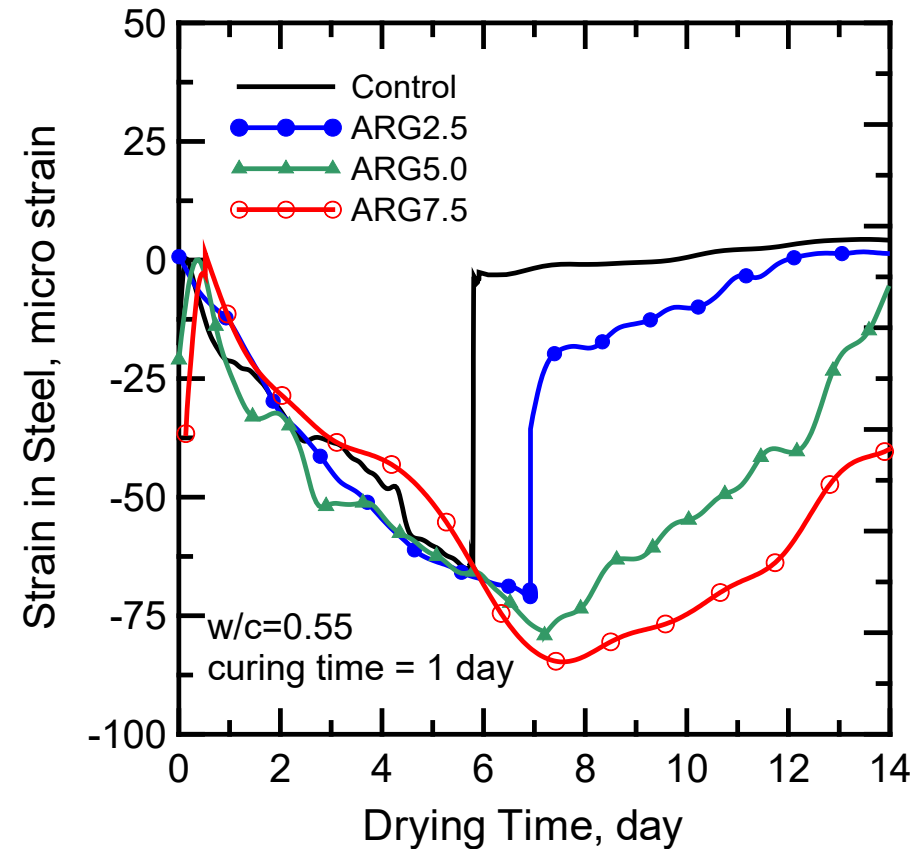


A typical result of a strain gage attached to steel ring

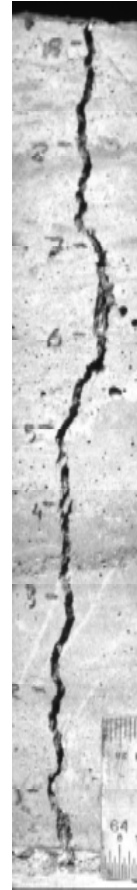
Image Analysis of 1D Restrained Shrinkage Cracks



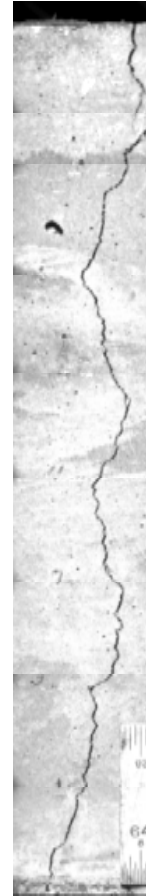
Effect of fiber addition on strain in the steel rings



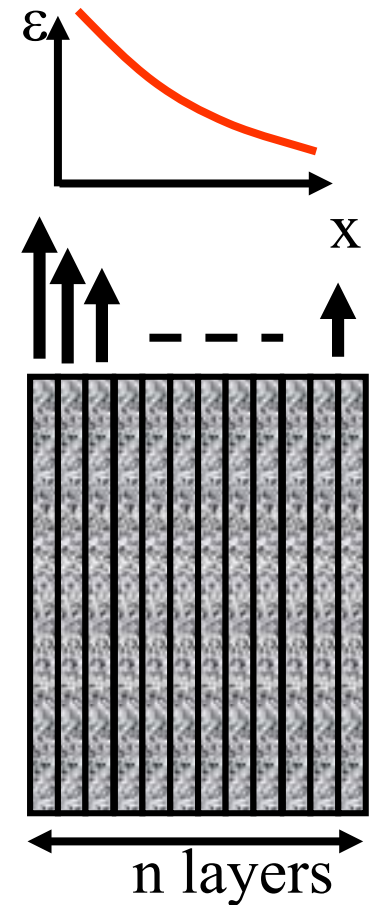
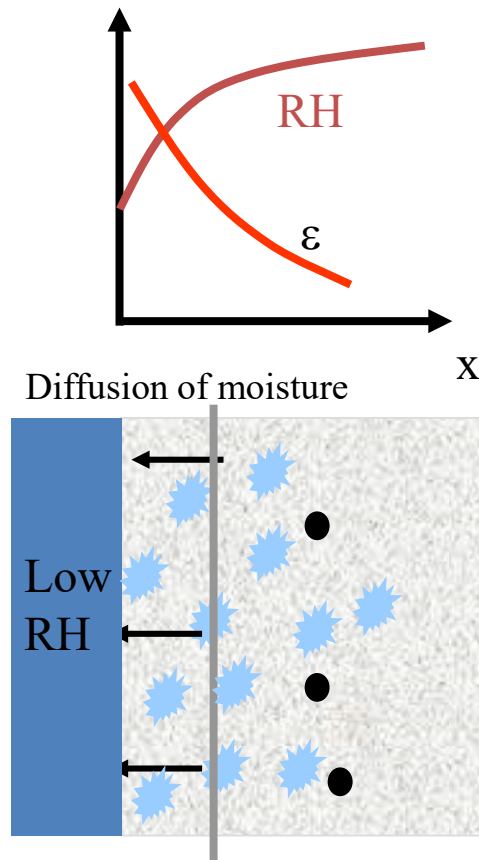
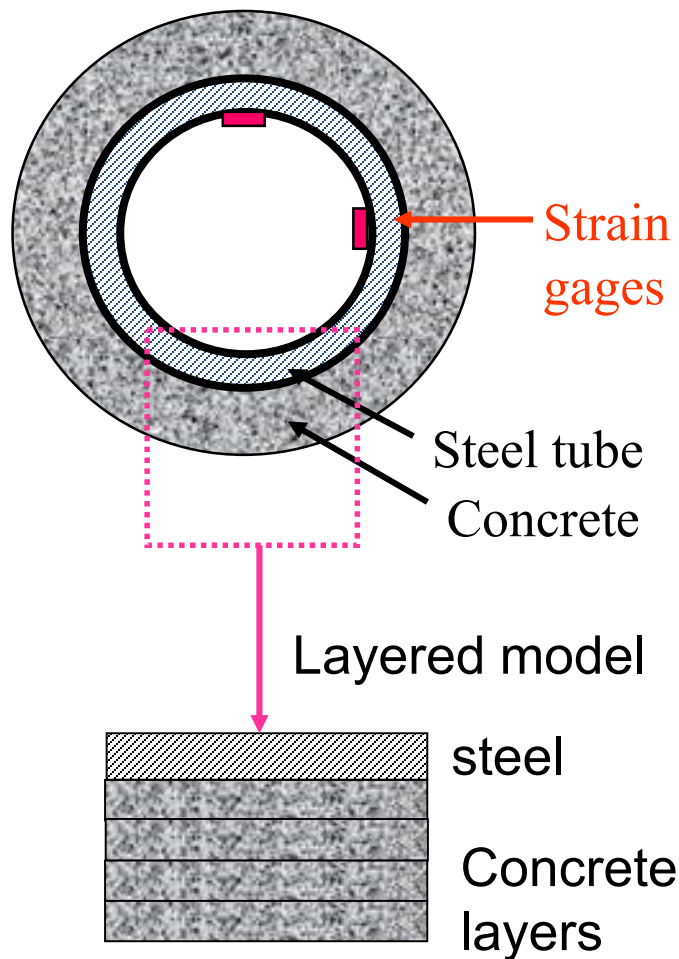
Average crack width: 1.15 mm
Standard deviation: 0.0787 mm



Average crack width: 0.341 mm
Standard deviation: 0.048 mm

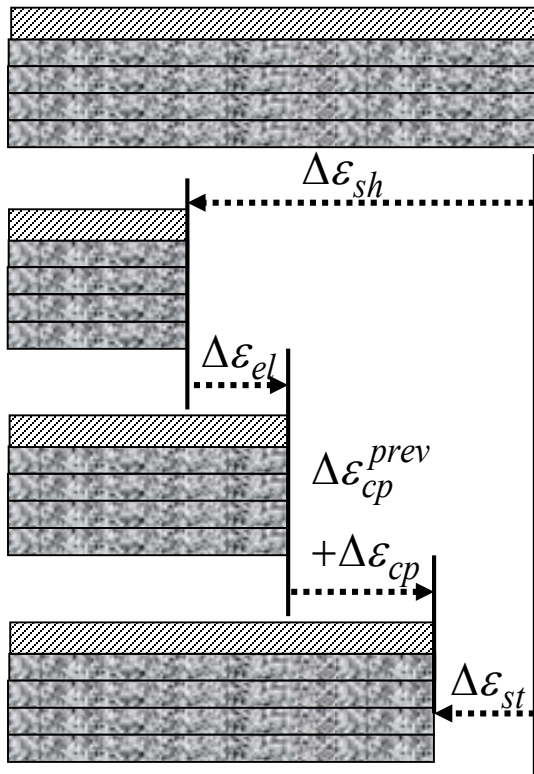


Induced Strain and Curvature in Layered Model Concept



- Moisture diffusion
- shrinkage and creep strain
- Development of early-age tensile strength

Restrained Shrinkage Test-Strain Compatibility Condition



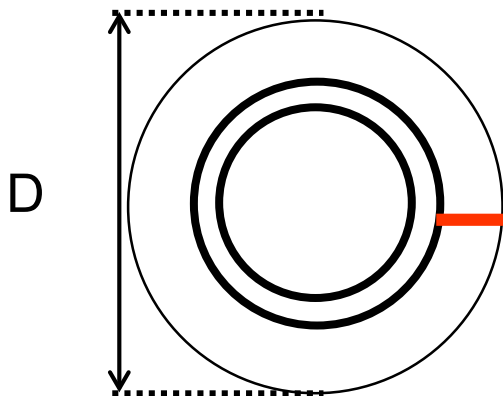
- Free shrinkage strain due to loss of moisture

- Tensile strain due to restraining force of steel

- Additional tensile strain due to creep effect

- Final incremental strain at equilibrium

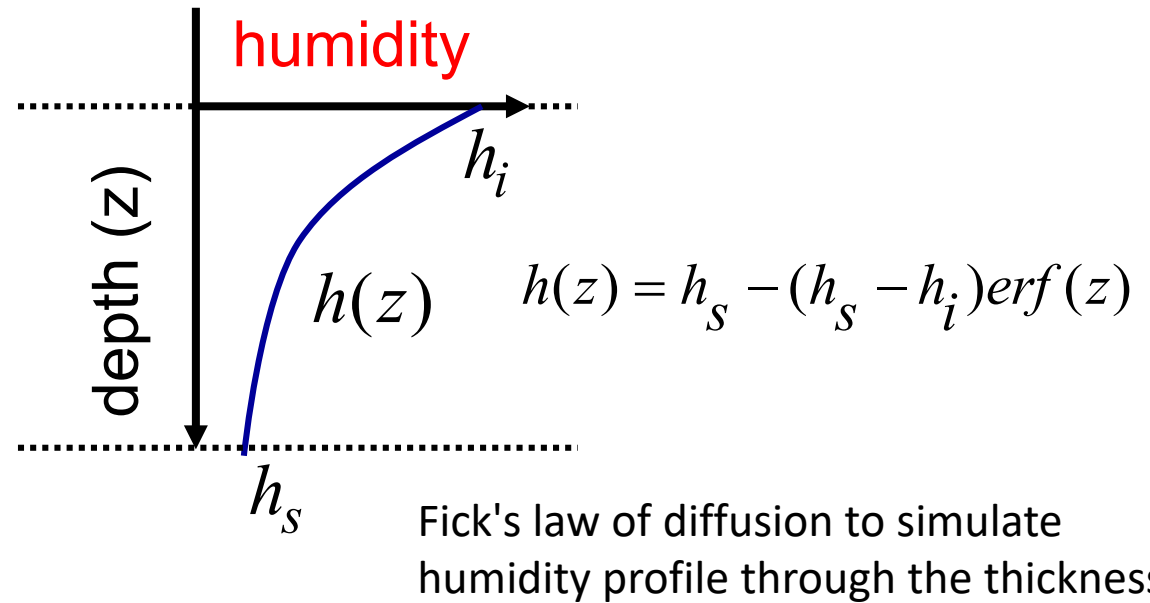
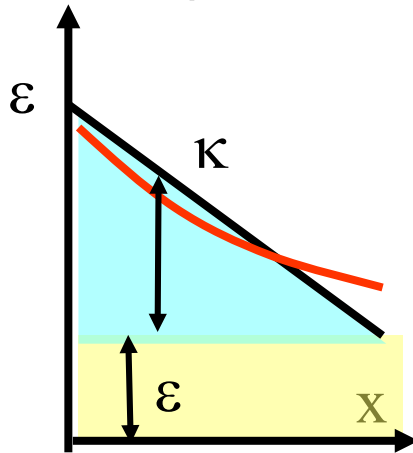
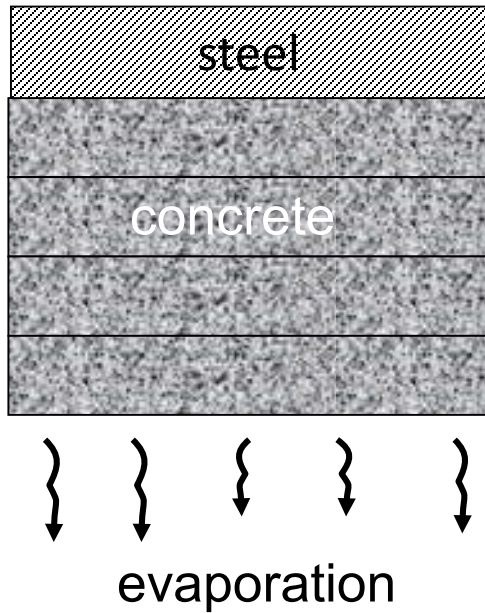
$$-\Delta\epsilon_{sh} = \Delta\epsilon_{el} + \Delta\epsilon_{cp} + \Delta\epsilon_{cp}^{(prev\sigma)} - \Delta\epsilon_{st}$$



- Crack width is a localization of smeared strain along circumference

$$w = (\epsilon_t - \epsilon_{t2}) \times \pi D$$

Moisture Diffusion, average strain and curvature as variables time and space

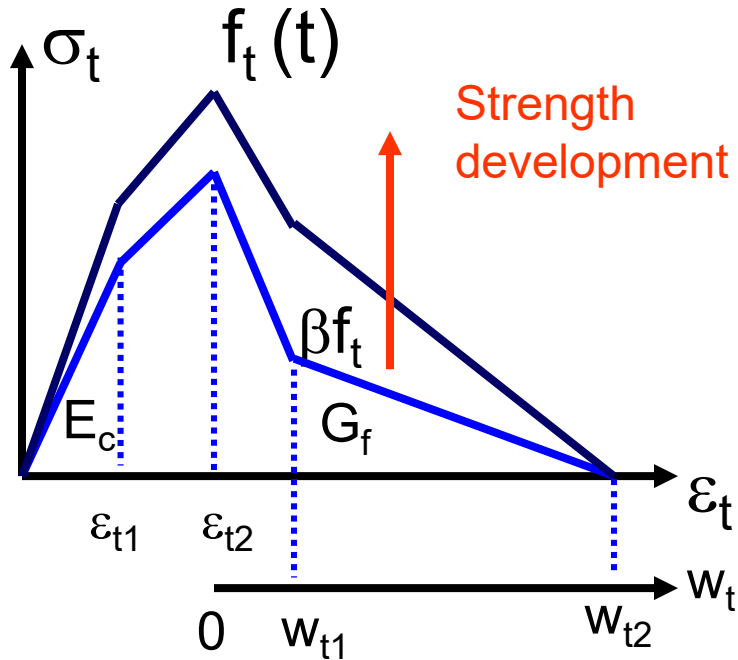


$$\epsilon_{sh}(t, x) = \bar{\epsilon}_{sh}(t) + \bar{K}_{sh}(t) x_i$$

$$\bar{\epsilon}_{sh}(t) = \frac{1}{n} \sum_{i=1}^n \epsilon_{sh}(t, i)$$

$$\bar{K}_{sh}(t) = \frac{1}{n} \sum_{i=2}^n [(\epsilon_{sh}(t, i) - \epsilon_{sh}(t, i-1)) / (x_i - x_{i-1})]$$

Early-Age Constitutive Relationship



Tensile strength $f_t(t) = f_{t28} \sqrt{\frac{f_c(t)}{f_{c28}}}$

Fracture energy $G_f(t) = G_{f28} \sqrt{\frac{f_c(t)}{f_{c28}}}$

Young's modulus $E_c(t) = E_{c28} \sqrt{\frac{f_c(t)}{f_{c28}}}$

Compressive strength $f_c(t) = \frac{t}{a + bt} f_{c28}$

Average Shrinkage and Curvature

$$\bar{\epsilon}_{sh}(t) = \bar{\epsilon}_{sh} * \frac{t}{35 + t}$$

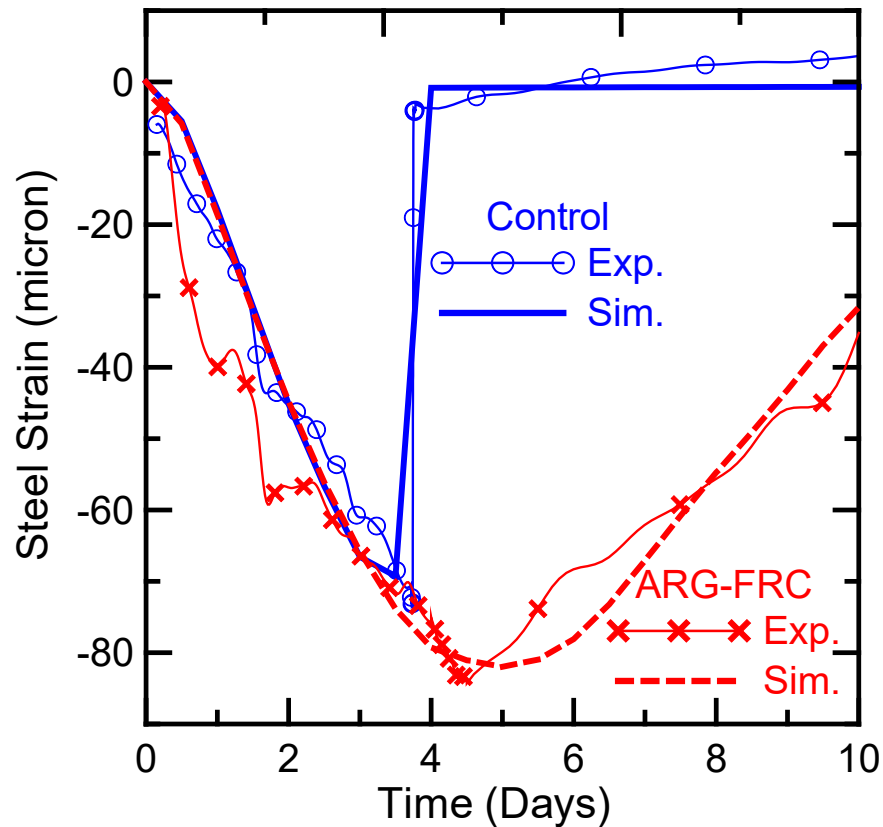
$$\bar{K}_{sh}(t) = \bar{K}_{sh} \frac{t}{35 + t}$$

Strength and Stiffness

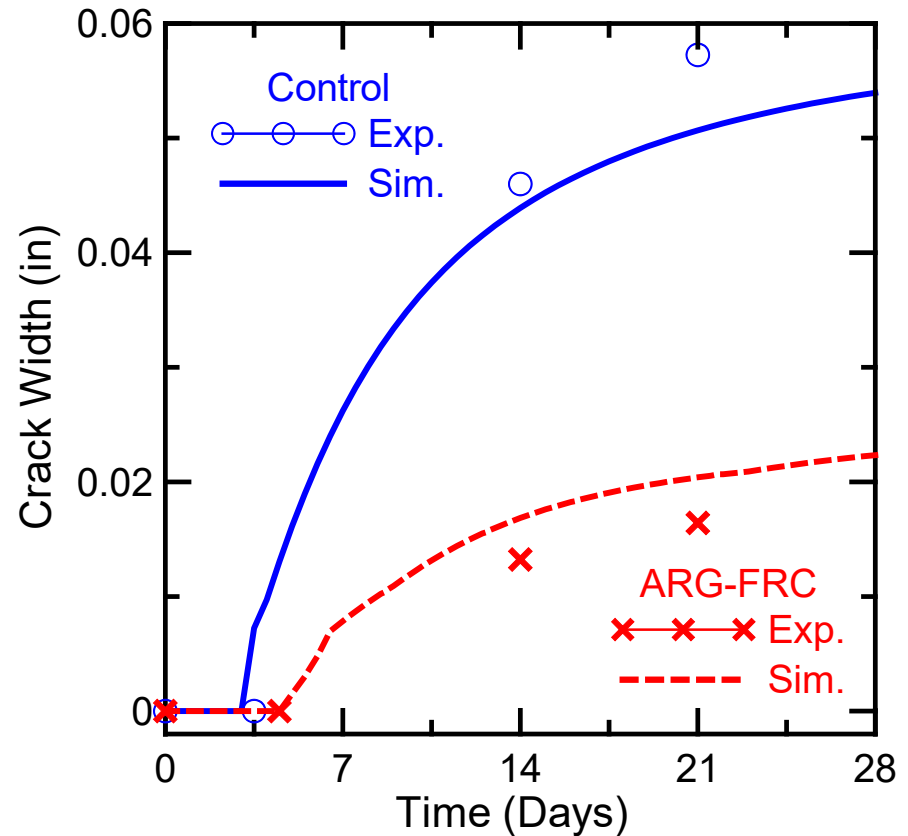
$$f_c(t) = f'_c \frac{t}{a + bt} \quad a = 4.0 \quad b = 0.85$$

$$E(t) = E \sqrt{\frac{f'_c(t)}{f'_c}}$$

Comparing Experimental Results and Model Simulation

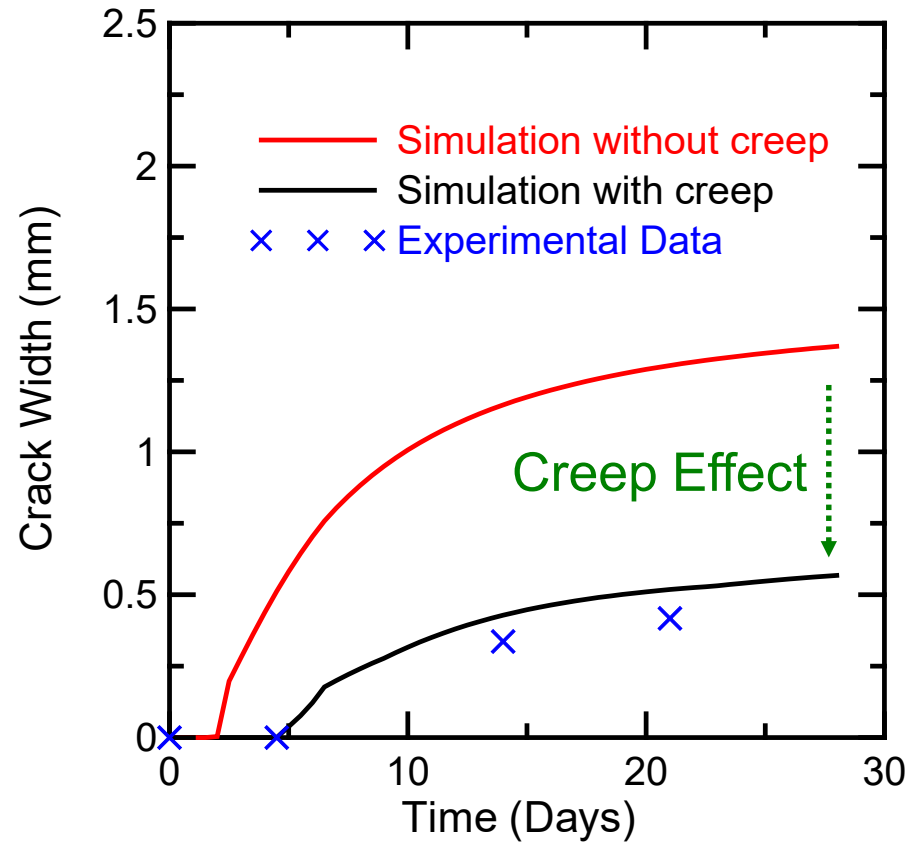
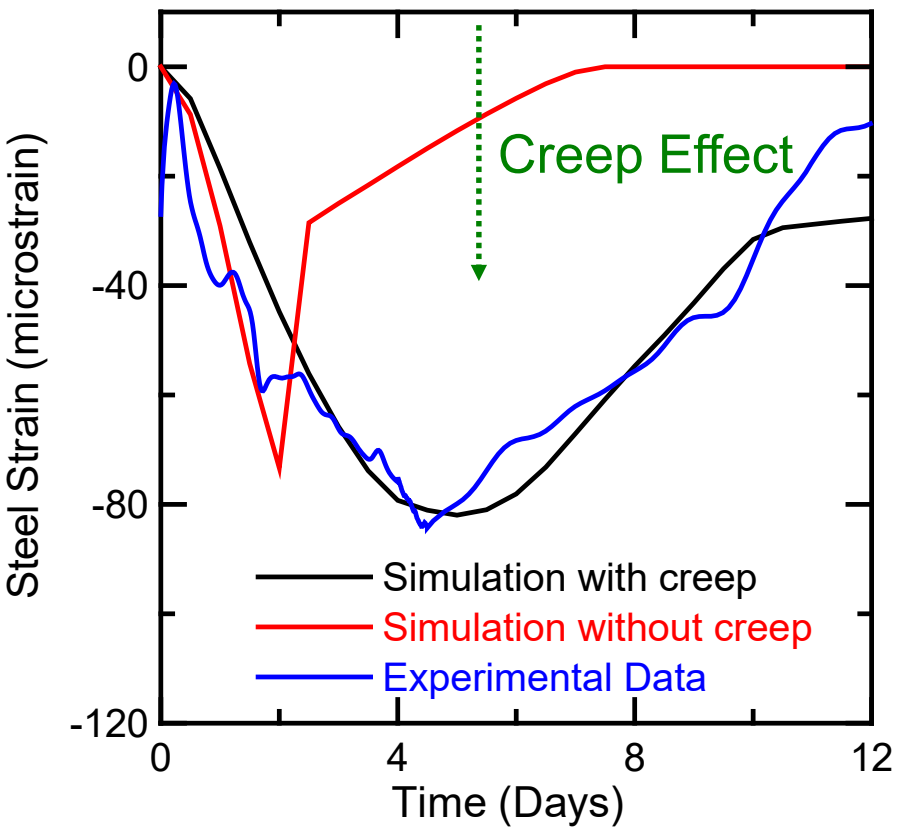


Strain history at steel ring



Crack width at concrete surface

Effect of creep in the simulation of AR-GFRC

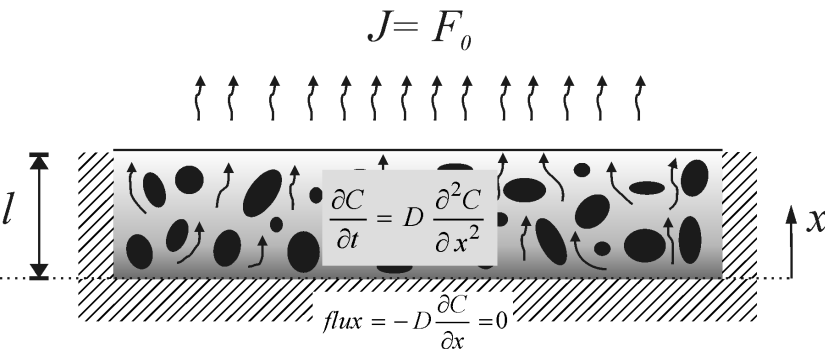
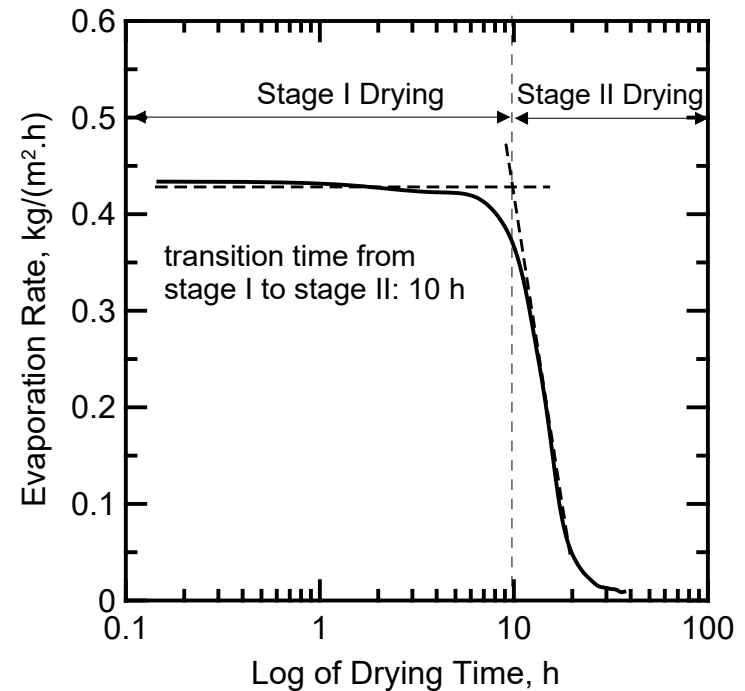


Recent Research on Early-Age Drying

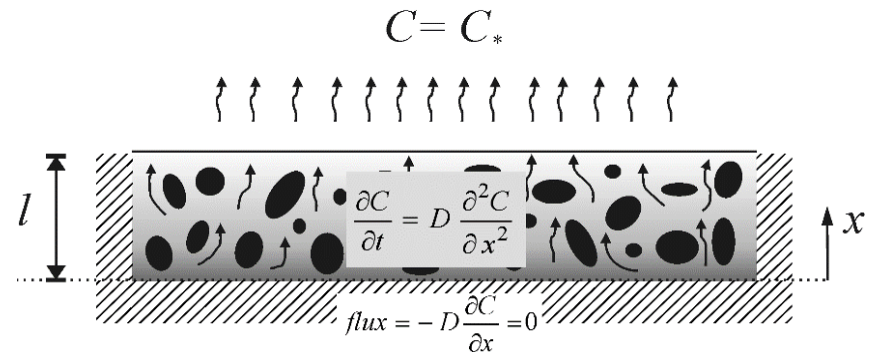
drying occurs in two distinct stages:

- Constant drying rate period (stage I)
- Falling rate drying period (stage II)

- In stage I drying, vapor phase diffusion process in surface is rate-determining, while in stage II drying, unsaturated flow process within porous medium is rate-determining.



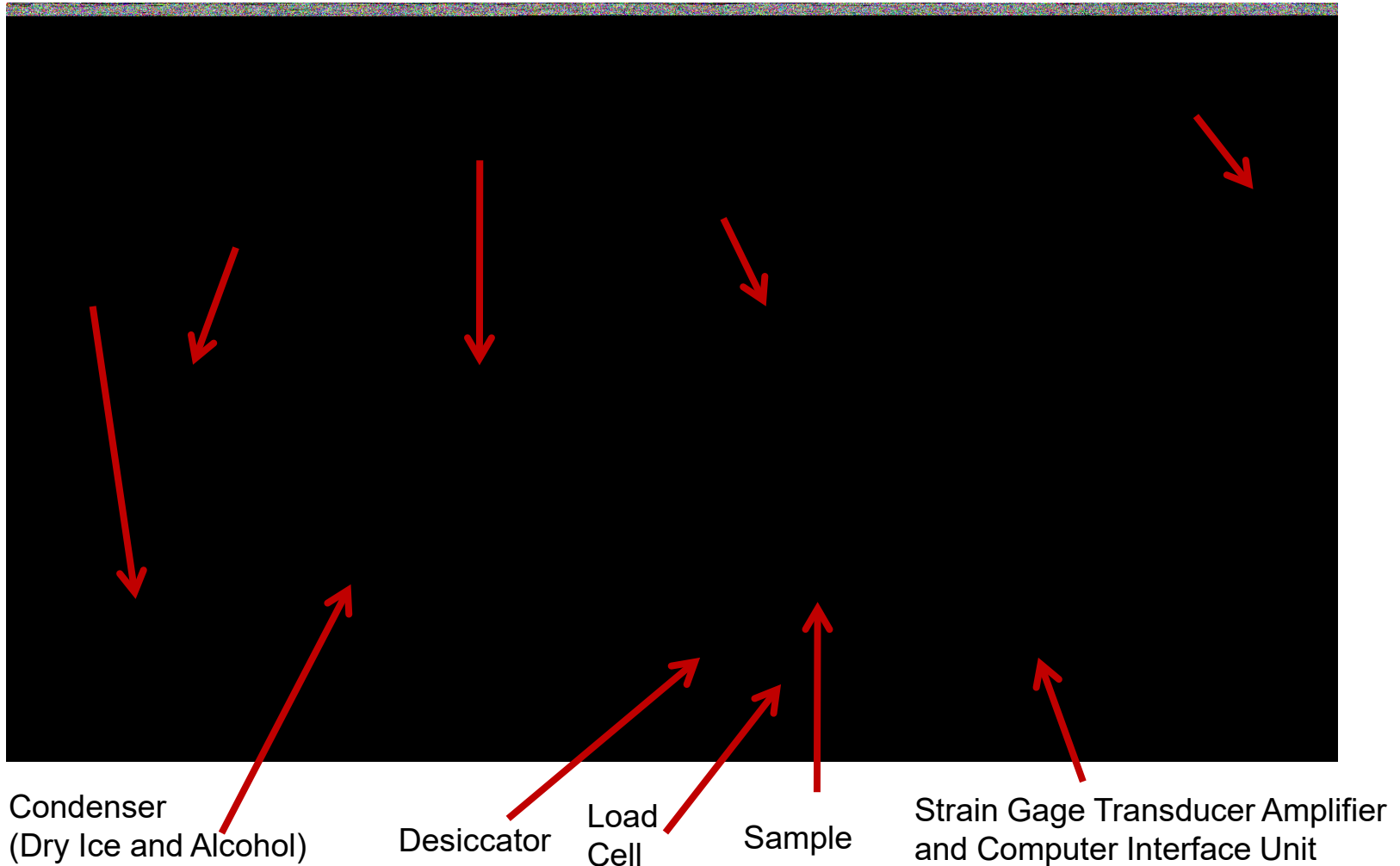
Stage I Drying



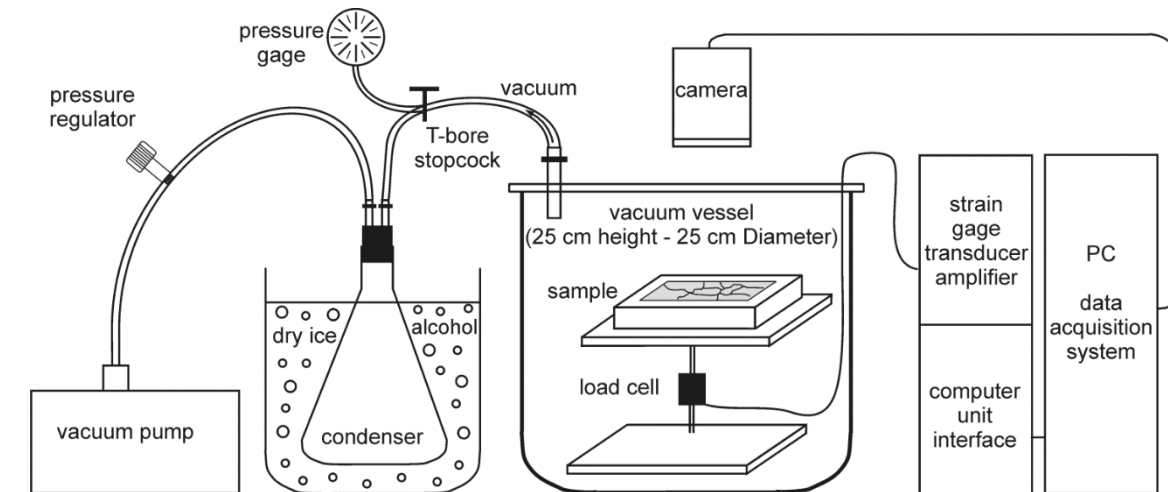
Stage II Drying

2-D Shrinkage Cracking

- Evaporation is simulated under low pressure condition.

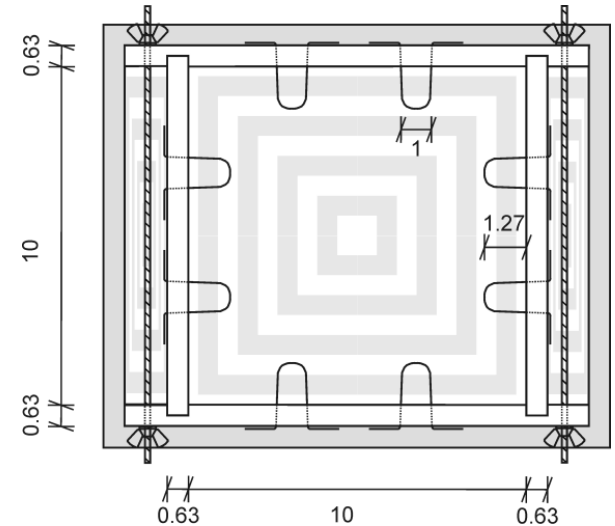


Development of a 2D Shrinkage Cracking Test

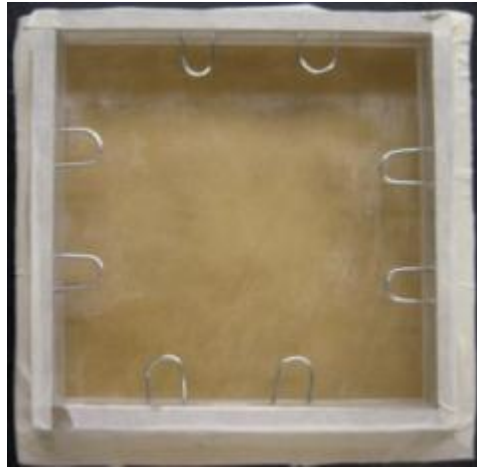


Load cell serves as a scale

Plan view of the mold (mm)

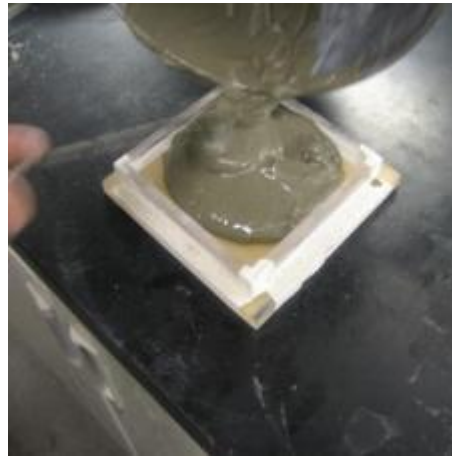


Test Method Procedures: Stages of Drying Cement Paste in Dessicator



Restrained Molds
(4"×4" ×0.5")

→
casting



Pouring cement paste

→
t =
30 min



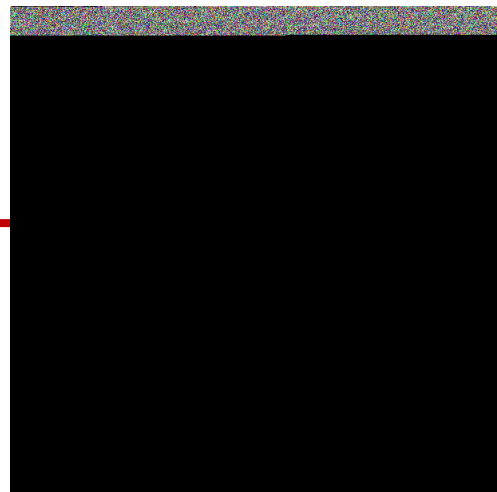
A film of water on surface

t = <1 hr



cracking on paste surface

t =
4 hr



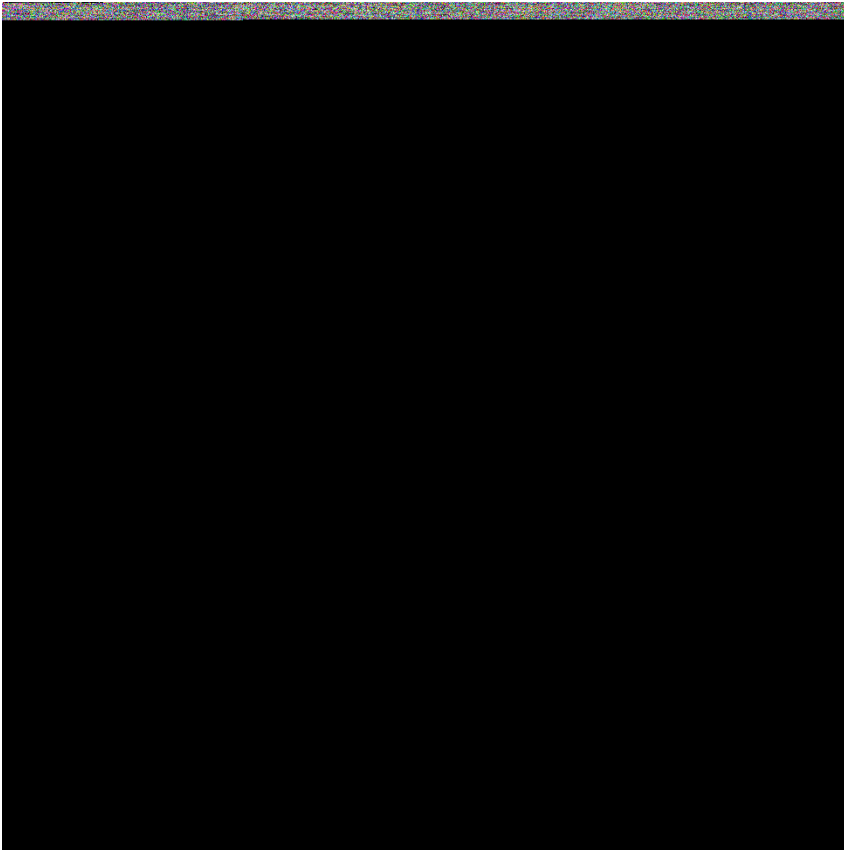
Formation of voids on paste surface

t =
2 hr



No layer of water on surface

Plastic Shrinkage Cracking



Control Specimen

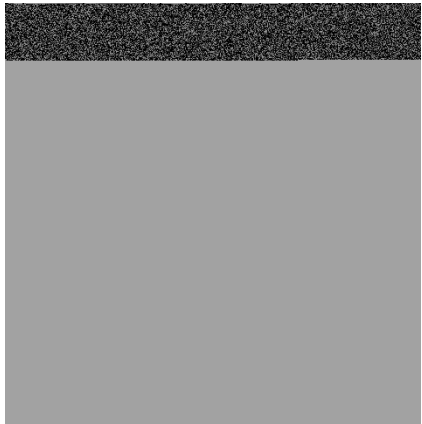


NYAD-G Specimen

Restrained Shrinkage Cracking : Crack Morphology



Control



C2000 (15%)



F55(15 %)

Comparison of Experimental Results

■ Compare Water and Concrete

■ Effect of Different Parameters

- w/c ratio
- Curing
- Fiber addition

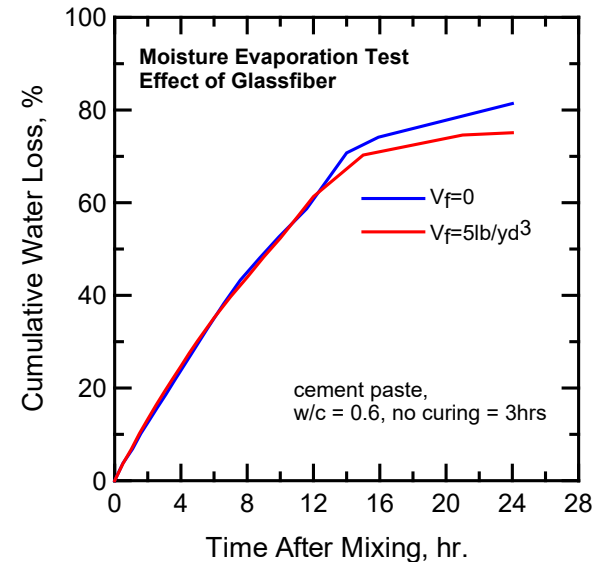
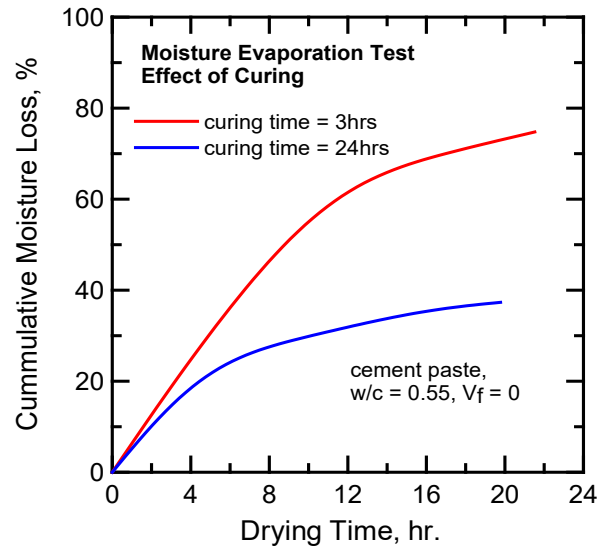
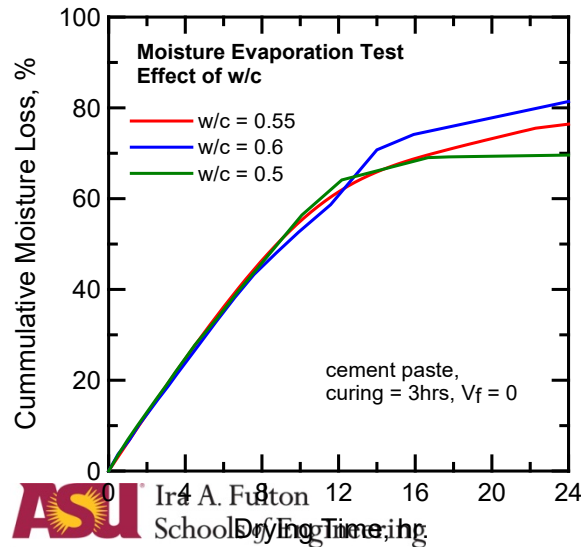
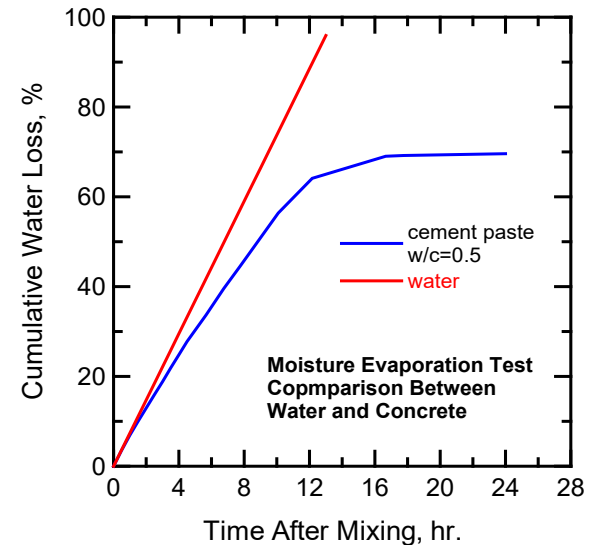
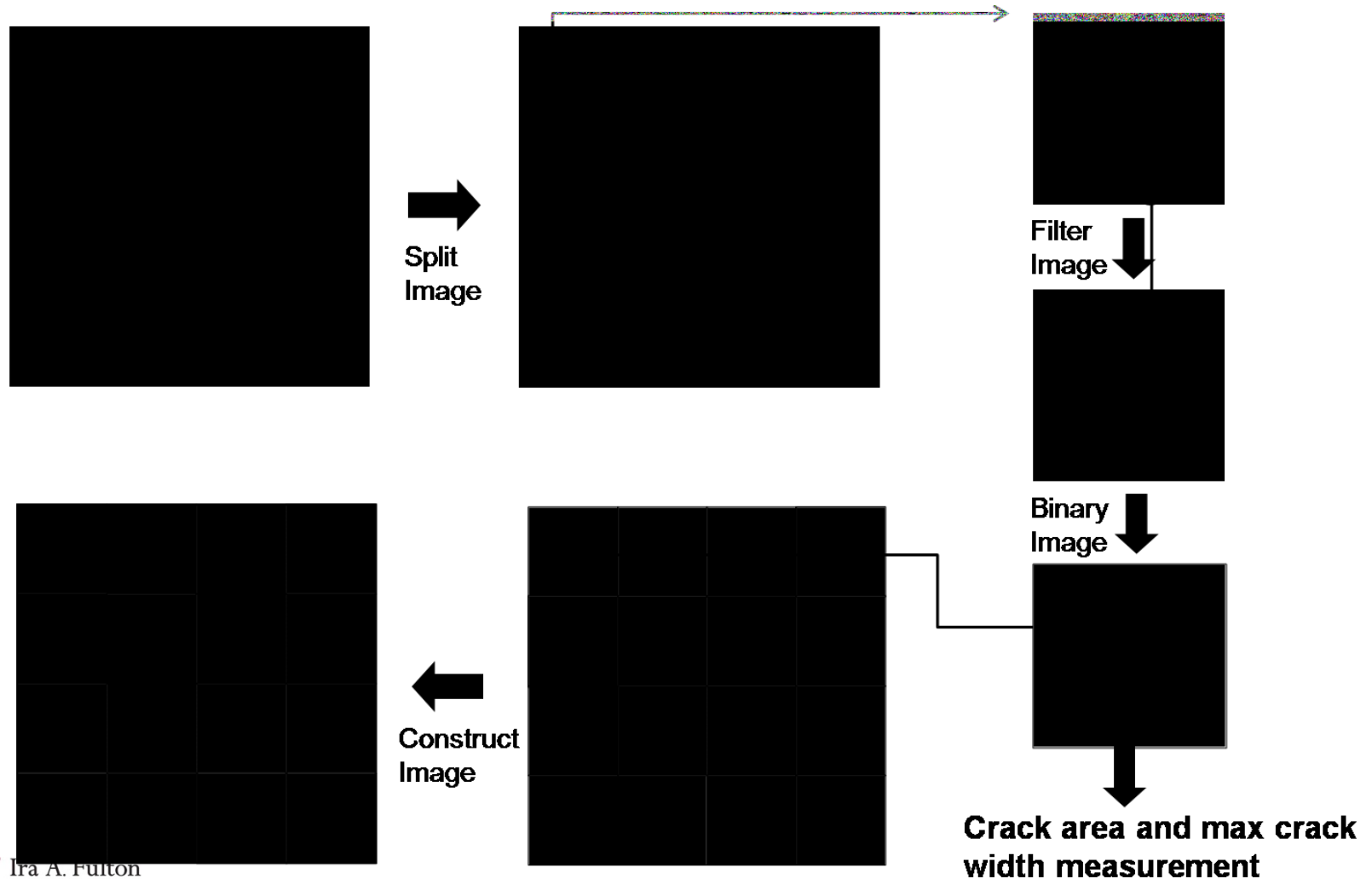
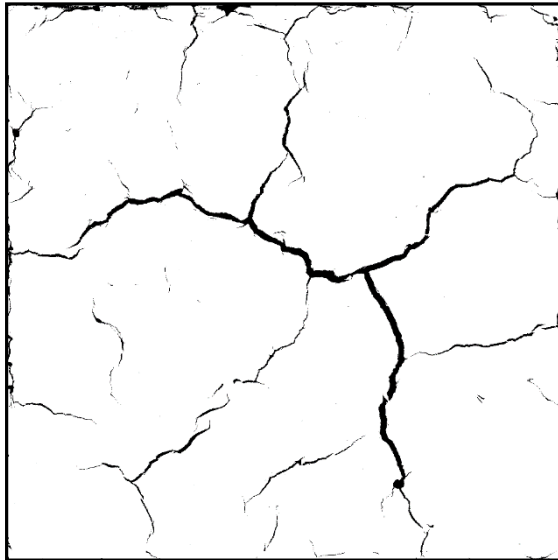


Image Analysis Process

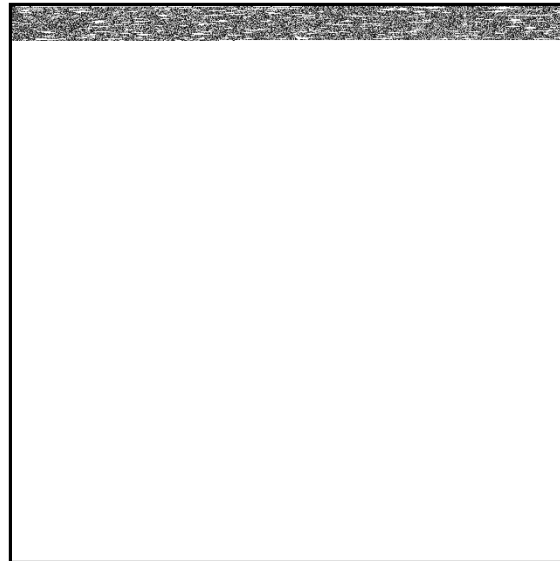


Results of image analysis on 2D cracks

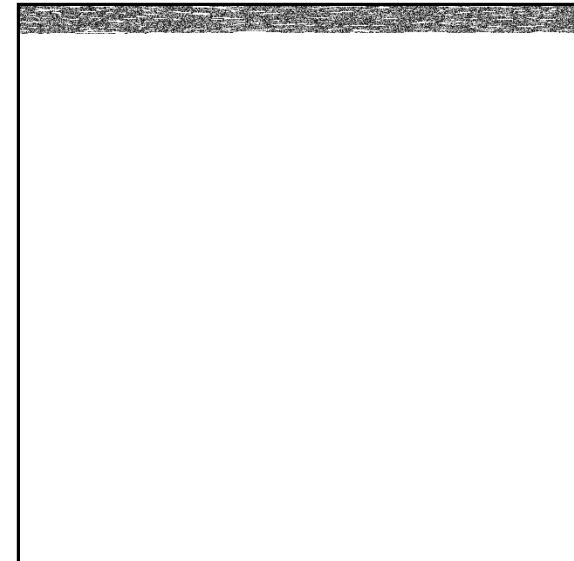
Plain Concrete



Fiber content = 5 lb/yd³



Fiber content = 10 lb/yd³

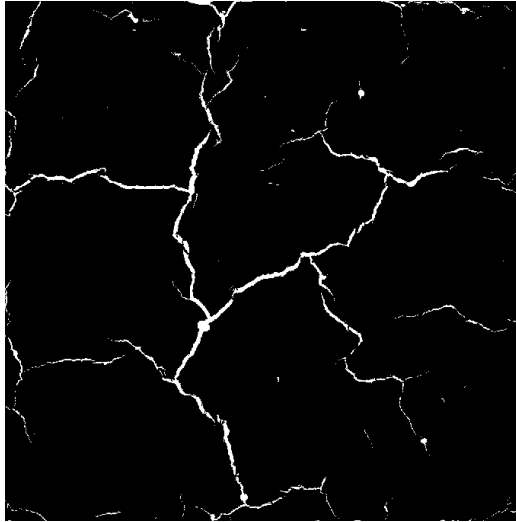


Sample ID	Average Max. Crack Width (in)	Reduction of Crack Max. Width by Adding Fiber (%)	Average Crack Density (in ² /in ²)	Reduction of Crack Density by Adding Fiber (%)
Control (Plain Concrete)	0.0504	-	0.0185	-
ARG5.0	0.0268	47%	0.0139	25%
ARG10	0.0144	134%	0.0068	84%

Analysis algorithm of shrinkage parameters

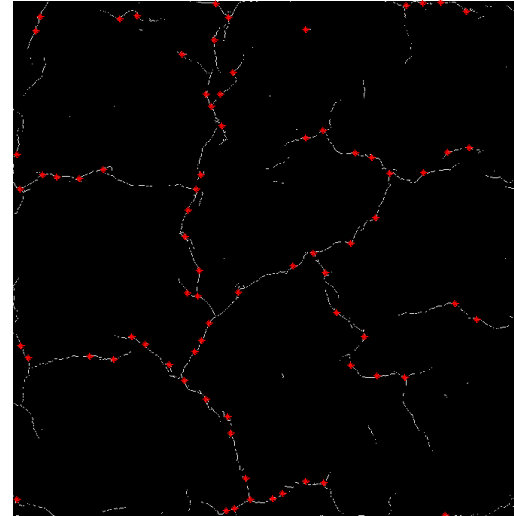
Step 1

Invert the
binary image of
crack
pattern



Step 2

Detect cracks
intersection
points from
skeletonized
image of
crack



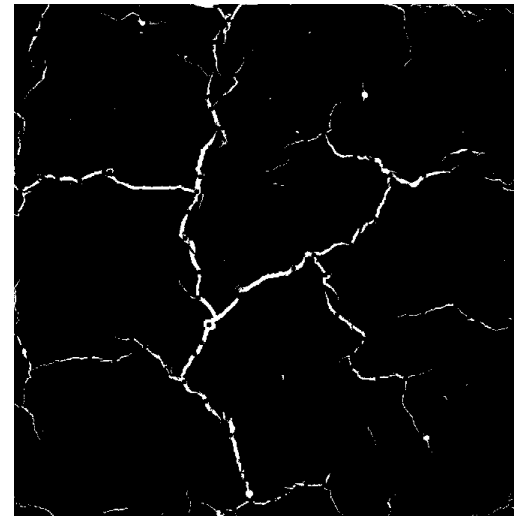
Step 3

Dilate the
intersection
n points

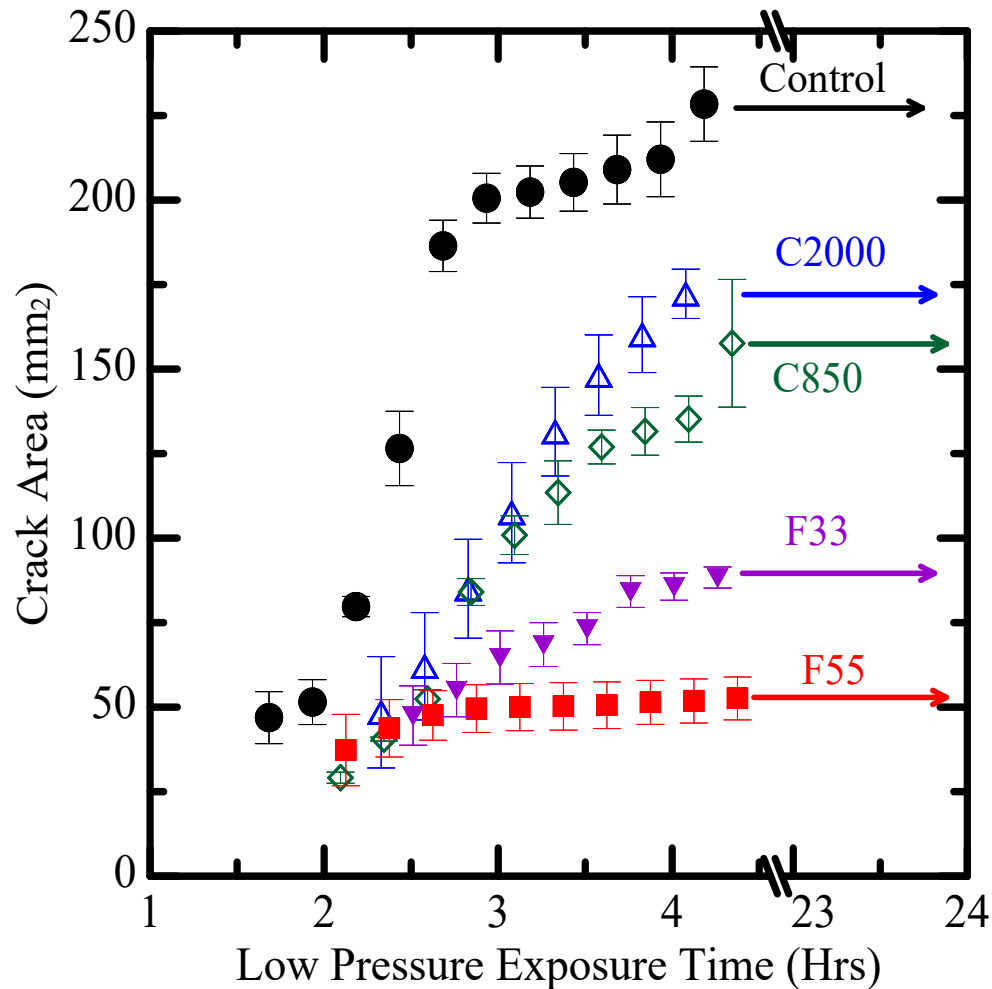


Step 4

Subtract
dilated
intersection
points from
the initial
binary image

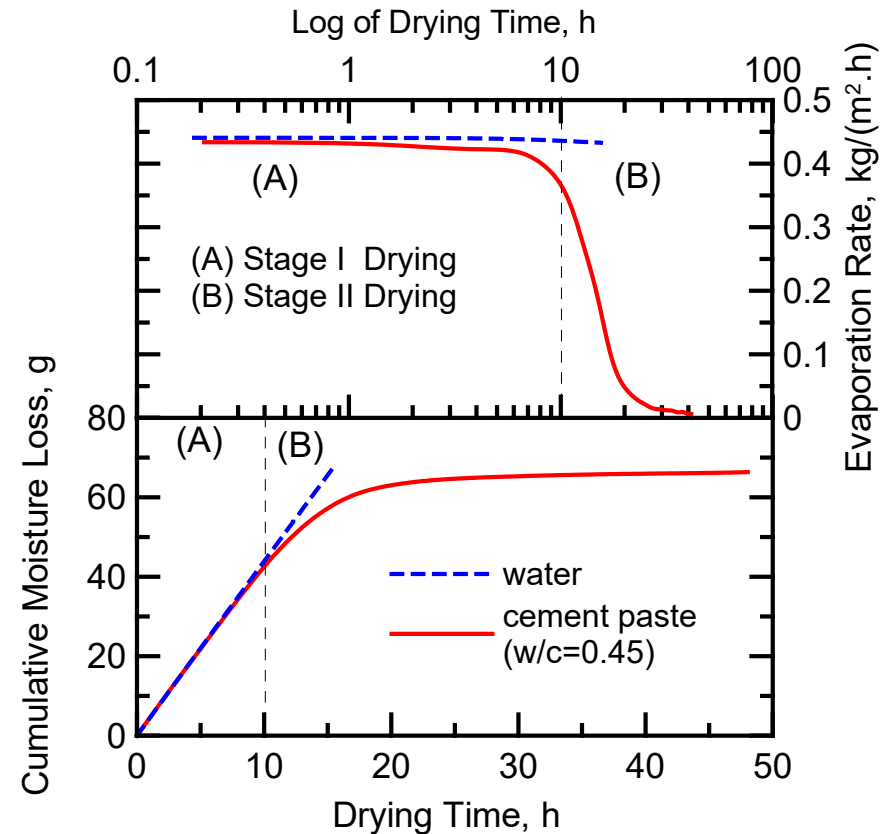
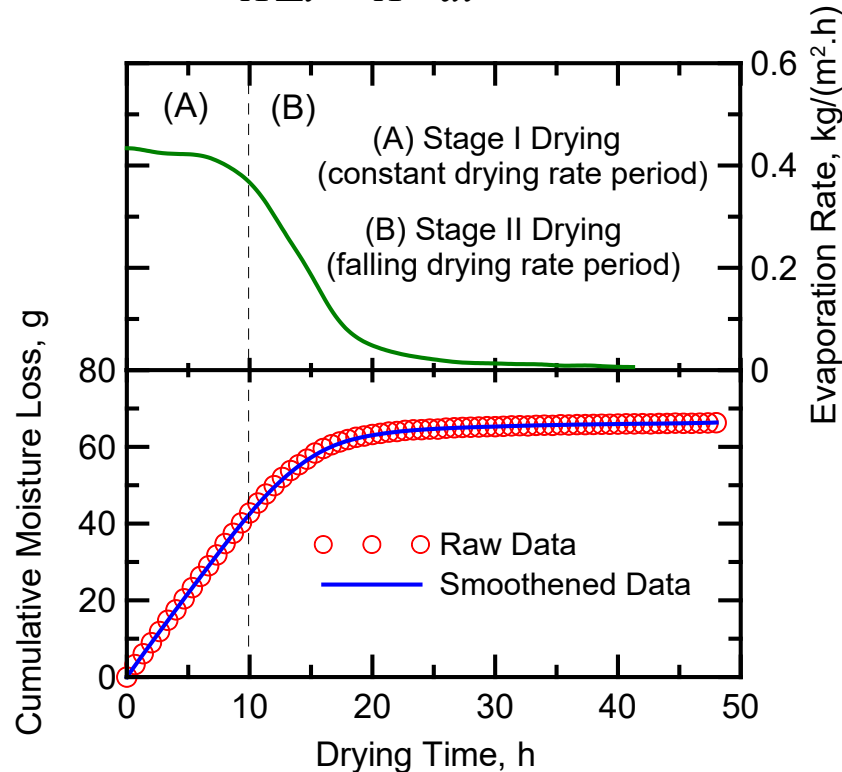


Shrinkage Crack Growth



Cumulative Loss and Evaporation Rate Analysis

$$J = \frac{\Delta M}{A \Delta t} = \frac{1}{A} \frac{dM}{dt}$$



- Similar to results obtained for wet capillary-porous materials such as clay, ceramics, paper, and leather.

Plastic Shrinkage Crack Modeling

- The model includes 4 major parts:
 1. Drying (Moisture Evaporation) Model
 - Gives moisture concentration profile at any time
 2. Desorption Isotherm
 - Relates moisture content to relative humidity (RH)
 3. Shrinkage Model
 - Relates RH to shrinkage strains using Kelvin's equation
 4. Material Strain-Stress and Crack Model
 - Predicts cracking time and crack width

1-D Drying Model and Analytical Solution

- Moisture transfer within the concrete described by Fick's 2nd law of diffusion;

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}, C(x, t); \quad 0 < x < L \quad \text{Diffusivity}$$

- Applied initial and boundary condition;

$$t = 0, \quad C = C_i; \quad 0 \leq x \leq L$$

$$t > 0, \quad \frac{\partial C}{\partial x} = 0; \quad x = 0$$

$$t > 0, \quad -D \frac{\partial C}{\partial x} = k \cdot (C(x = L, t) - C_\infty); \quad x = L$$

- Analytical Solution;

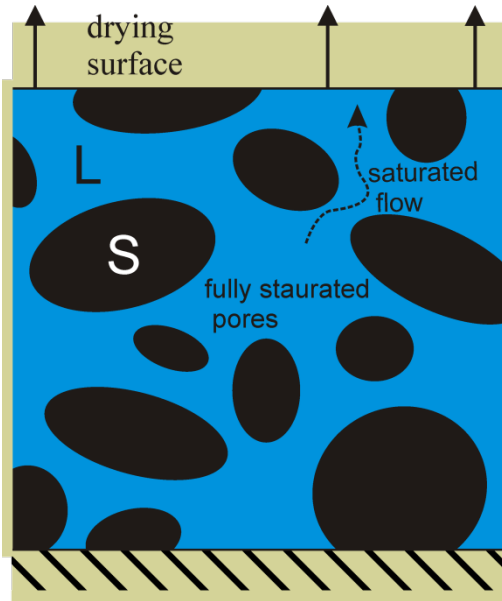
Convective moisture transfer coefficient

$$C(t, x) = C_\infty + (C_i - C_\infty) + \sum_{n=1}^{\infty} \frac{2 \sin(\lambda_n)}{\lambda_n + \sin(\lambda_n) \cos(\lambda_n)} \cos(\lambda_n \frac{x}{L}) e^{\left(-\lambda_n^2 \frac{Dt}{L^2}\right)}$$

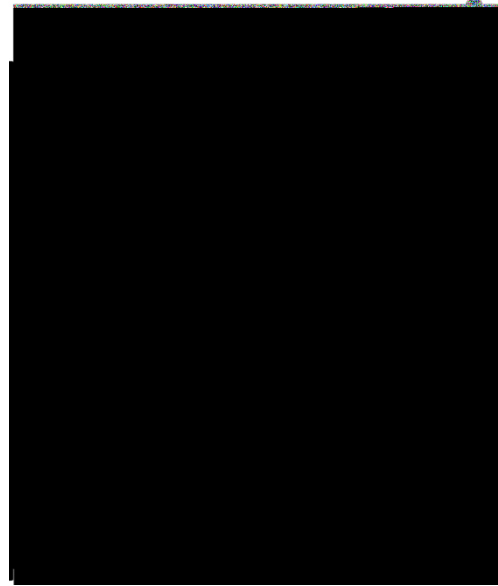
$$\lambda_n \tan(\lambda_n) = \frac{k_c L}{D}$$

Moisture Transport During Evaporation

Initial Condition
(full saturation)



Capillary Saturation
(continuous liquid phase)



hygroscopic state
(continuous vapor phase)



External Moisture Transfer Control

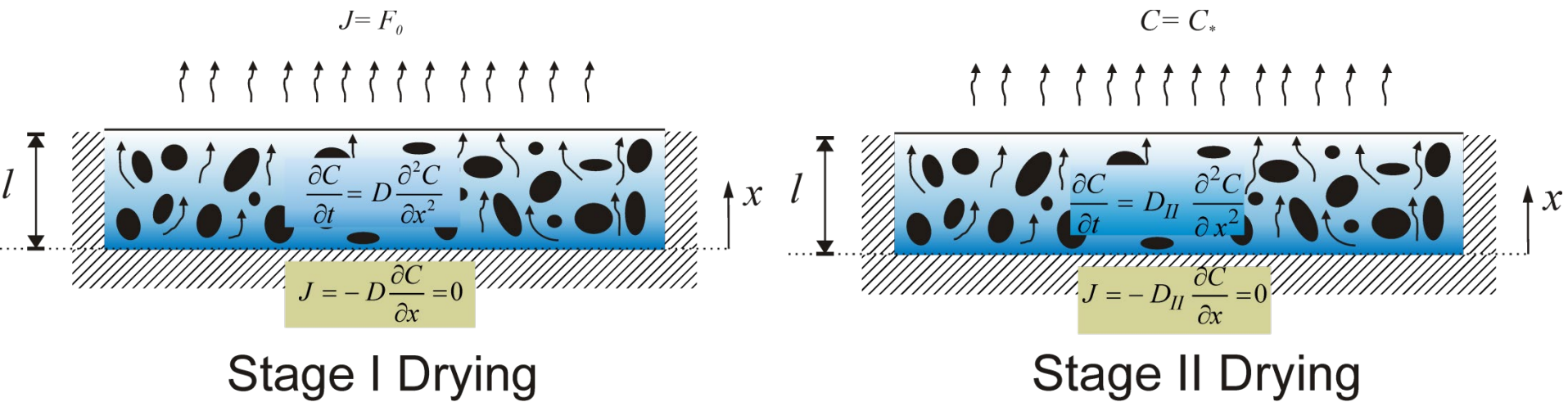
Internal Moisture Transfer Control

Black network: Solid matter (hydrates and anhydrous cement)

Blue network: Liquid phase

White network: Vapor phase

Modeling: 1D Moisture Transport During Drying



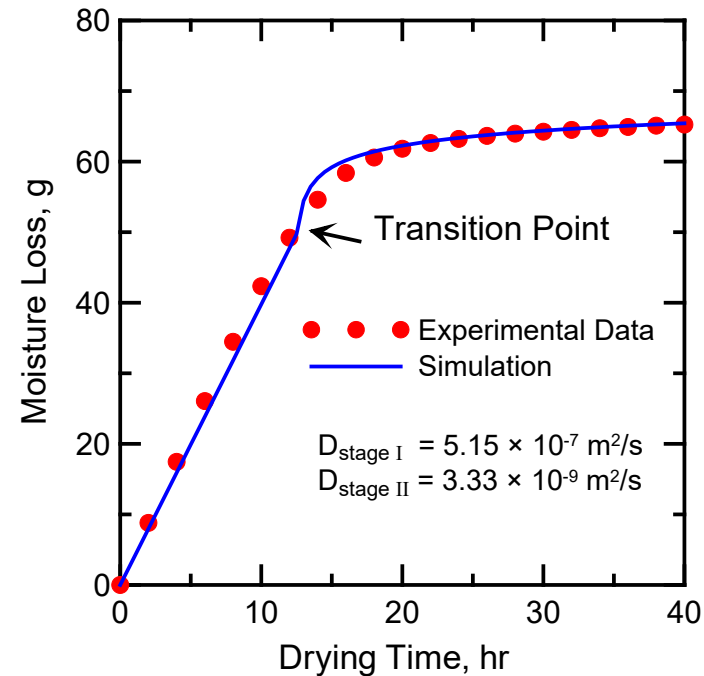
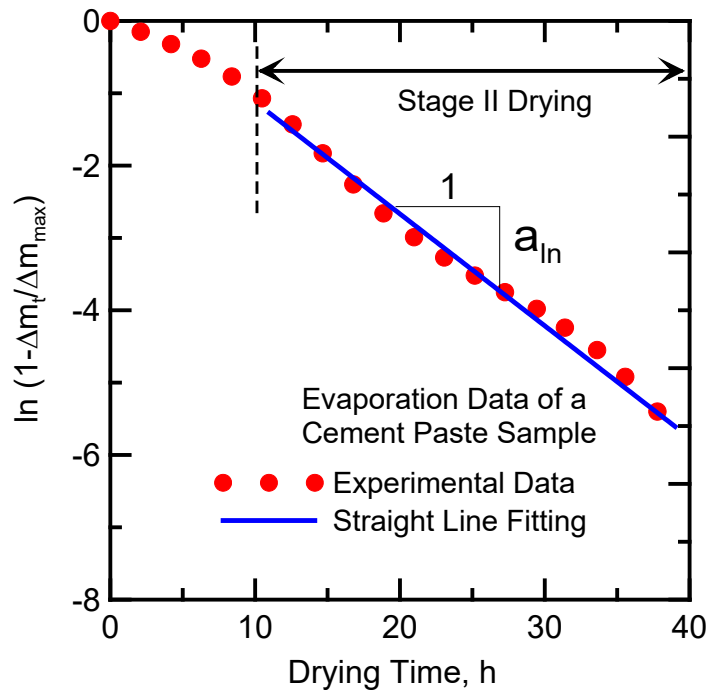
- Solution to Stage I:

$$C(t, x) = C_i + \frac{F_0 L}{D_I} \left\{ \frac{D_I t}{L^2} + \frac{3x^2 - L^2}{6L^2} - \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \exp\left(-\frac{D_I n^2 \pi^2 t}{L^2}\right) \cos \frac{n\pi x}{L} \right\}$$

- Solution to stage II:

$$\frac{C - C_i}{C_* - C_i} = 1 - \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \exp\left\{-D_{II} (2n+1)^2 \pi^2 t / 4l^2\right\} \cos \frac{(2n+1)\pi x}{2L}$$

Analysis: Calculating Diffusivity During Stage II Drying



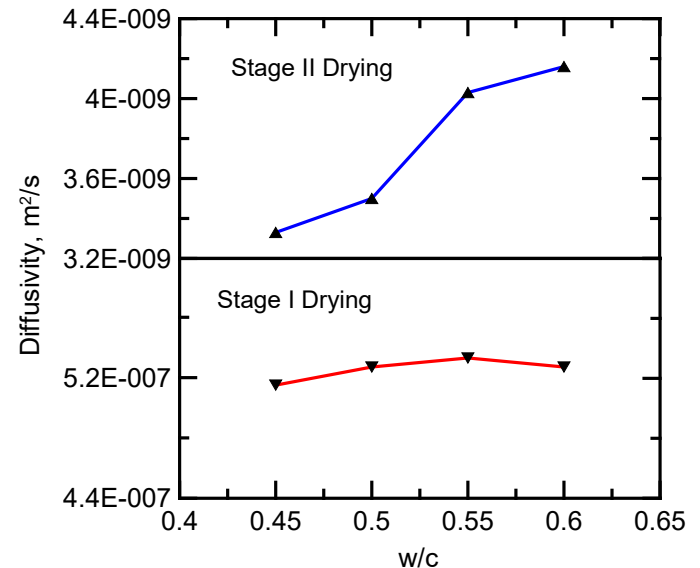
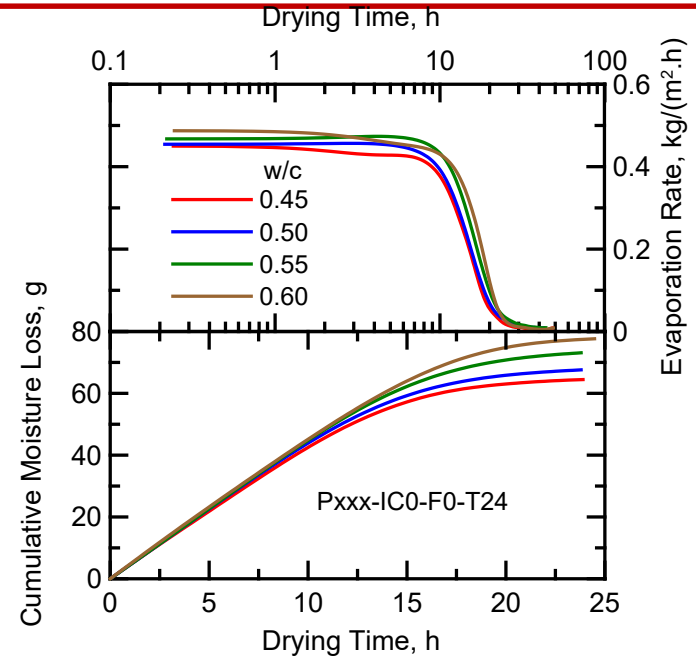
- By integration of concentration:
- Assuming RHS are following a linear relationship with the time:

$$D_{II} \cdot \frac{\pi^2}{-4L^2} = \frac{\ln\left(1 - \frac{M_t}{M_\infty}\right) - \ln \frac{8}{\pi^2}}{t}$$

$$D_{II} = \frac{-4L^2}{\pi^2} \cdot a_{ln}$$

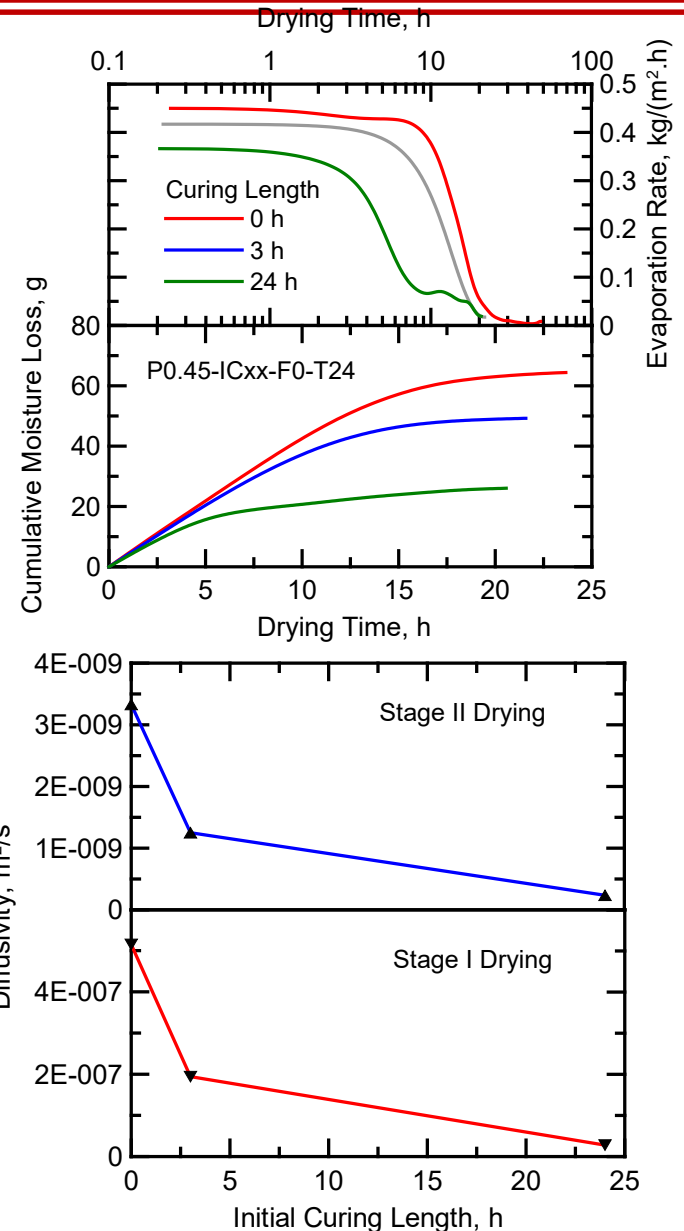
Effect of w/c Ratio

- Cumulative moisture loss:
Increased from 65 to 68, 74 and 78g
Translates into 20% increase from 0.45 to 0.6
- Initial evaporation rate:
No significant change.
Evaporation from water surface is the upper limit.
- Transition time:
Changed from 9.7 to 9.8, 11 and 12.3 h
- Moisture diffusivity: 



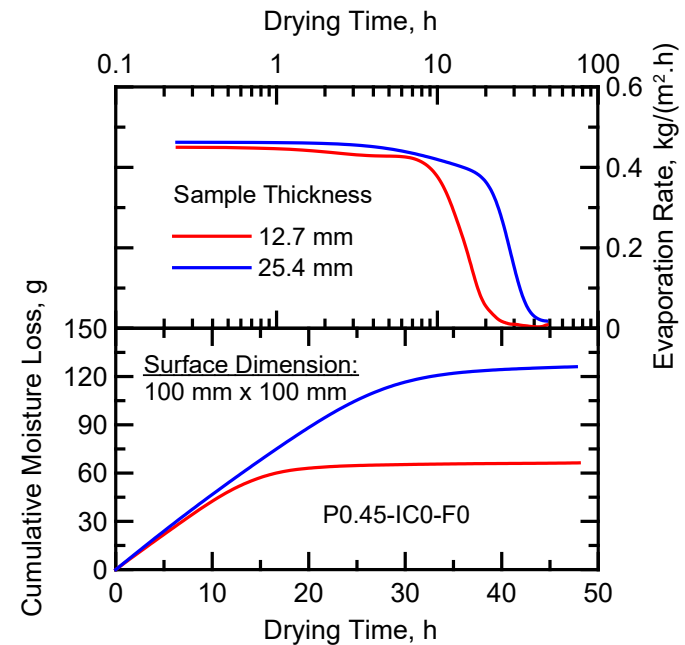
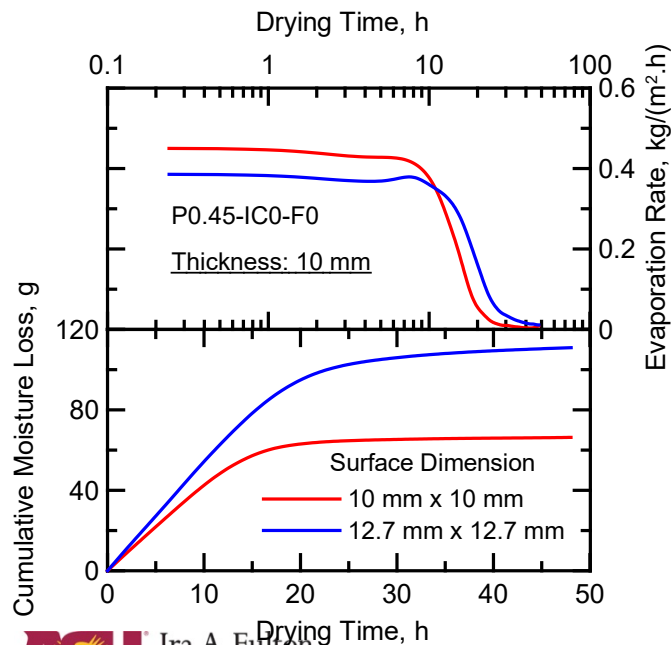
Effect of Initial Curing Duration

- Cumulative moisture loss:
 - Decreased from 65 to 49 and 26 g
 - Translates into 24% and 60% reduction
- Evaporation rate:
 - Initial Evaporation rate:
decreased from 0.42 to 0.39 and 0.34
 - Evaporation rate at 12 h:
decreased from 0.3 to 0.2 and 0.07 kg/(m²h)
- Transition time:
Changed from 9.7 h to 7.3 h and 3 h
- Moisture diffusivity:
 D_{II} decreased by as much as 62% and 93%



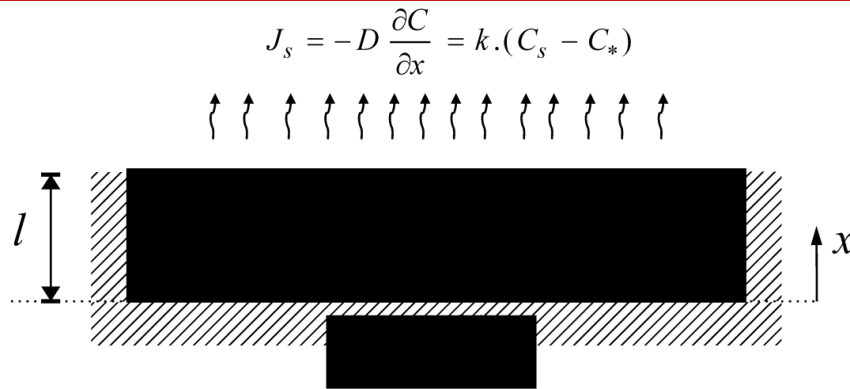
Effect of Sample Thickness and Surface Area

- Initial evaporation rate were not affected.
- Cumulative moisture loss Increased by 95%.
- Transition time changed from 9.7 to 18.9 h.
- D_I : changed from 5.1×10^{-7} to $5.3 \times 10^{-7} \text{ m}^2/\text{s}$.
- D_{II} : changed from 3.3×10^{-9} to $3.6 \times 10^{-9} \text{ m}^2/\text{s}$.



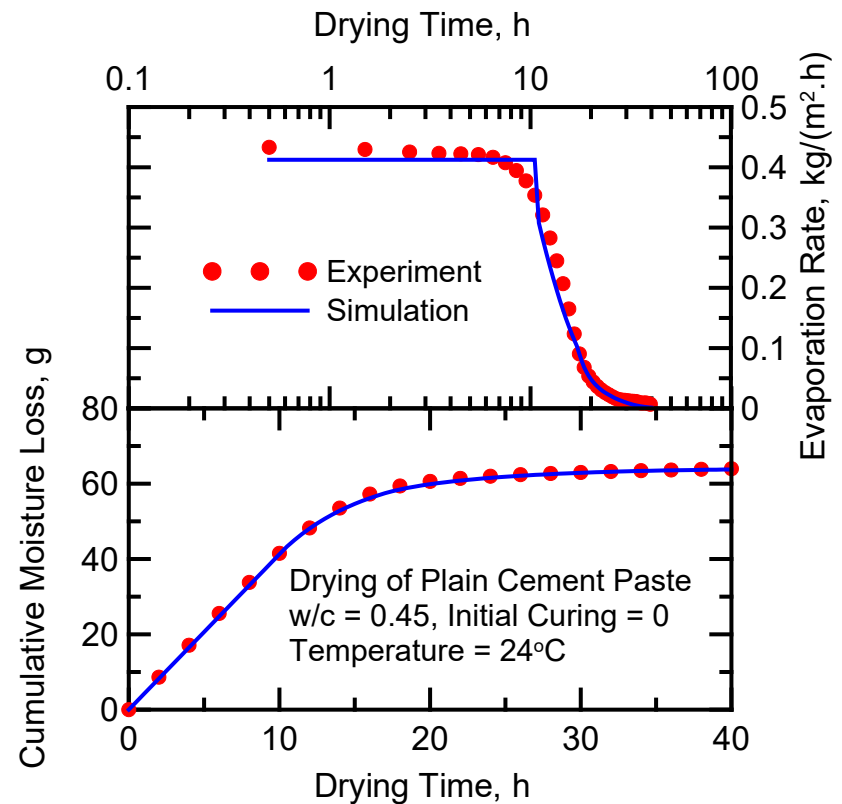
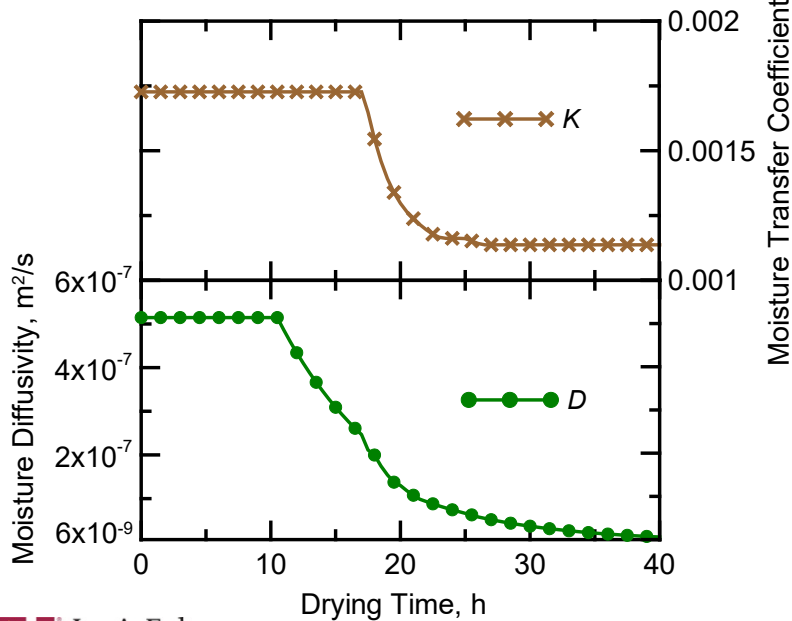
- Cumulative moisture loss Increased by 66%.
- Initial evap. rate reduced from 0.42 to 0.39 $\text{kg}/(\text{m}^2 \cdot \text{hr})$.
- Transition time changed from 9.7 to 12.18 h.
- D_I : changed from 5.1×10^{-7} to $4.9 \times 10^{-7} \text{ m}^2/\text{s}$.
- D_{II} : changed from 3.3×10^{-9} to $3.2 \times 10^{-9} \text{ m}^2/\text{s}$.

Convective BC for Modeling of Stage II Drying

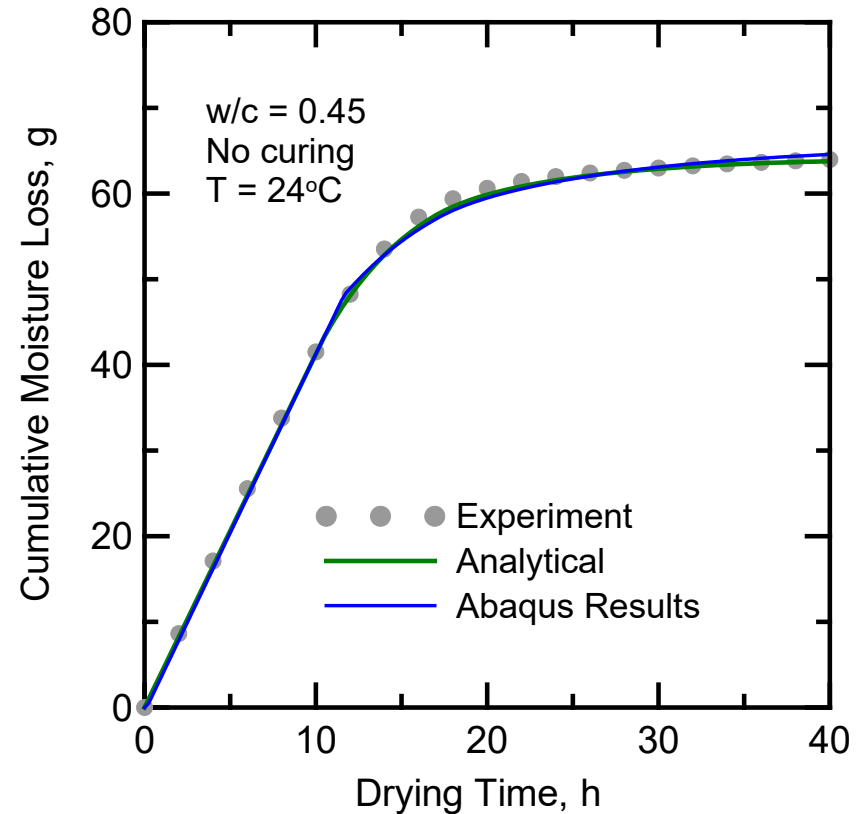
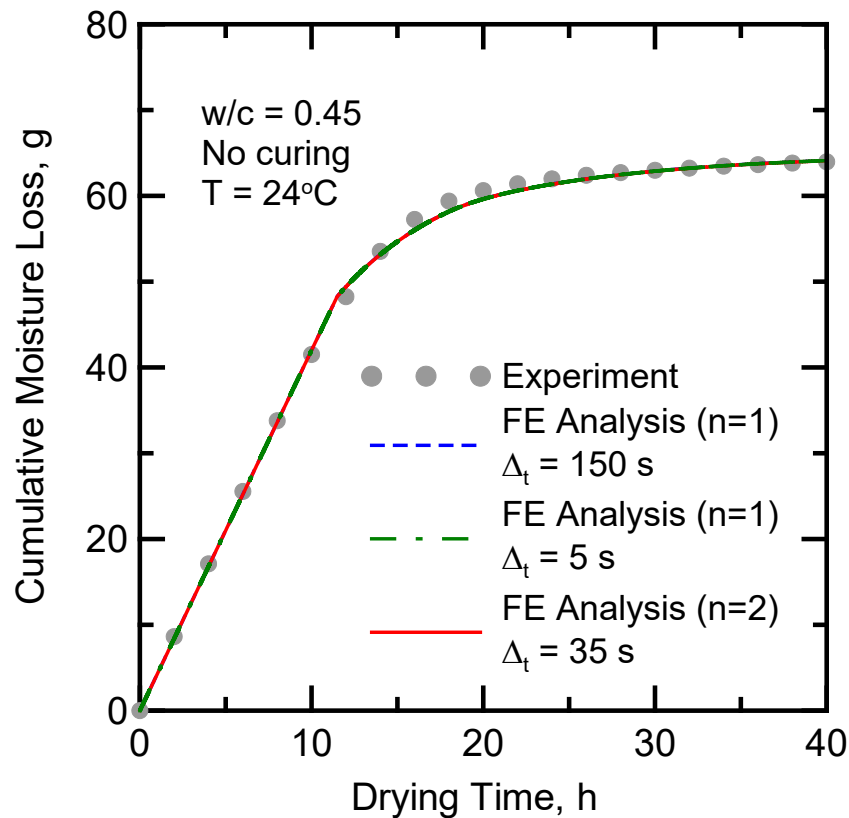


Stage II Drying

$$C(t, x) = C_* + (C_i - C_*) + \sum_{n=1}^{\infty} \frac{2 \sin(\lambda_n)}{\lambda_n + \sin(\lambda_n) \cos(\lambda_n)} \left(\cos(\lambda_n \frac{x}{L}) \exp\left(-\lambda_n^2 \frac{Dt}{L^2}\right) \right)$$



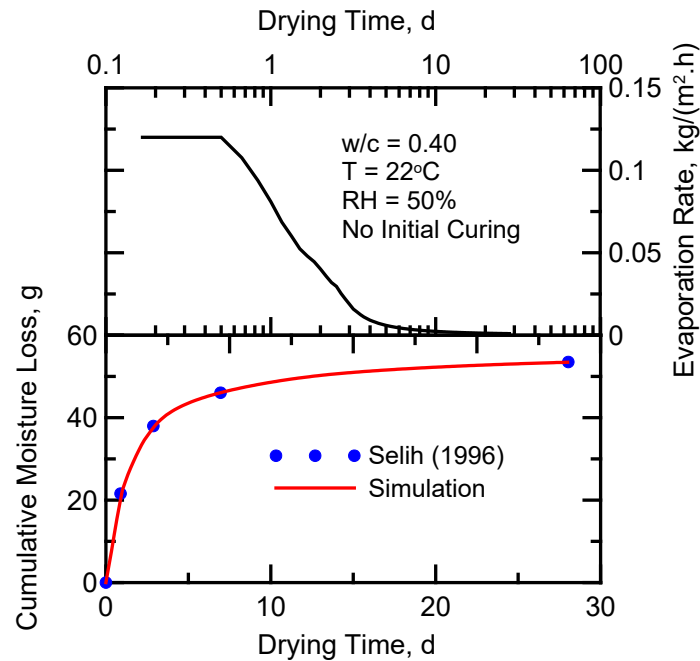
Verification with FE Analysis



- Forward Difference Method

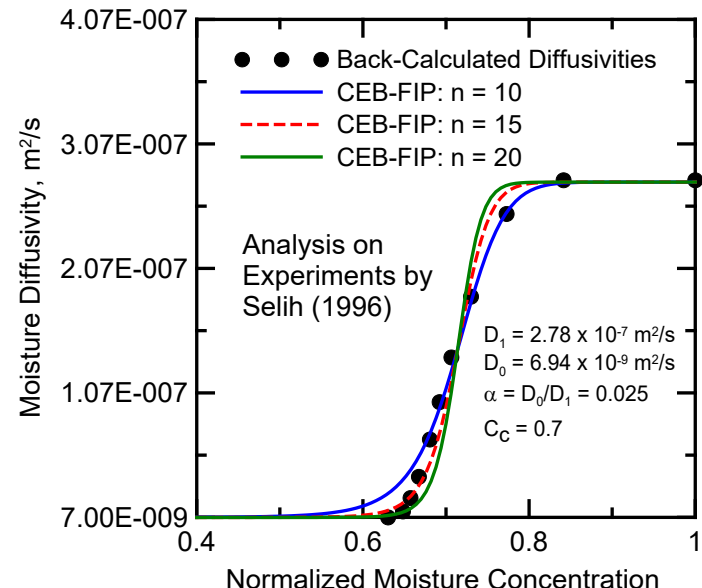
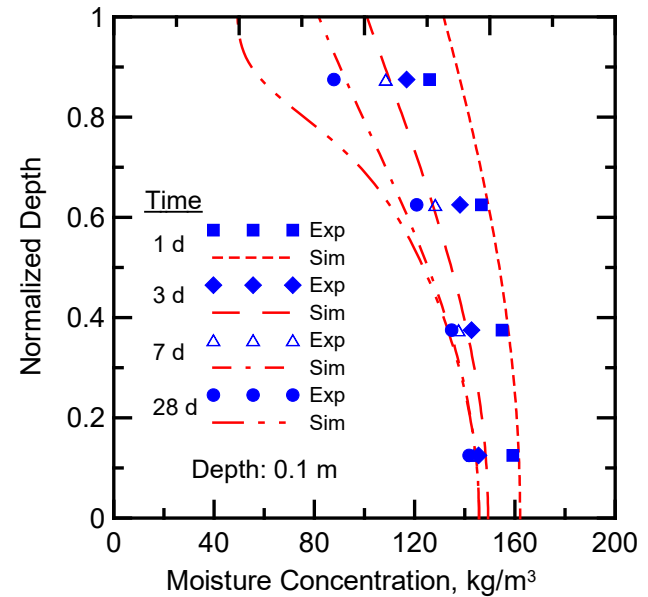
$$a_n = a_{n-1} + \Delta t_n [CL]^{-1} (F_{n-1} - K a_{n-1})$$

Verification of Predicted Profiles with Experiment



$$D(C_{norm}) = D_1 \left(\alpha + (1 - \alpha) \left(1 + \left(\frac{1 - C_{norm}}{1 - C_c} \right)^n \right)^{-1} \right)$$

M. Bakhshi, B. Mobasher, C. Soranakom, "Drying of Early-age Cementitious Materials: Theory and Modeling", in Review, Construction and Building Materials, 2011.

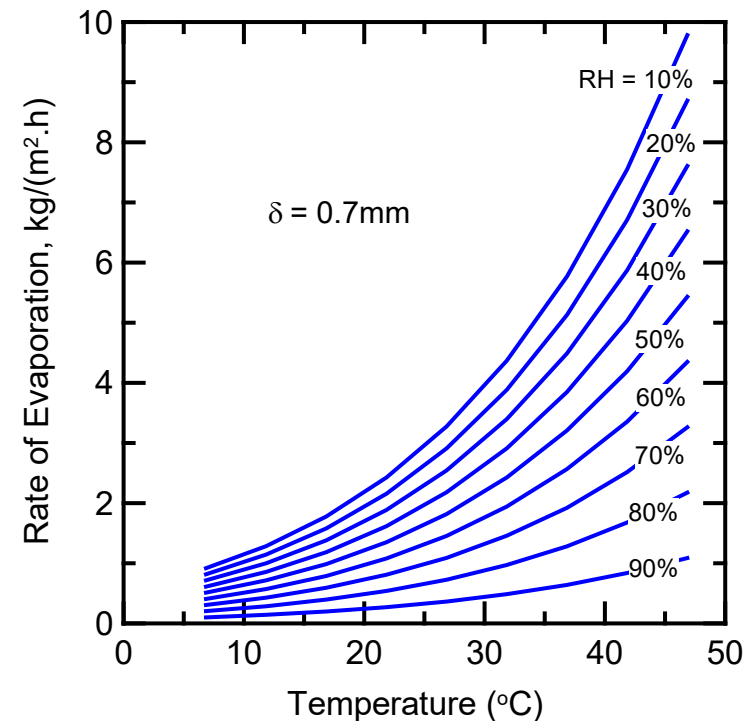
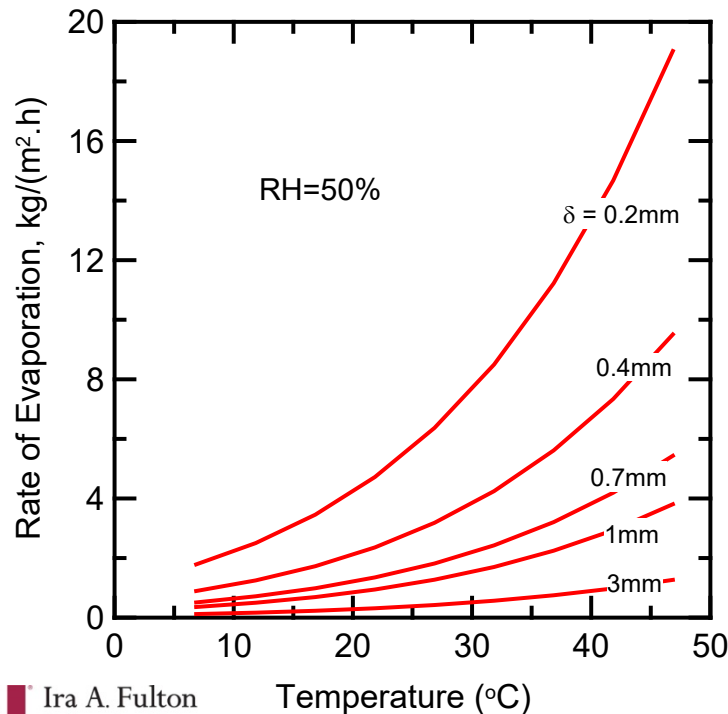
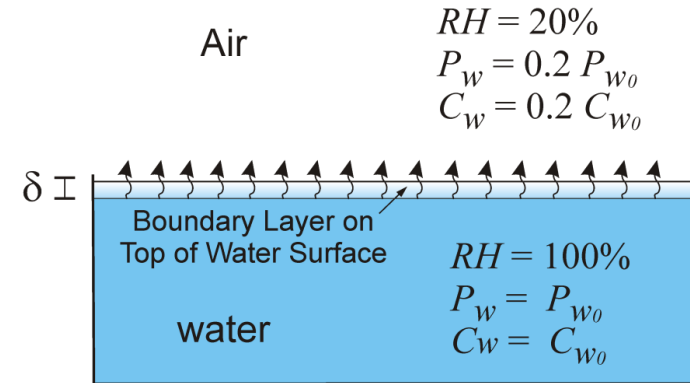


Evaporation as a Boundary Layer Mass Transfer

Simplified stagnant boundary layer:

$$J = \frac{D_w M}{RT} \frac{p_{w0} (1-H)}{\delta}$$

$$p_{w0} = 611.21(1.0007 + 3.46 \times 10^{-8} p) e^{\frac{17.502(T-273.15)}{T-32.18}}$$

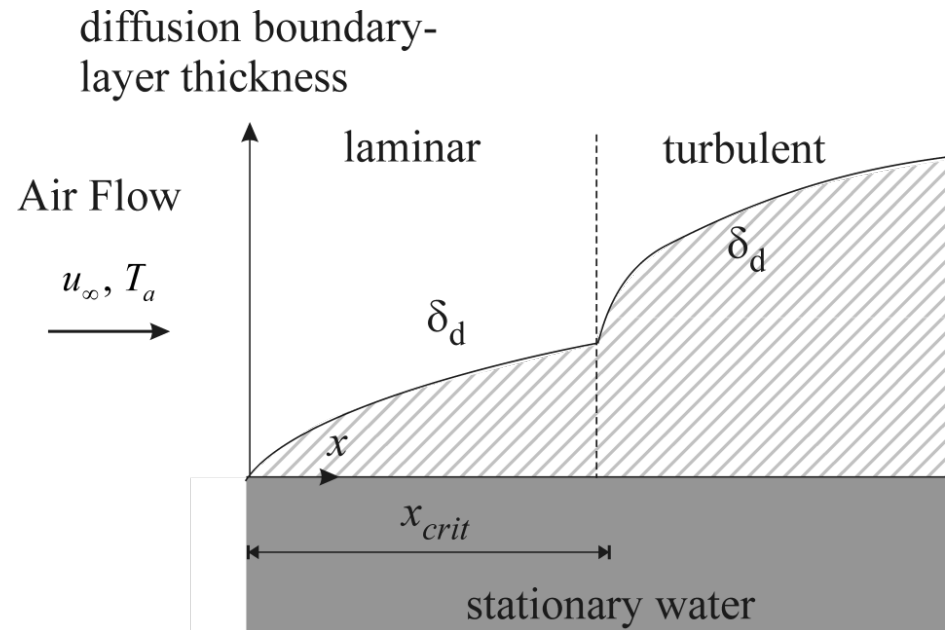


Advanced Modeling: Evaporation as a Boundary Layer Mass Transfer

$$J_s = -D \left. \frac{\partial c}{\partial y} \right|_{y=0} \approx -\frac{(c_e - c_s)D}{\delta_d} \approx h_m (c_s - c_e)$$

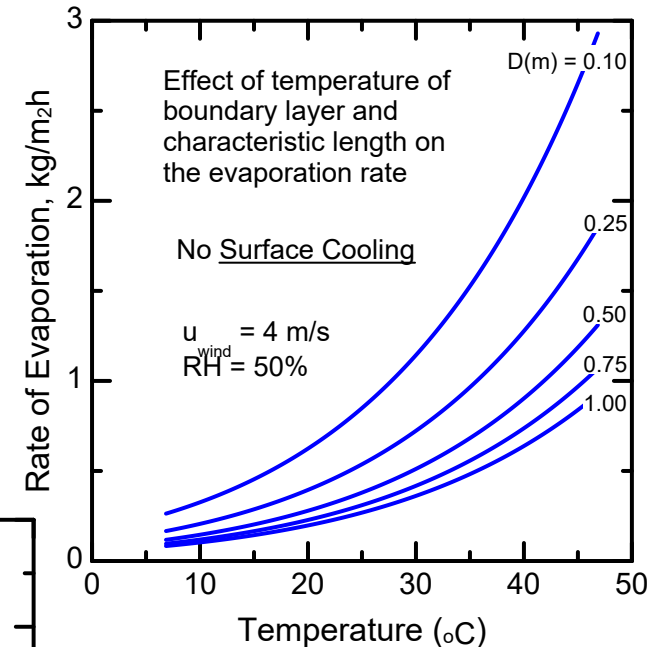
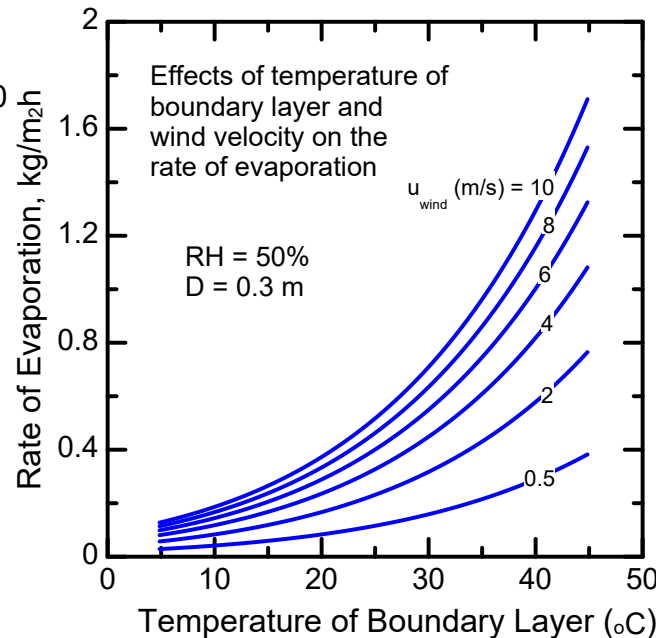
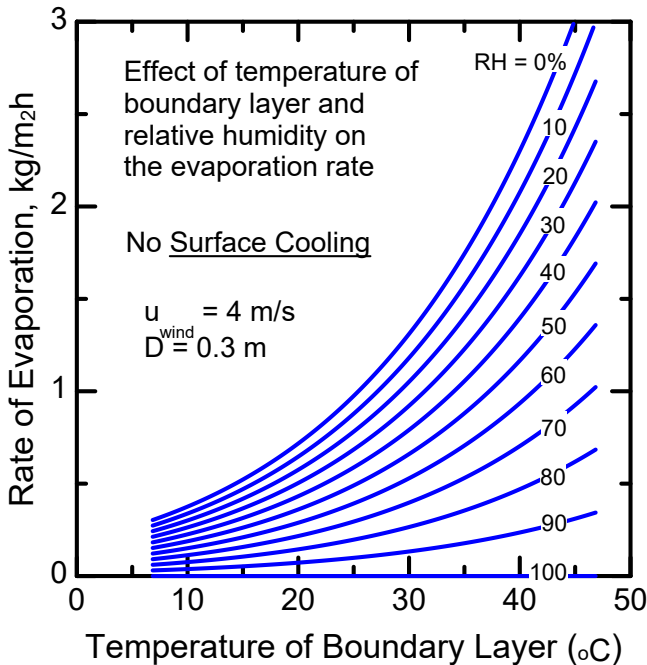
$y \uparrow$ $y \uparrow$

concentration distribution near interface



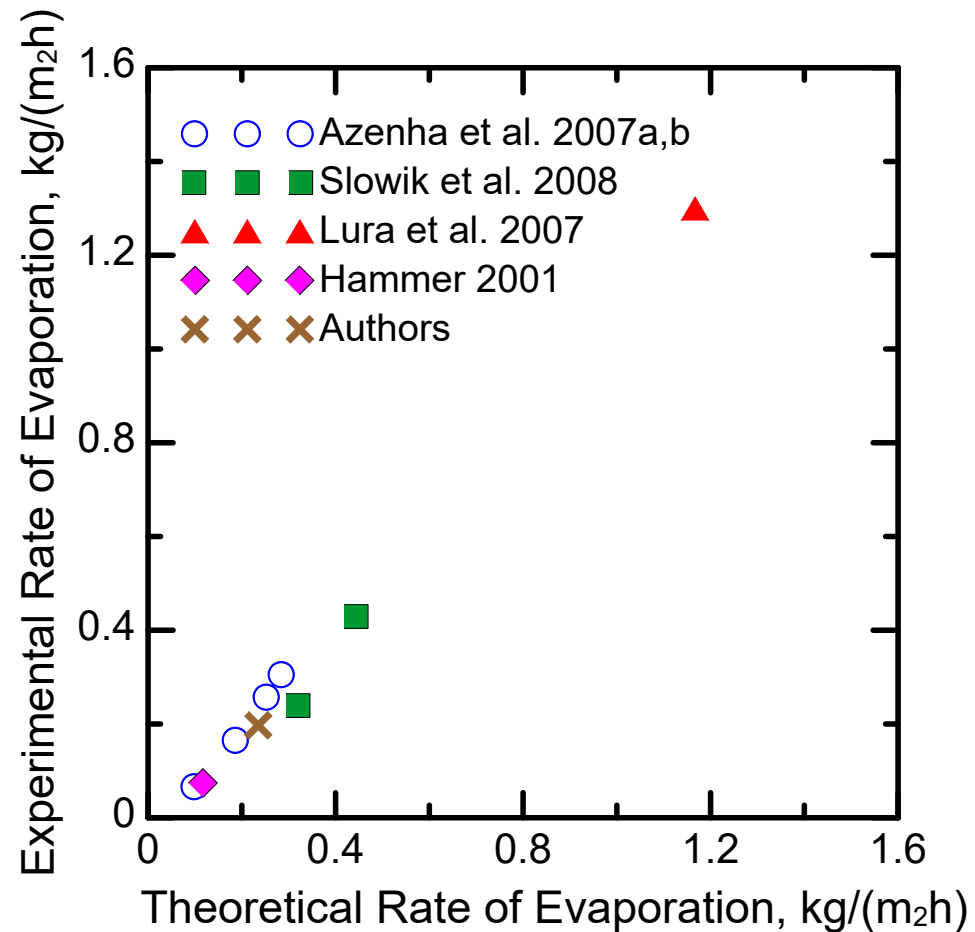
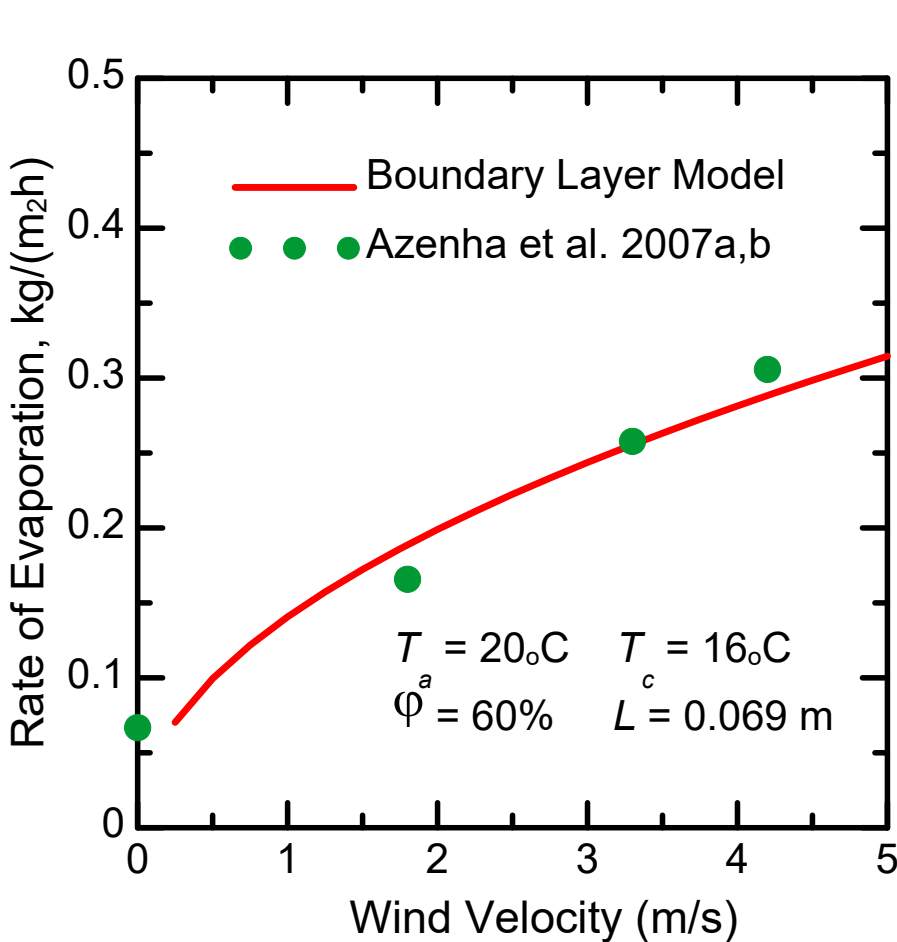
Diffusion boundary layer thickness in a mixed laminar and turbulent regions for free stream flow over a flat plate

Modeling: Parametric Studies on Advanced Mass Transfer



“Model for early age rate of evaporation of cement-based materials”, M. Bakhshi, B. Mobasher, M. Zenouzi., *ASCE Journal of Engineering Mechanics*, 2011.

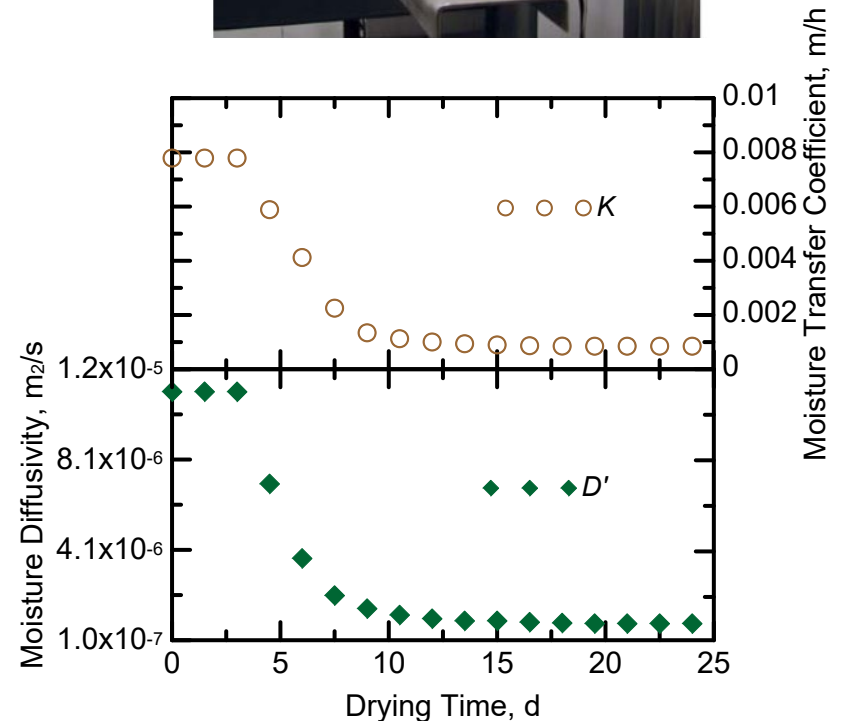
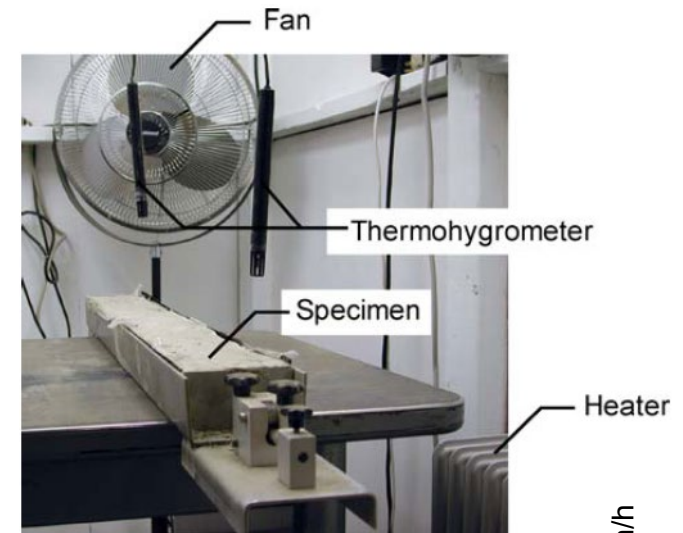
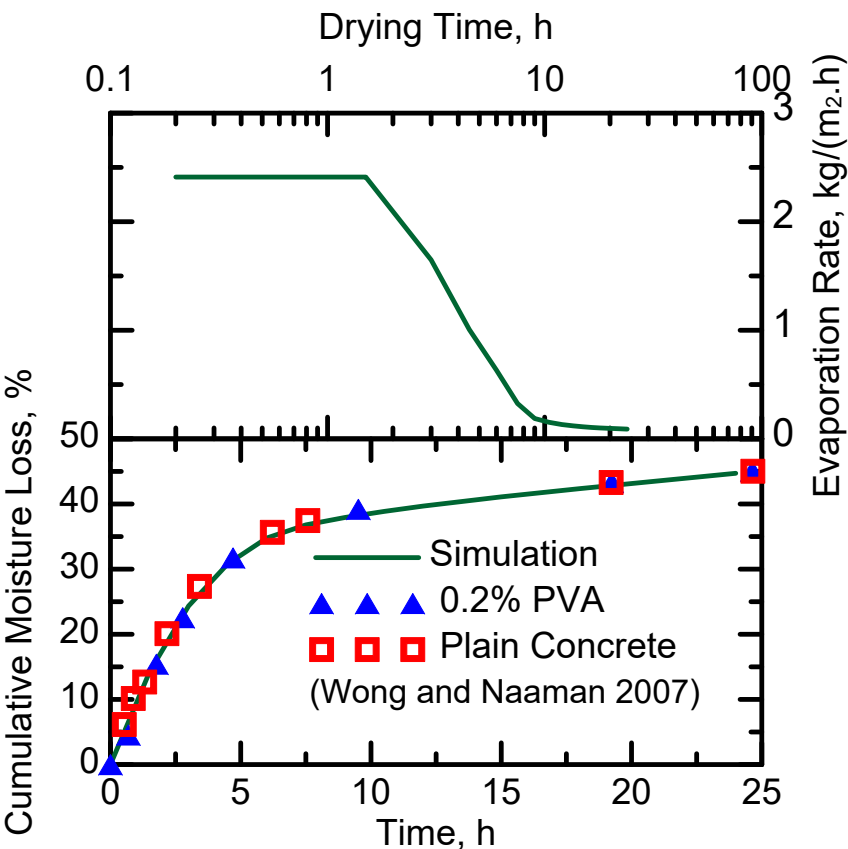
Modeling: Verification with Experimental Data



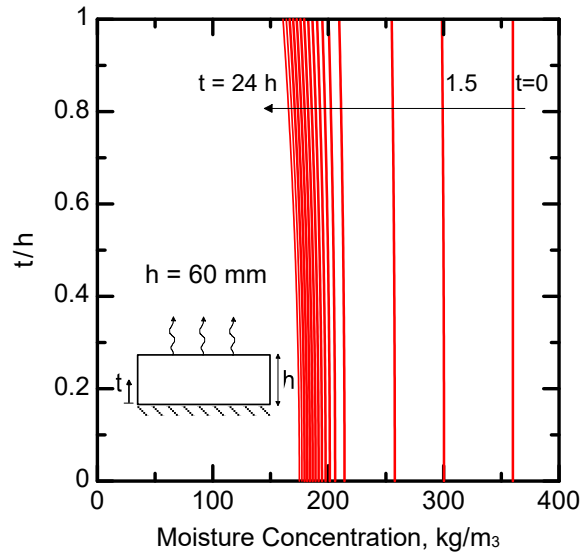
“Model for early age rate of evaporation of cement-based materials”, M. Bakhshi, B. Mobasher, M. Zenouzi., *ASCE J. Eng. Mech.*, 2011.

Verification of Shrinkage with Experimental Data

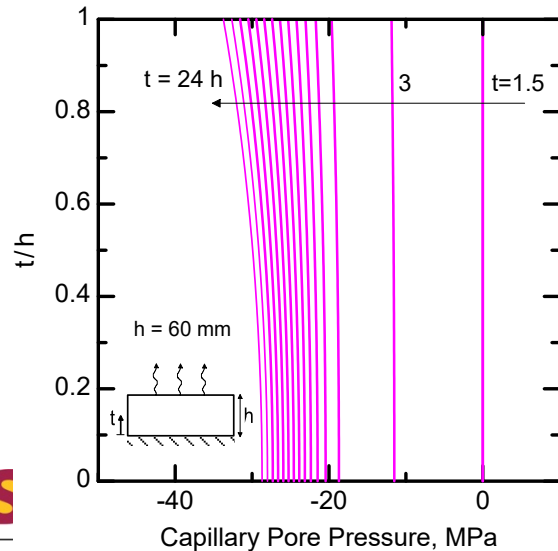
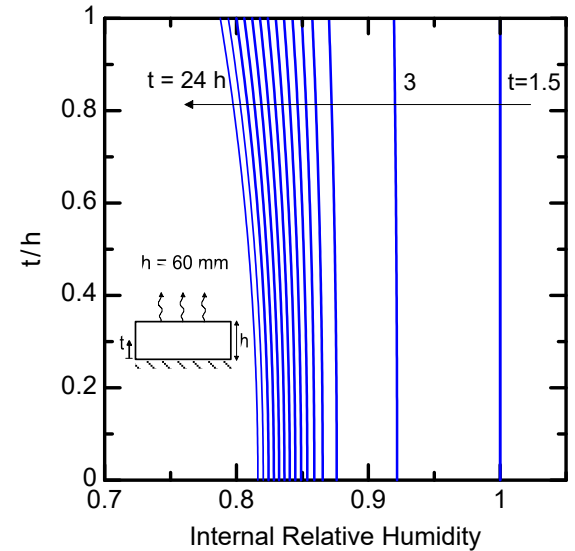
Unrestrained early-age shrinkage test
(Wongtanakitcharoen and Naaman 2007)



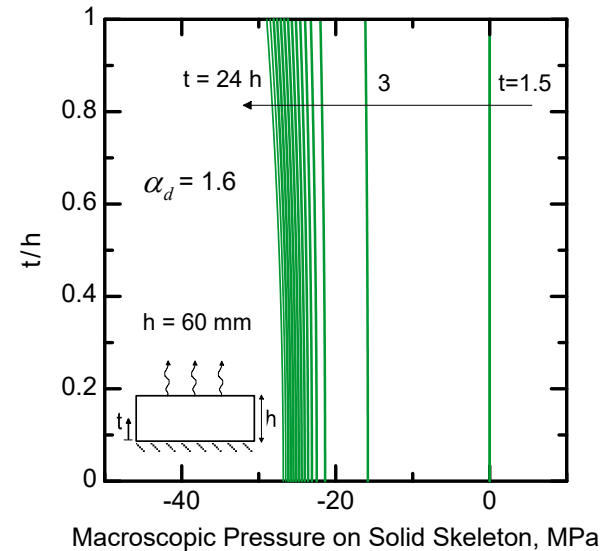
Simulation of Internal Humidity & Capillary Pressure



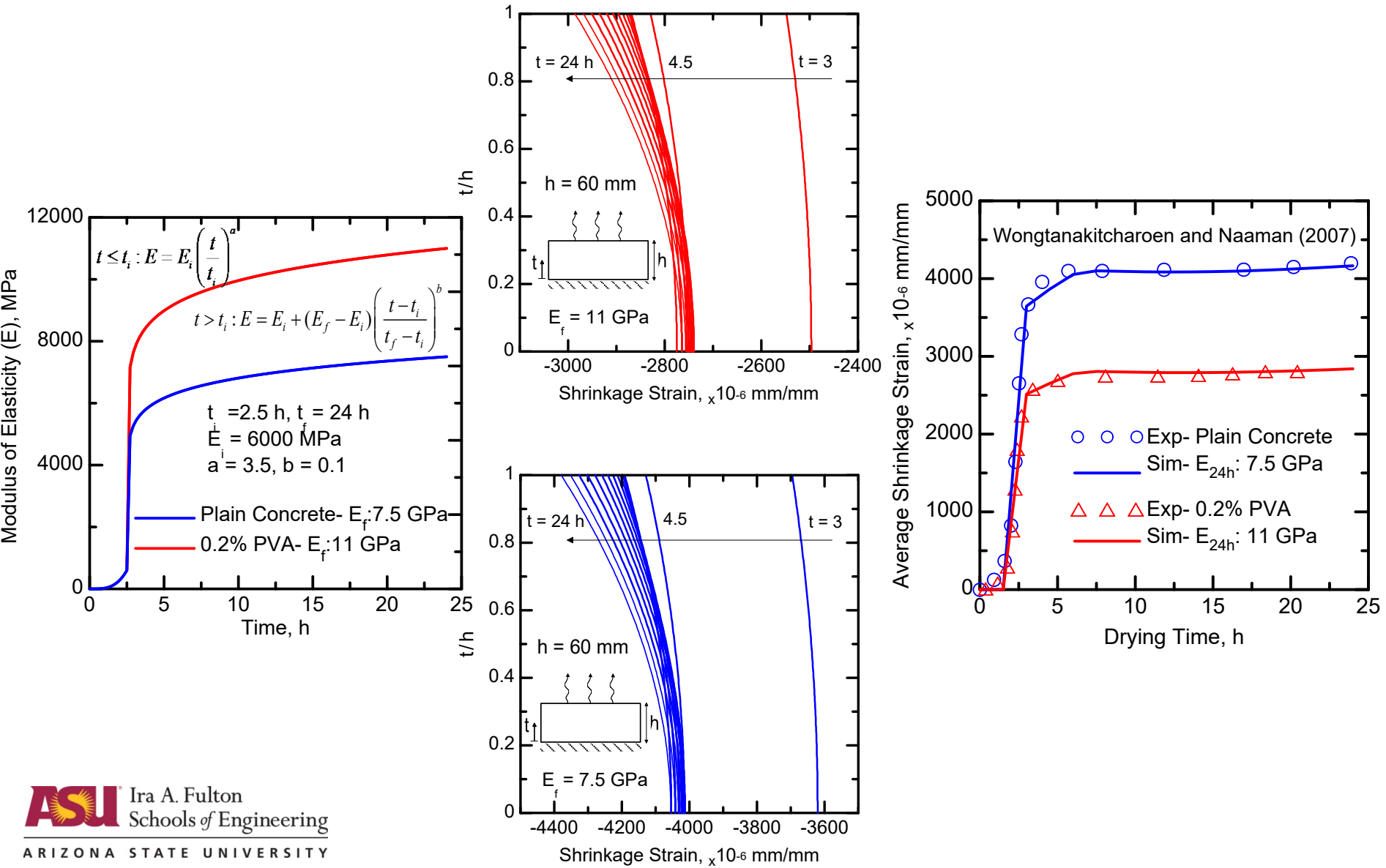
$$C^* = \frac{AkV_m H}{(1 - kH)[1 + (A - 1)kH]}$$



$$p_s = \alpha_d S_l p_c$$



Fitting Experimental Shrinkage Strains



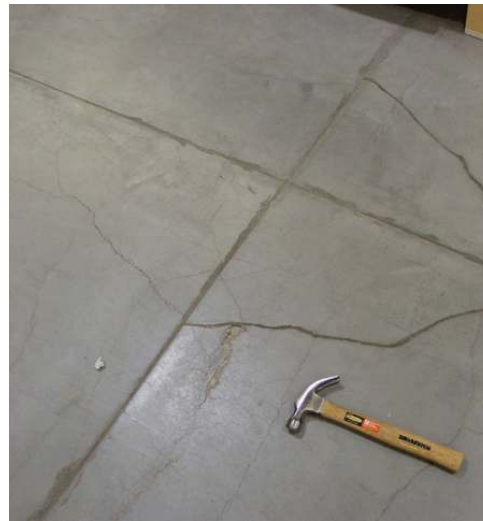
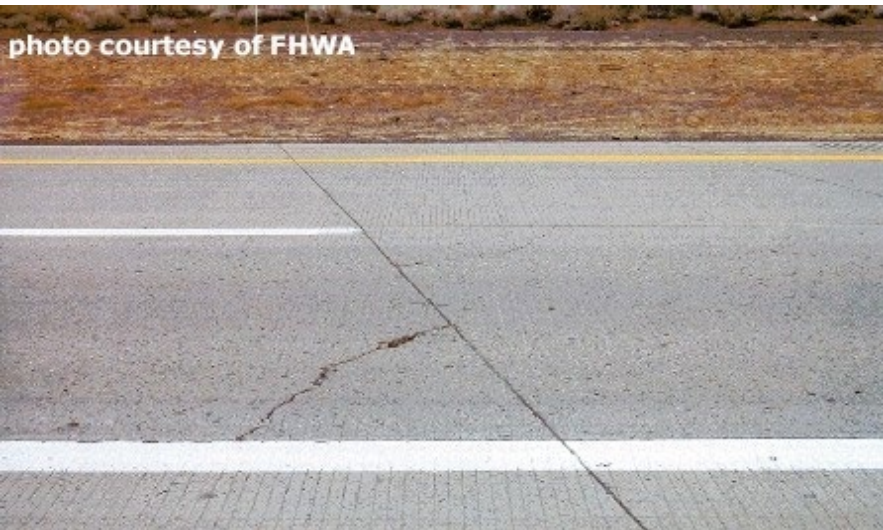
Shrinkage Induced Transverse Cracking

- Reduce load carrying capacity
- Accelerate deterioration
- Increase maintenance costs
- Reduce service life
- Plastic Shrinkage: Water evaporation at early age
- Drying Shrinkage: Loss of excessive water from hardened concrete



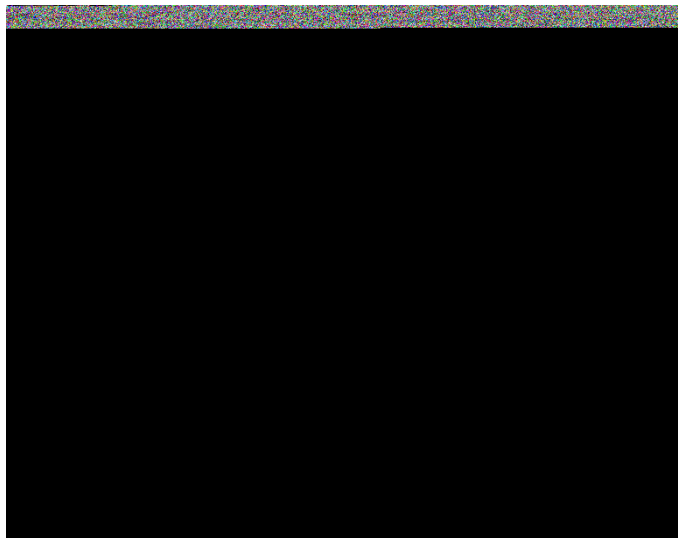
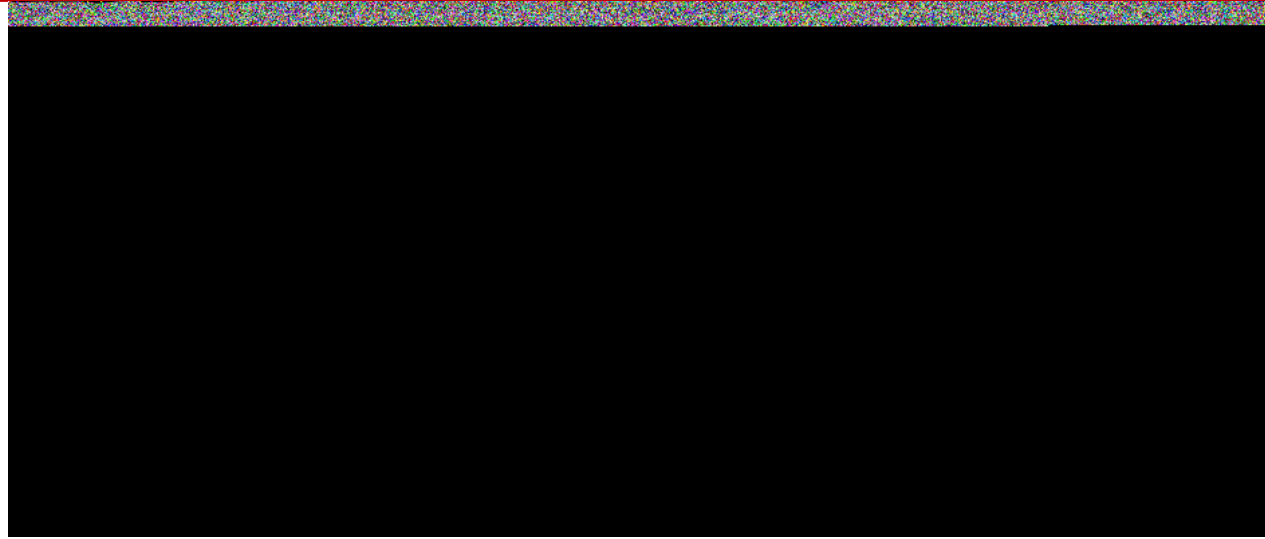
Joint failure

- Intended to control the cracks
- Presence of joints and corners aggravates the potential cracking as well as long term durability
- Joint edge chipping, uplift, and corner flexural cracks
- Potentially induced due to curling of the slab
- Point load such as truck wheel



Use of fibers in joint-free slab

- Crack bridging
- Load transfer
- Distributed fine cracks
- Non-critical crack pattern

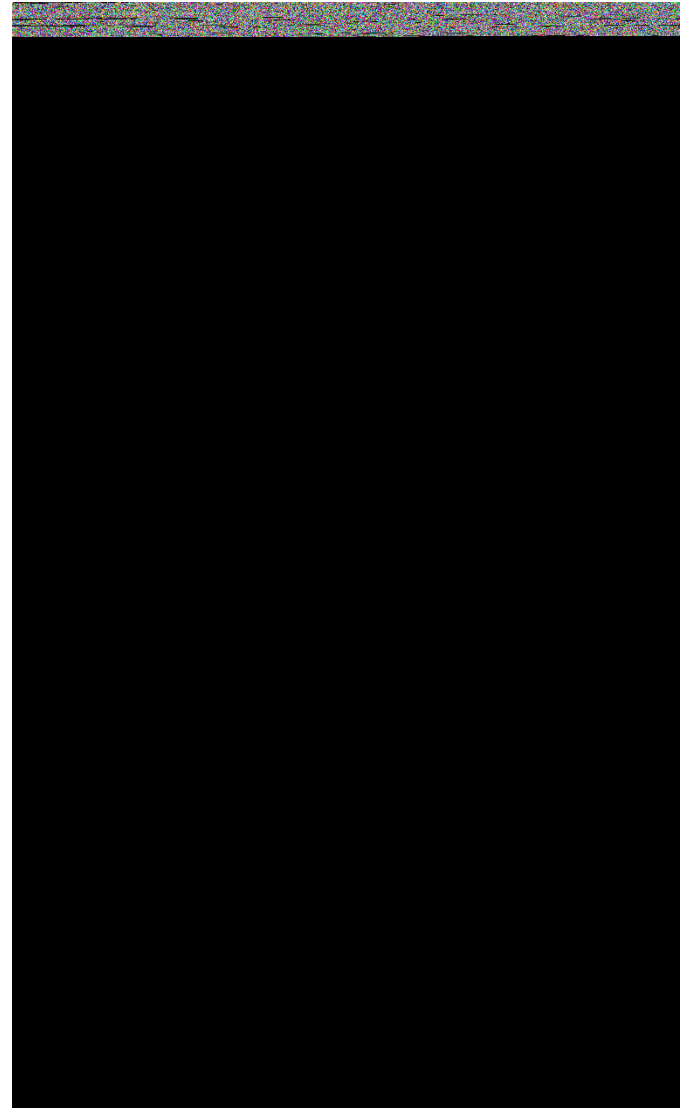


Parameters Affecting Drying Shrinkage

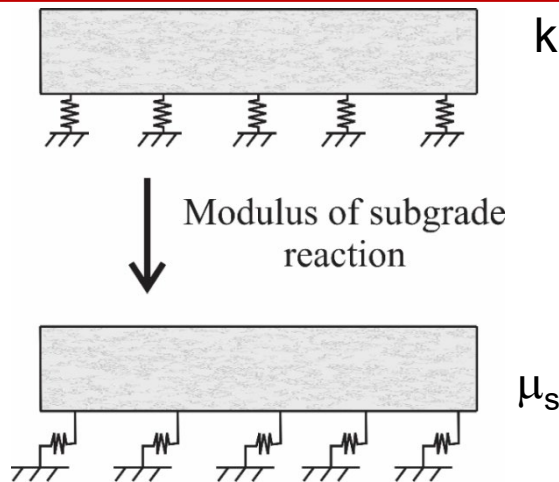
- Concrete Matrix
 - Water/cement ratio
 - Superplasticizer

- Internal Restriction due to Fibers
 - Type: straight, hooked end, undulated
 - Fiber shape
 - Bond

- Slab Geometry and External Boundary Conditions
 - Evaporation rate
 - Temperature
 - Degree of restraint due to base friction

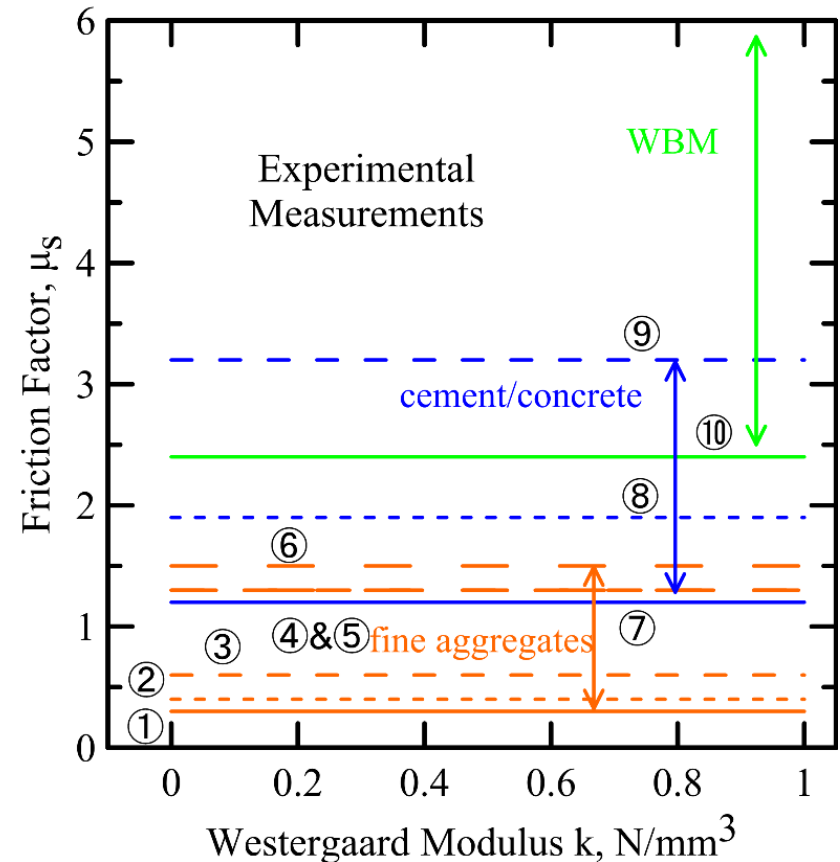


Parameters affecting the Shrinkage cracking



■ Main Variables

- Base friction coefficient
- Fiber length, volume, and diameter
- Fiber-matrix
- Combined effect of w/c and superplasticizer
- Slab length and thickness
- free shrinkage parameters



Experimentally measured frictional coefficients (Maitra et. al 2009)

Proposed Empirical Model

- Proposed Base friction coefficient

$$\mu_s = \alpha + \ln(k) \quad k > 0.03 \text{ N/mm}^3$$

- Proposed Empirical equation

$$\omega = \frac{L\lambda\varepsilon_R}{\frac{V_f L_f}{d} \mu_s (1 + \mu_m)^{2.5}}$$

where :

ω = crack opening

L = slab length

λ = combined effect of w/c ratio and HRWRA

ε_R = free shrinkage strain

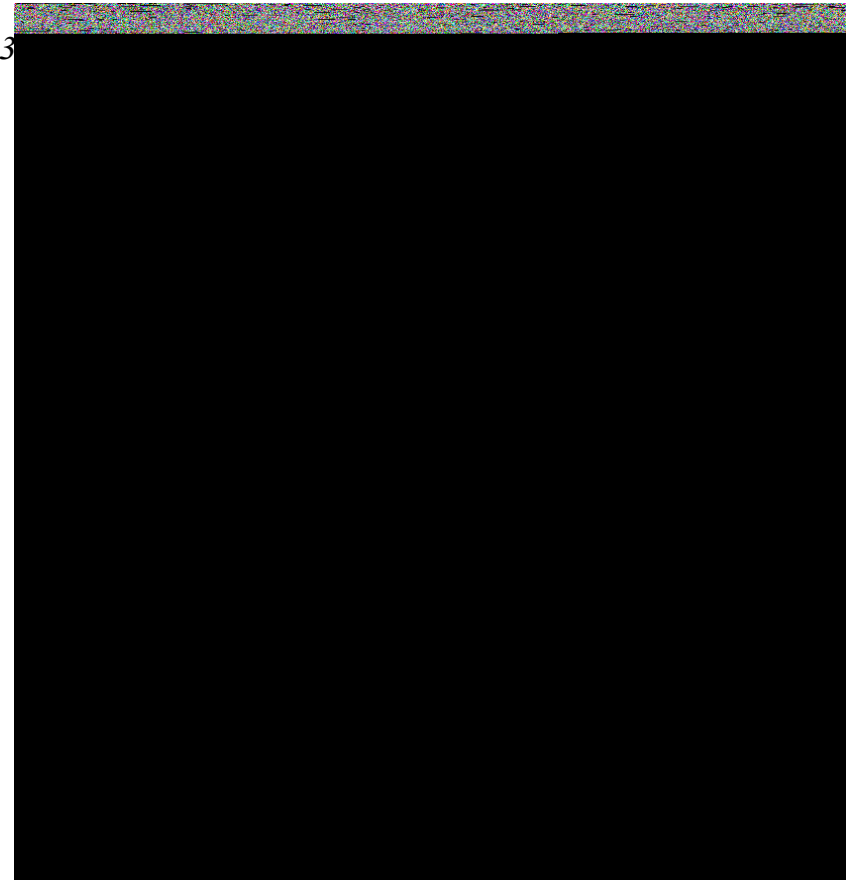
V_f = fiber volume fraction

L_f = fiber length

d = fiber diameter

μ_s = friction coefficient

μ_m = fiber shap parameter

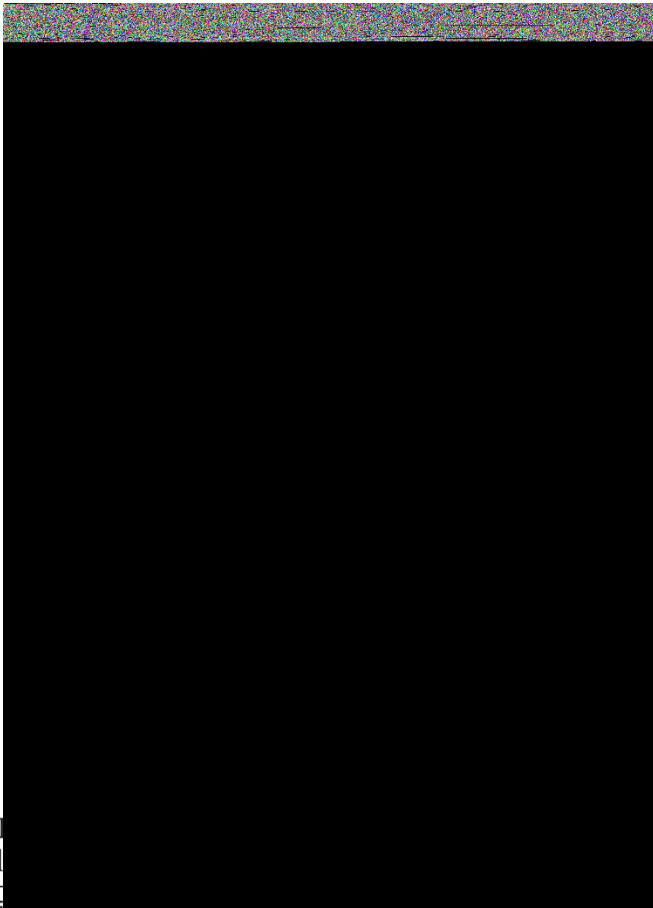


Correlation of base subgrade reaction modulus k and base friction coefficient

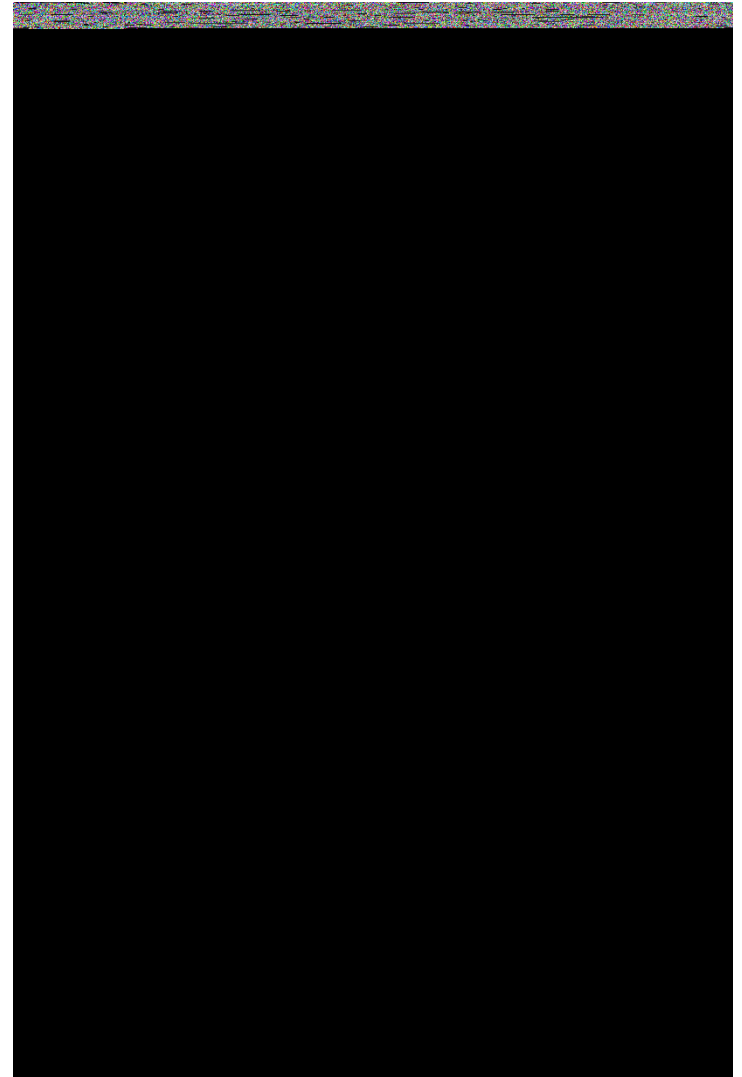
Finite Difference Model

- Matrix stiffness and strength
- Stress-crack width relationship
- Base friction assumed constant

Free Slab

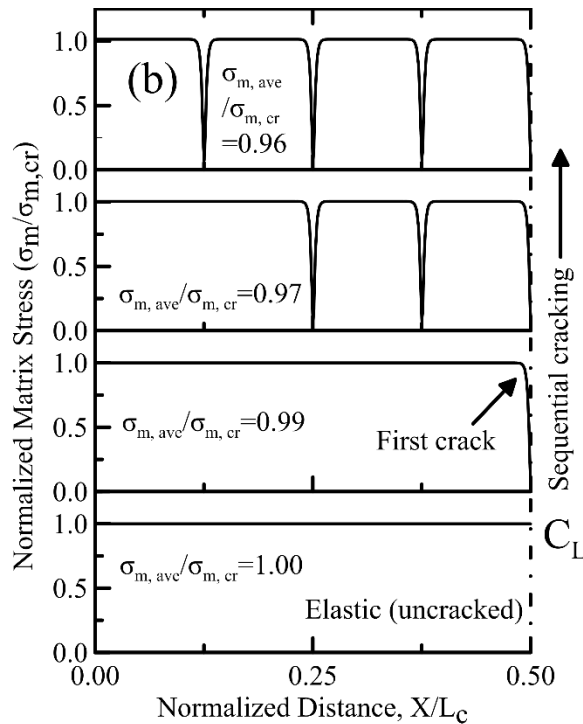


Restrained Slab

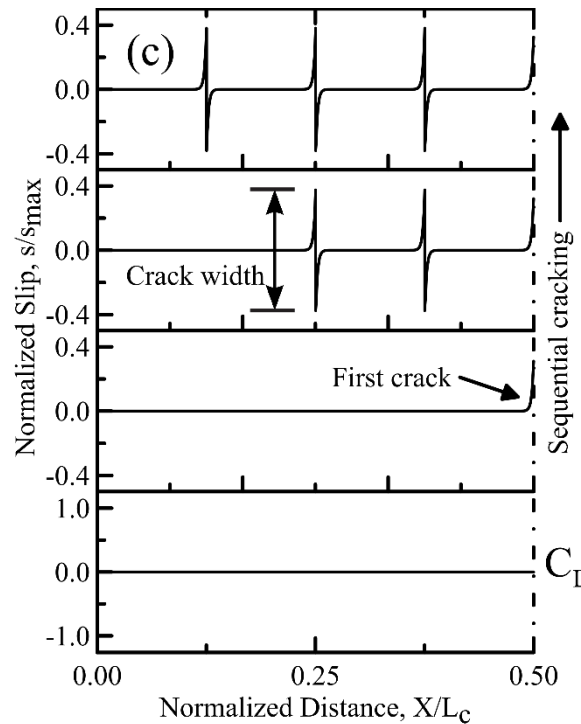


Sequential Cracking

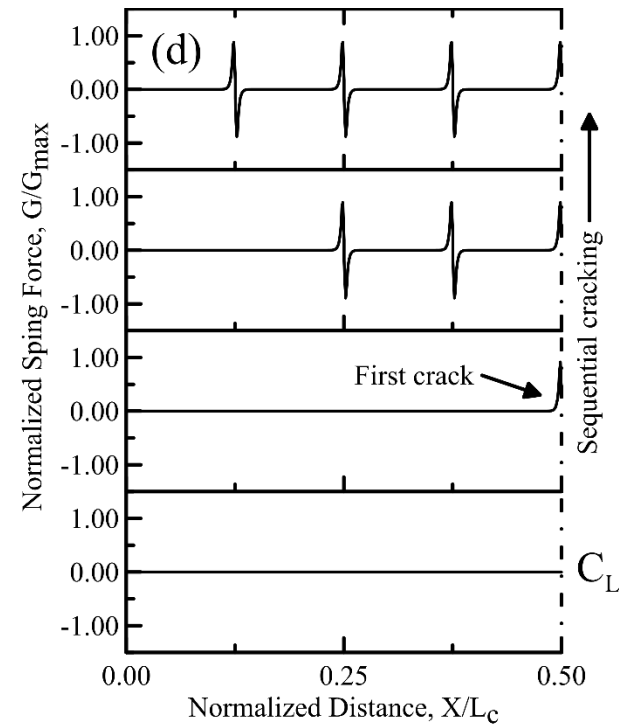
Matrix stress



Slip displacement



Spring force



Distribution of matrix stress, slip and frictional force as cracks sequentially form

Comparison of Model Parameters

■ Methodology #1

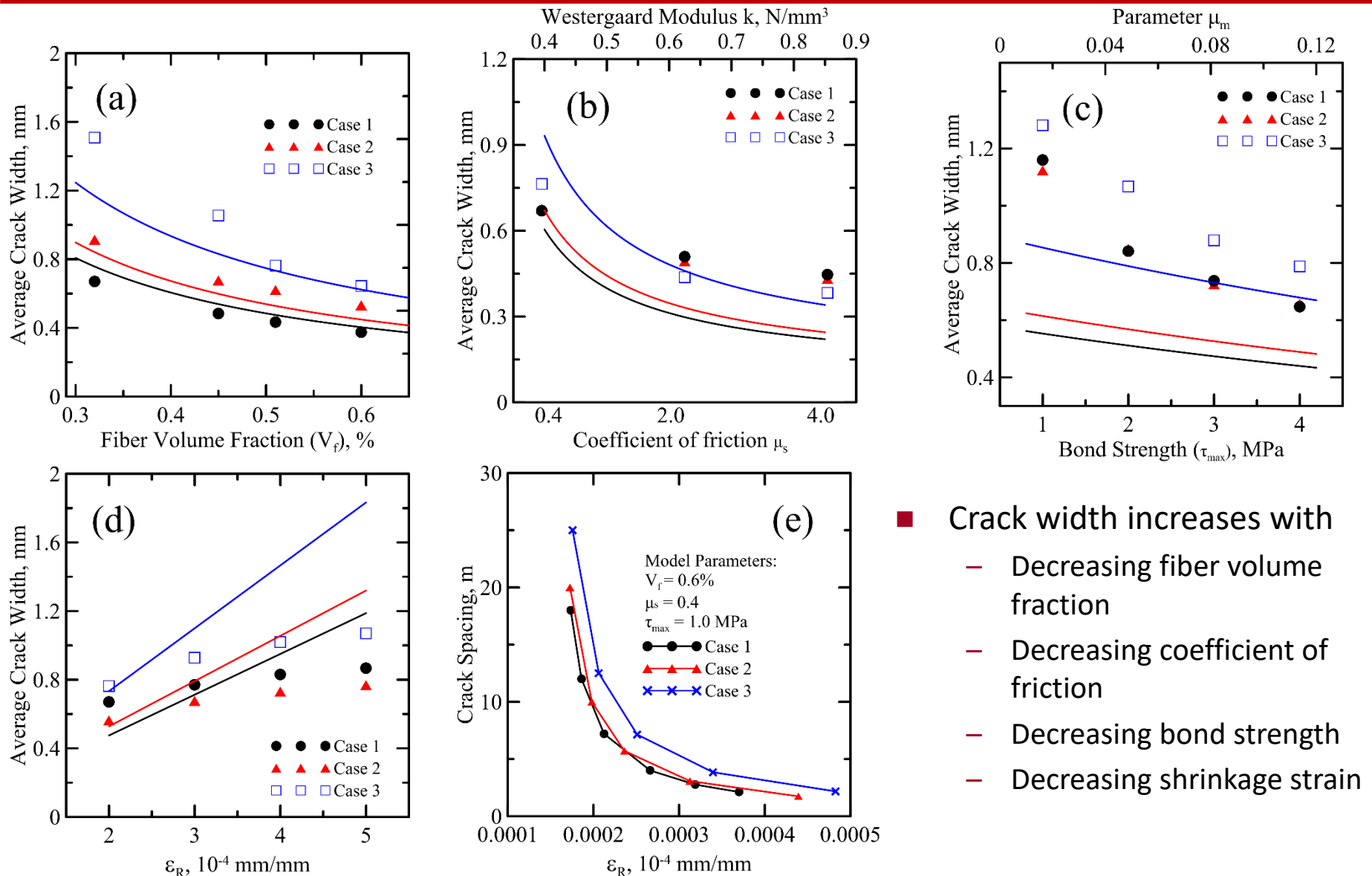
- Filed data collected from case studies were used in fitting the empirical model and obtain the model parameters

■ Methodology #2

- Use the material properties to calibrate the finite difference model and correlate with the empirical equation

Effect	Empirical Model	Finite Difference Model
Fiber volume fraction	V_f	
Free shrinkage strain	ϵ_R	
Effect of W/C and HRWRA	λ	N/A
Sub-base friction	k	μ_s
Fiber matrix interaction	μ_m	Maximum bond strength

Parametric Study



Comparison between Predictions and Field Measurements

- Slab 1: 36m x 36m x 0.2m, C25-30 mix design, 25 kg/m³ of 35 mm length by 0.75 mm diameter, Poznan, Poland.
- Slab 2: 40m x 40m x 0.15m, C25-30 mix design, 35 kg/m³ of 60 mm length
- Slab 3: 50m x 50m x 0.18mm, C30-37 mix design, 40 kg/m³ of 54 mm long by 1.00 mm diameter, Bornem, Belgium.

Case	Measurements	Empirical Model						Finite Difference Model		
	ω (mm)	V_f (%)	ϵ_R ($\mu\epsilon$)	l	k	μ_m	ω (mm)	Bond strength (MPa)	μ_s	ω (mm)
Slab 1	0.5 -0.7	0.32	200	1.05	0.5	0.08	0.636	3.5	0.4	0.735
Slab 2	0.65	0.45	200	1.35	0.45	0.04	0.658	3.25	0.4	0.620
Slab 3	0.8	0.51	200	1.35	0.45	0.04	0.806	2.5	0.4	0.838

Slab Curling

- Upward lift at the corner or edges
- Due to differences in moisture content or temperature
- Critical above 0.5 mm (Daiutolo 2008)
- Cause flexural cracks if a point load applied
- Additional maintenance cost

Fig. 1

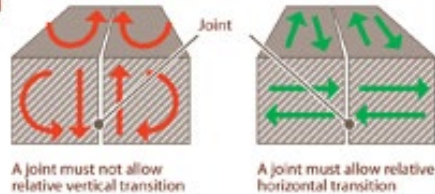


Fig. 2

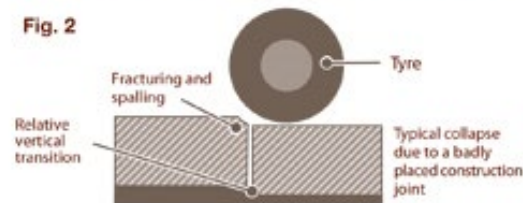
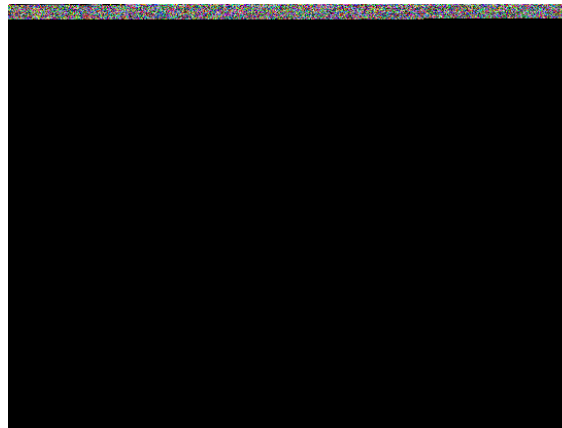


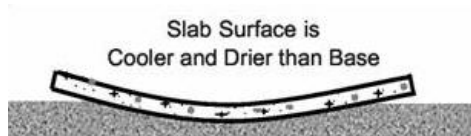
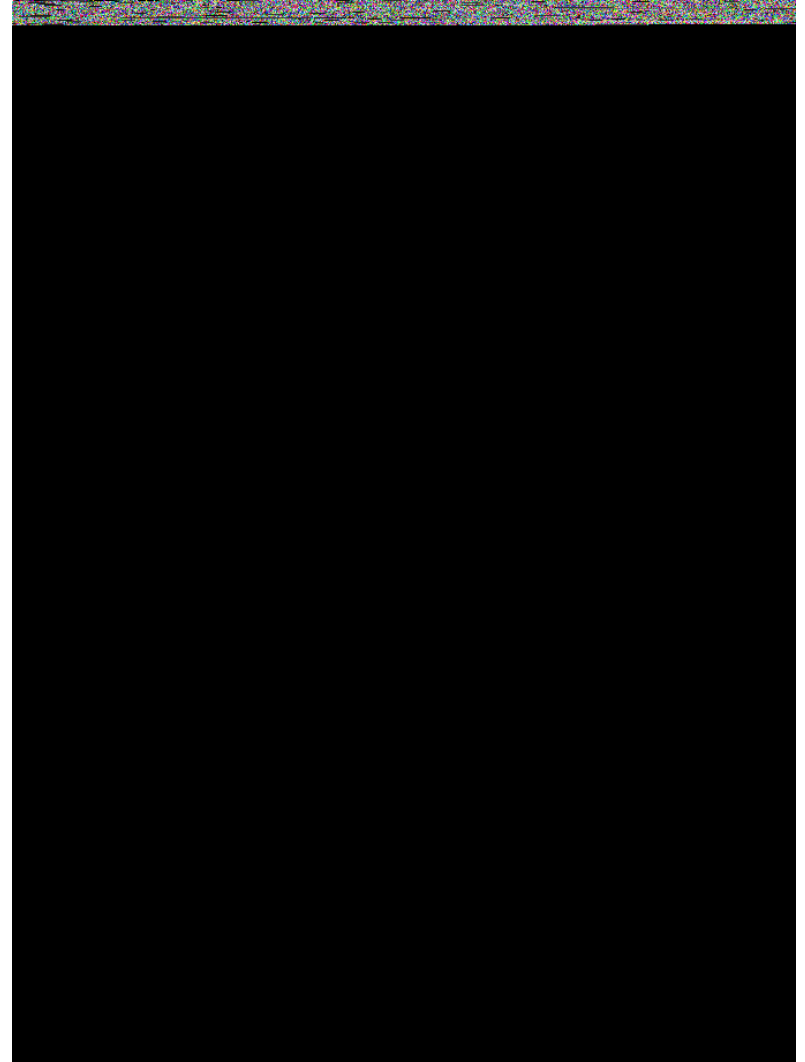
Fig. 3



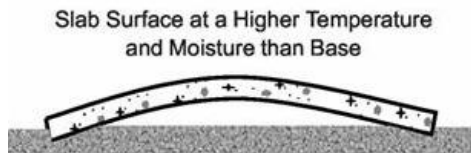
Example of construction joint damaged after a few months of life



Conventional Approach



A. Upward Curling – Typical in Internal Slabs



B. Downward Curling

Curling of Concrete Slabs

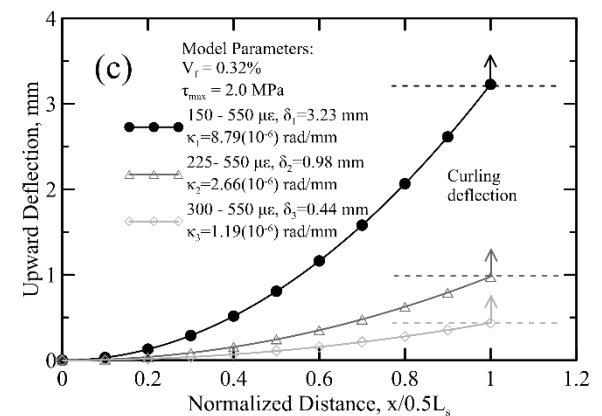
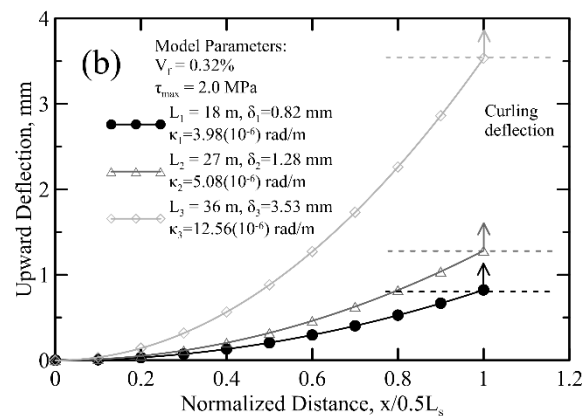
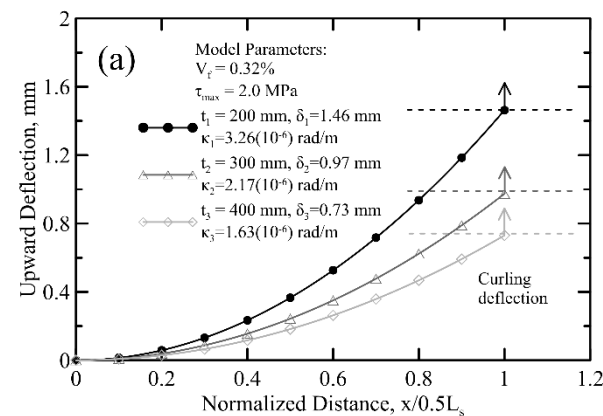
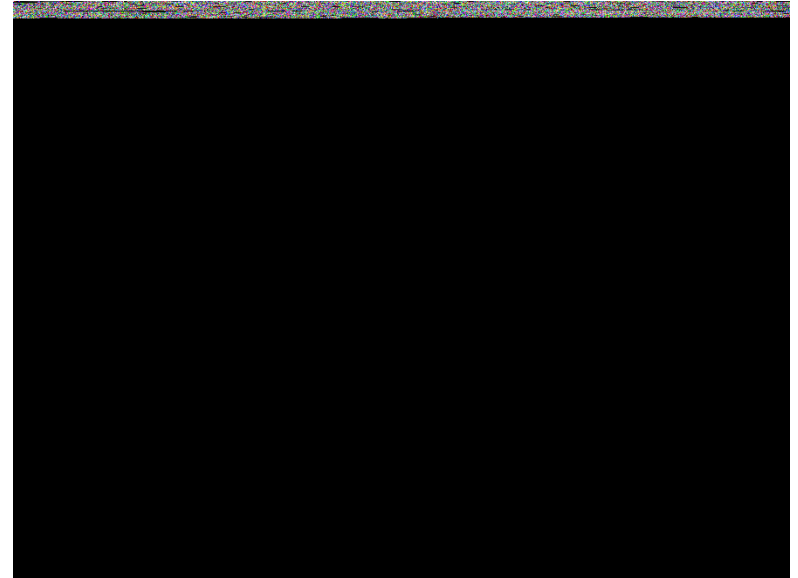
Estimation of Curling Deflection

■ Modelling procedure

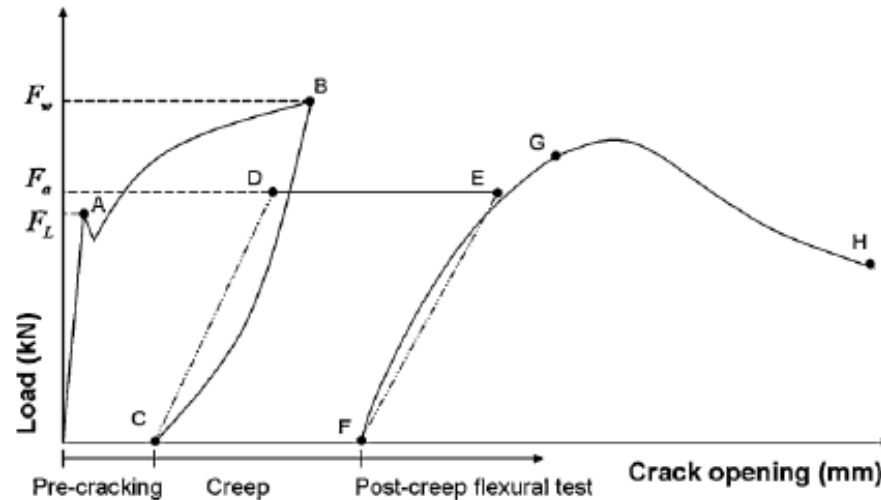
- Discretized into N layers
- Impose strain gradient
- Constant strain in each layer
- Distribution of total strain
- Curvature and tip deflection

■ Deflection increase with

- Smaller thickness
- Longer length
- Steeper free shrinkage strain gradient



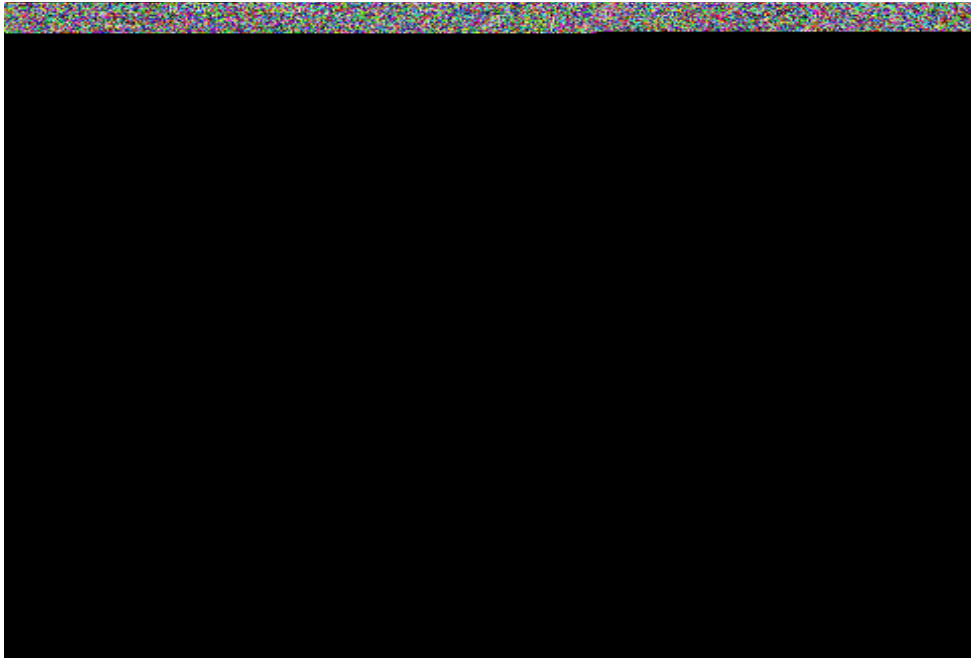
Creep Model Response



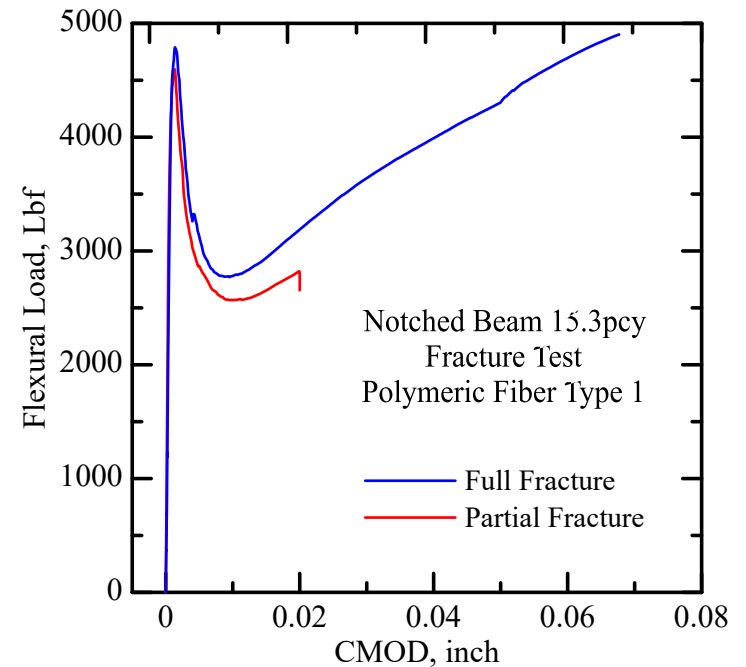
Load versus CMOD response for the entire test procedure

- Typical response of a specimen after the complete test procedure.
- Develop a model to generate the same response with the changing strains.
- Prediction of Deflection and CMOD with respect to time under sustained loads is essential from the design perspective.

Fracture Tests



Full Fracture Test Set up on MTS machine



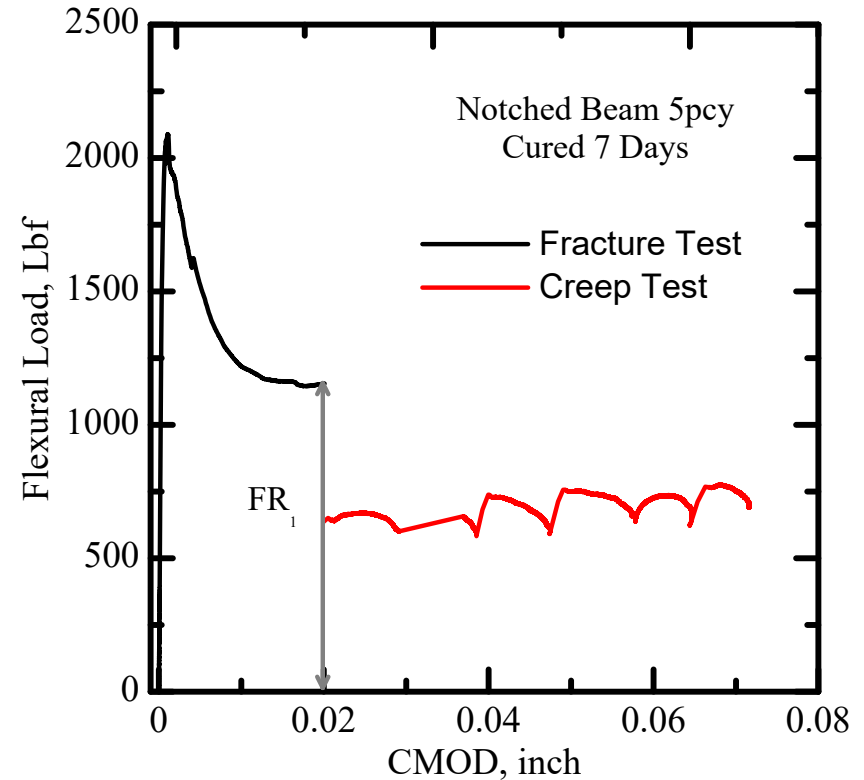
Typical Response for Partial and Full Fracture of Fiber Reinforced Beams

For Partial Fracture tests, the samples were loaded under monotonic closed loop of a CMOD controlled flexural test up to a $CMOD_1$ of 0.02 inch and respective FR_1 was noted.

Fracture Test

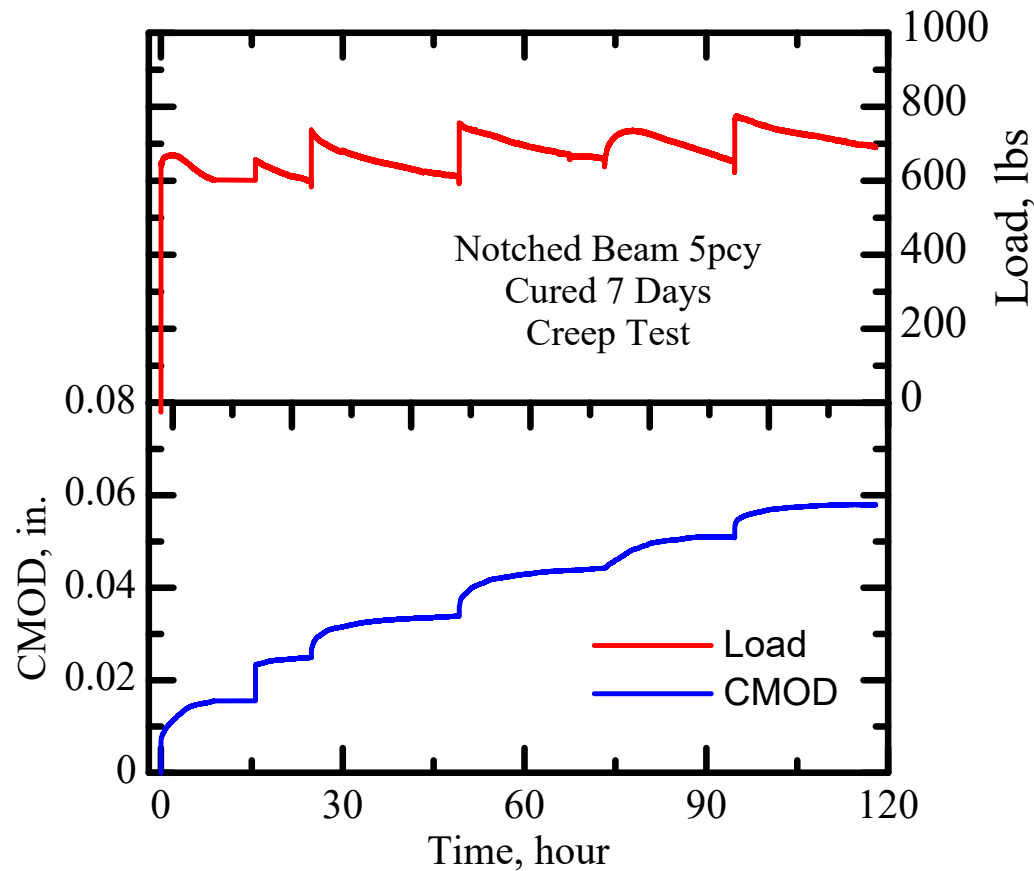
Mix Design

Materials	Mixture Proportion by weight (%)
Cement	15.5
Aggregate	39.0
Water	11.6
Sand	33.8
SAP	217 ml
Fiber	5 lb/yd ³



Load-CMOD responses obtained
from fracture and creep tests

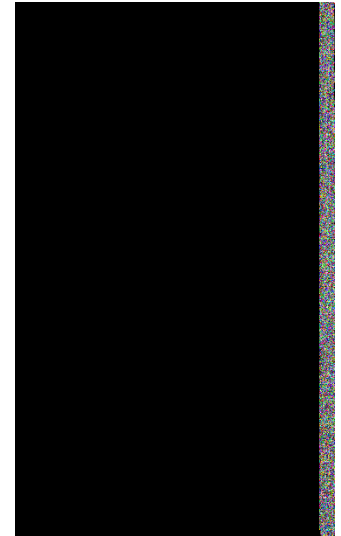
Creep Test



Load and CMOD versus time histories obtained from creep tests



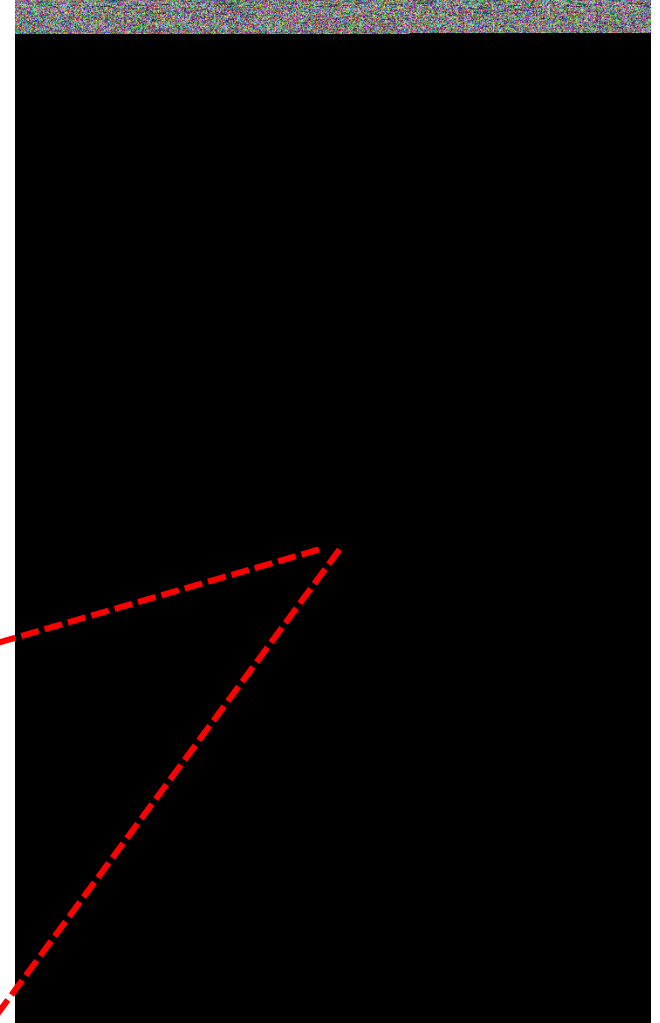
Crack at the beginning of creep



Crack on the 5th day

Testing Procedure

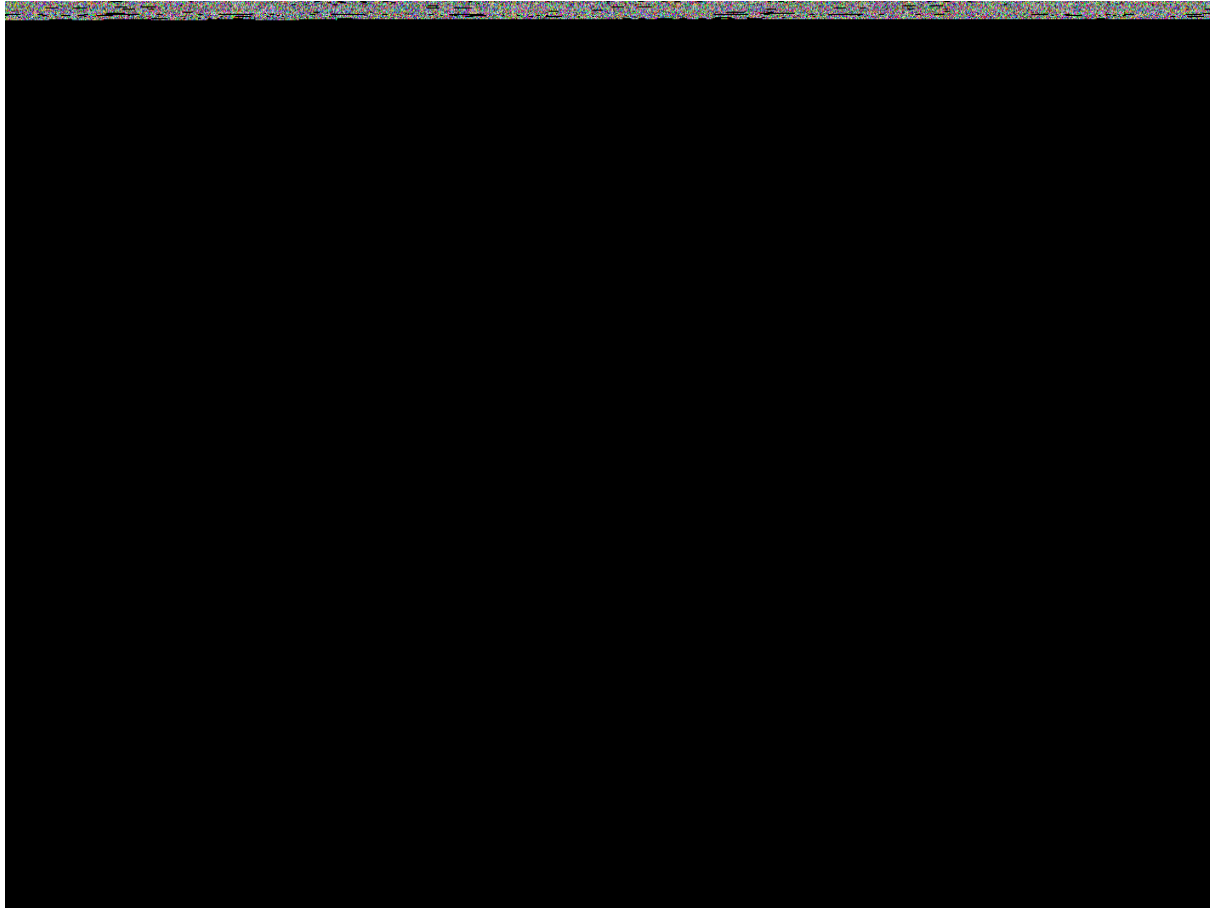
1. Fracture test up to FR_1 (CMOD = 0.5 mm/0.02 in.) using MTS machine
2. Load the cracked beam on creep testing fixture with 50% of FR_1



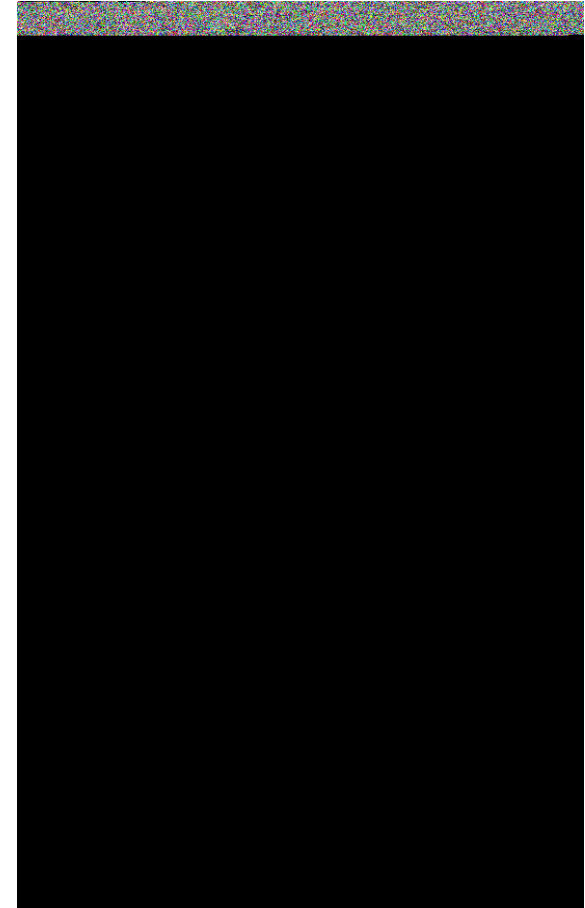
Creep Frame



Creep Test Set up

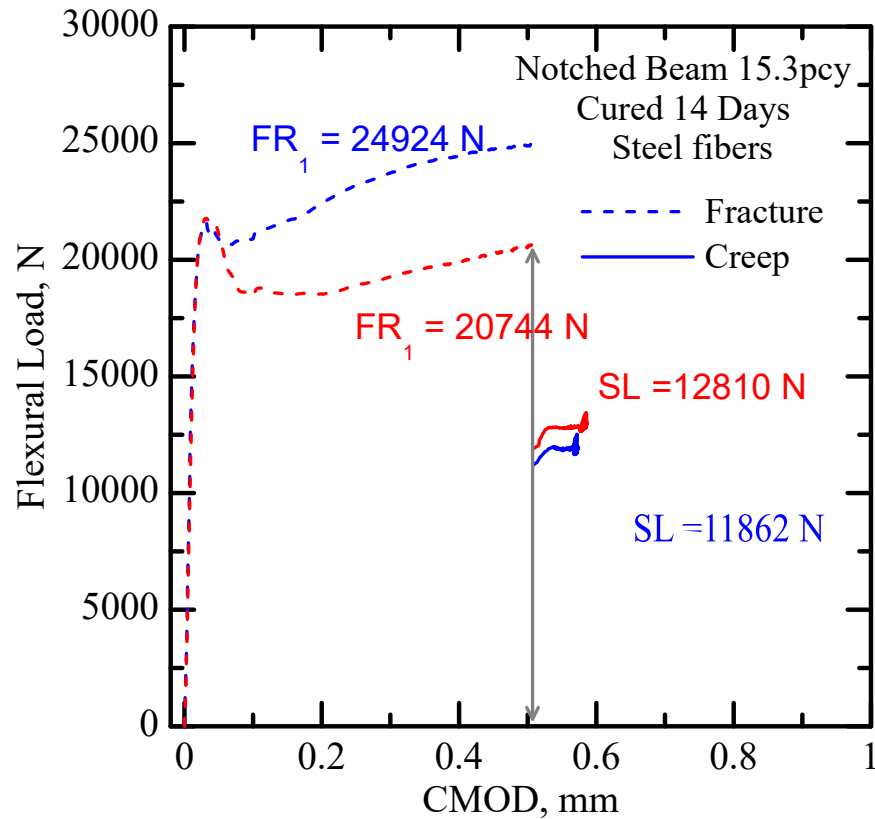


*Creep Testing Set up Schematic
Drawing*

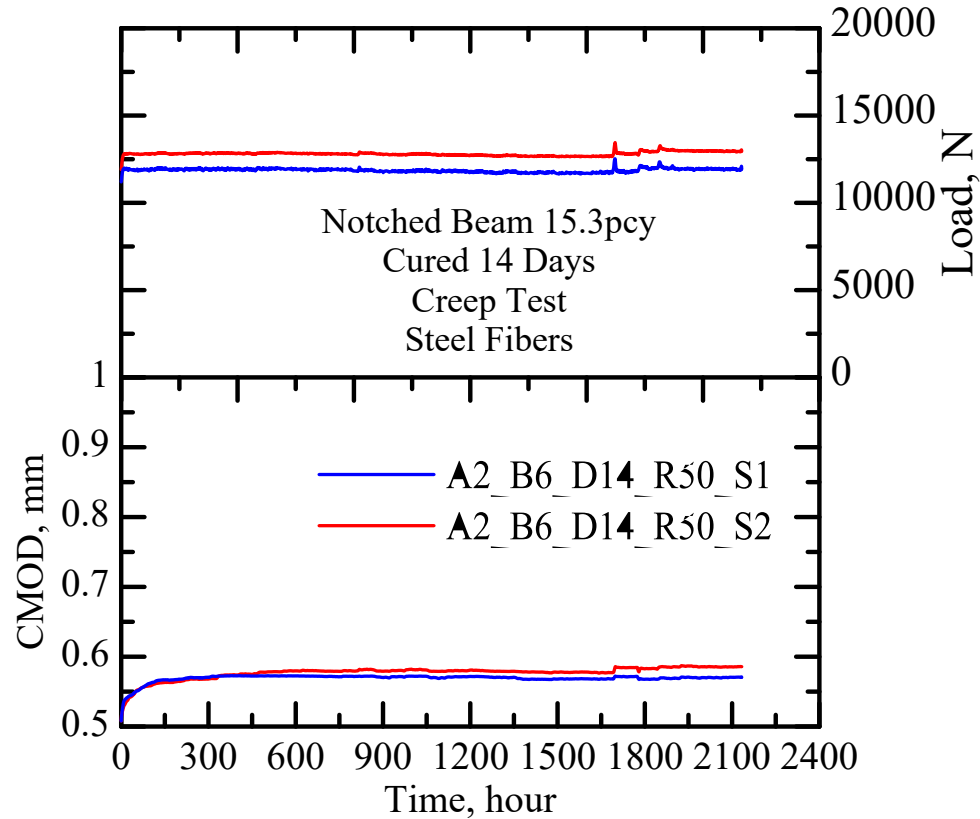


*Test Frame with High
Capacity Piston*

Typical Creep Response

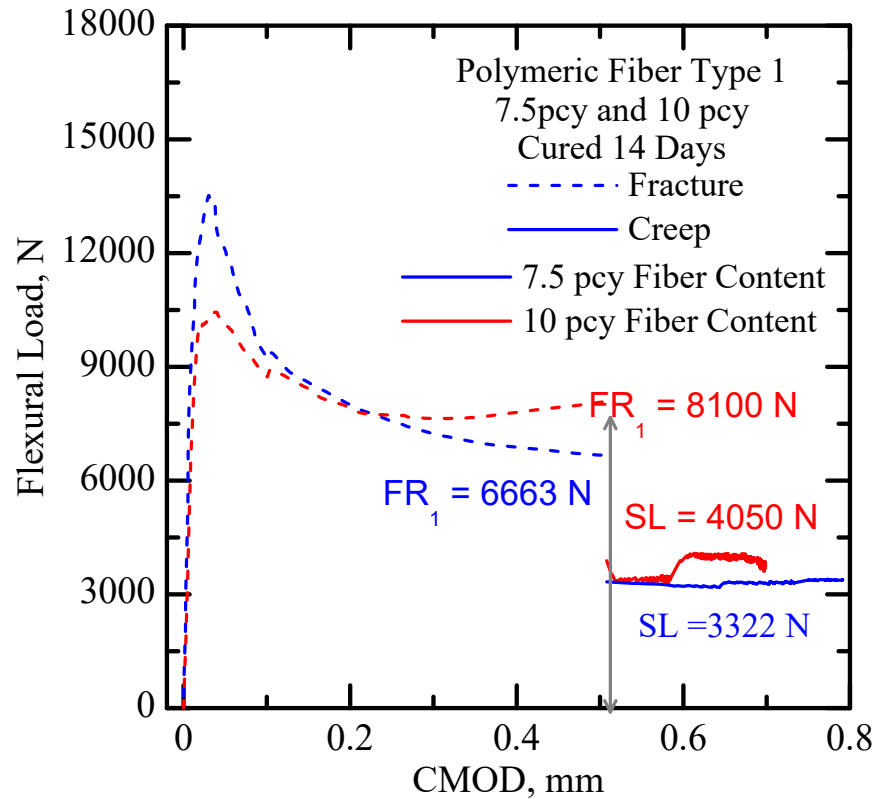


*Combined Fracture and Creep
versus CMOD response for a
Typical Creep Test*

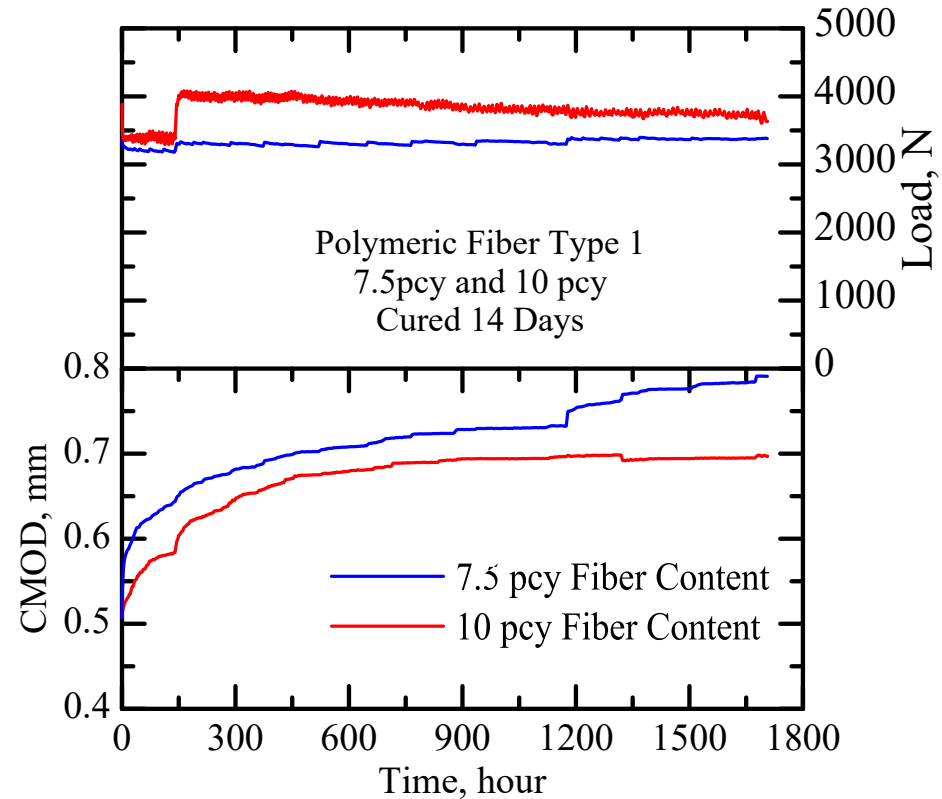


*Load and CMOD versus Time
response for a Typical Creep Test*

Effect of Fiber Volume Fraction

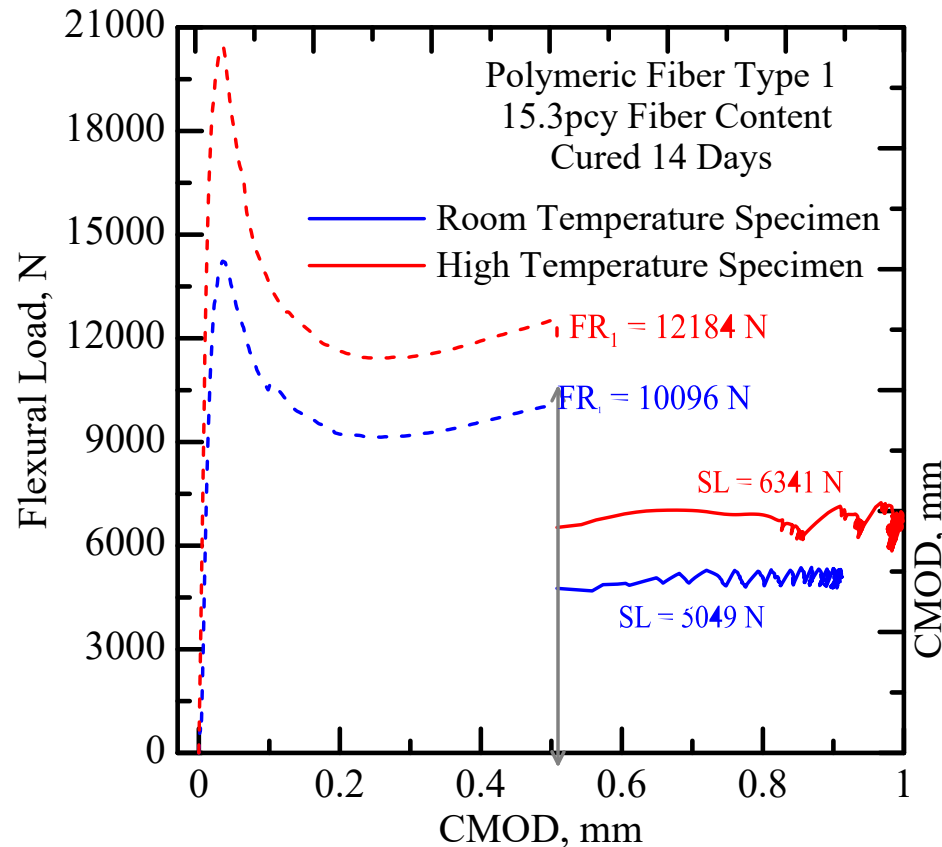


Combined Fracture and Creep versus CMOD response for a Typical Creep Test

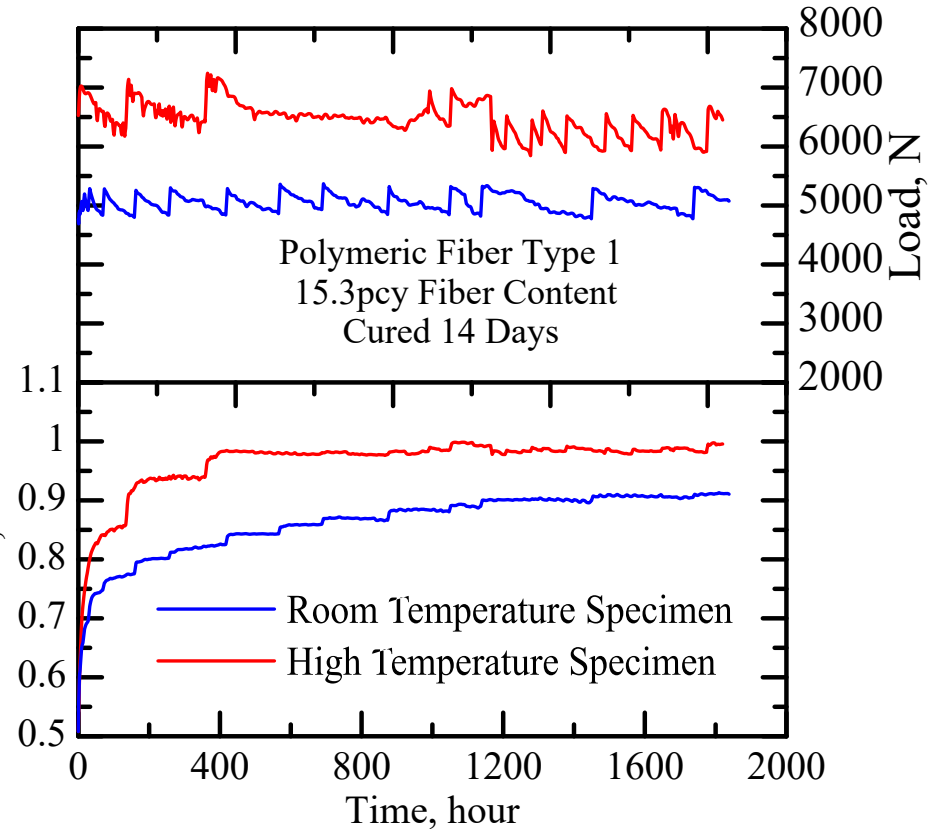


Load and CMOD versus Time response for a Typical Creep Test

Effect of Temperature

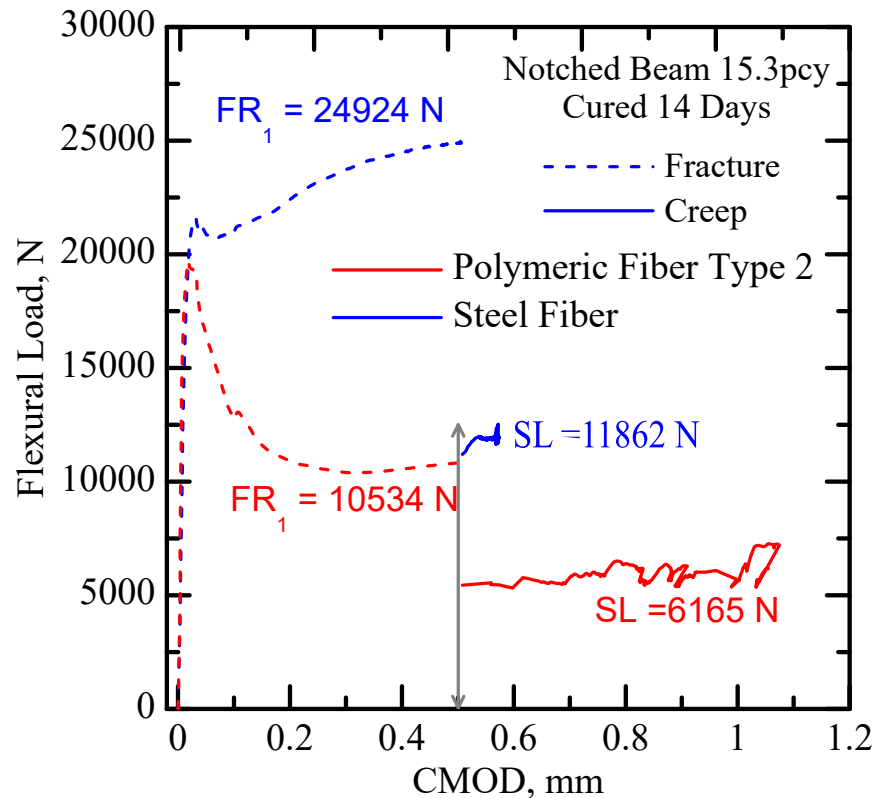


Combined Fracture and Creep versus CMOD response for a Typical Creep Test

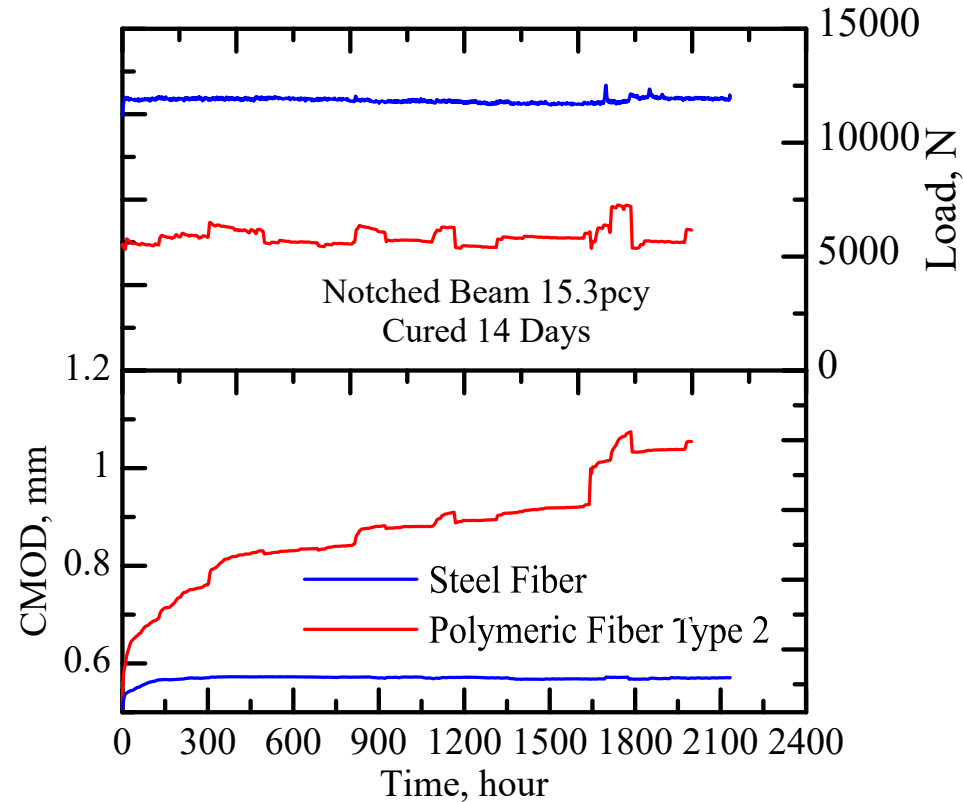


Load and CMOD versus Time response for a Typical Creep Test

Effect of type of fiber



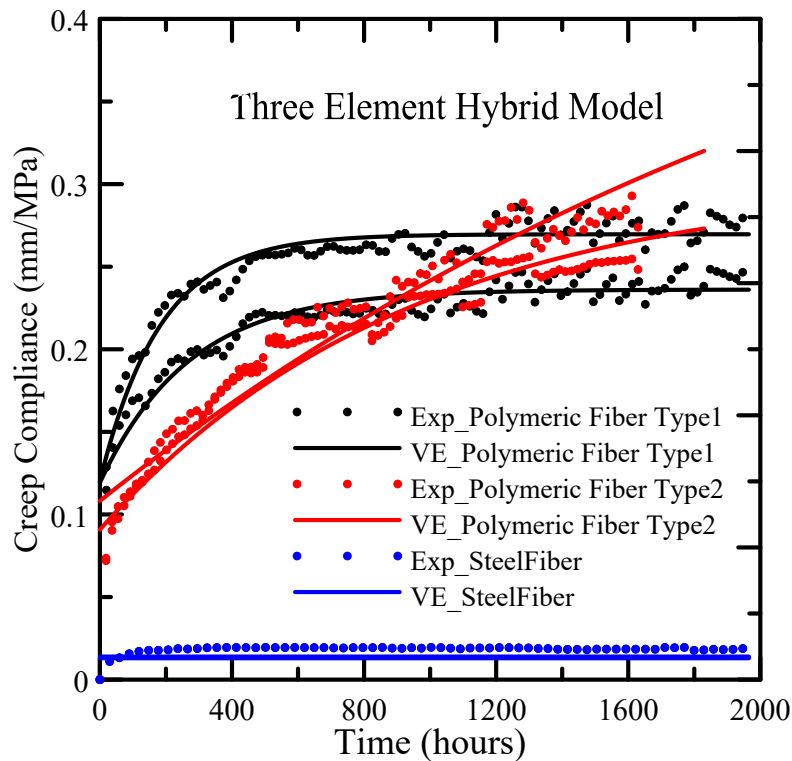
*Combined Fracture and Creep
versus CMOD response for a
Typical Creep Test*



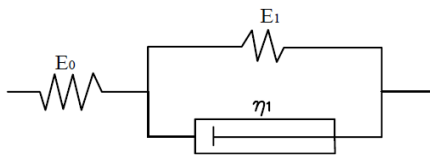
*Load and CMOD versus Time
response for a Typical Creep Test*

Hybrid Viscoelastic Model Fits

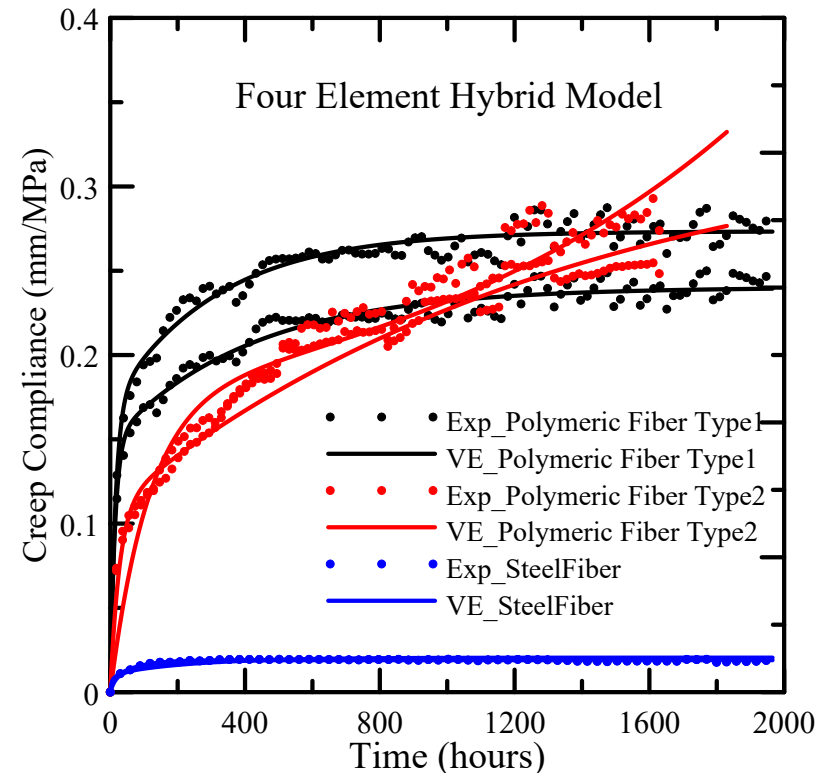
$$J(t) = \frac{1}{E_0} + \frac{1}{E_1}(1 - e^{-t/\tau_1})$$



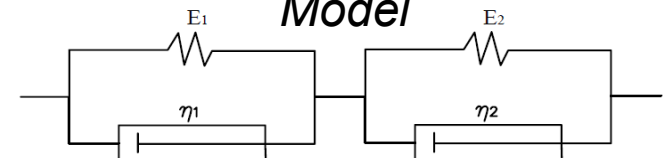
Three Element Hybrid Model



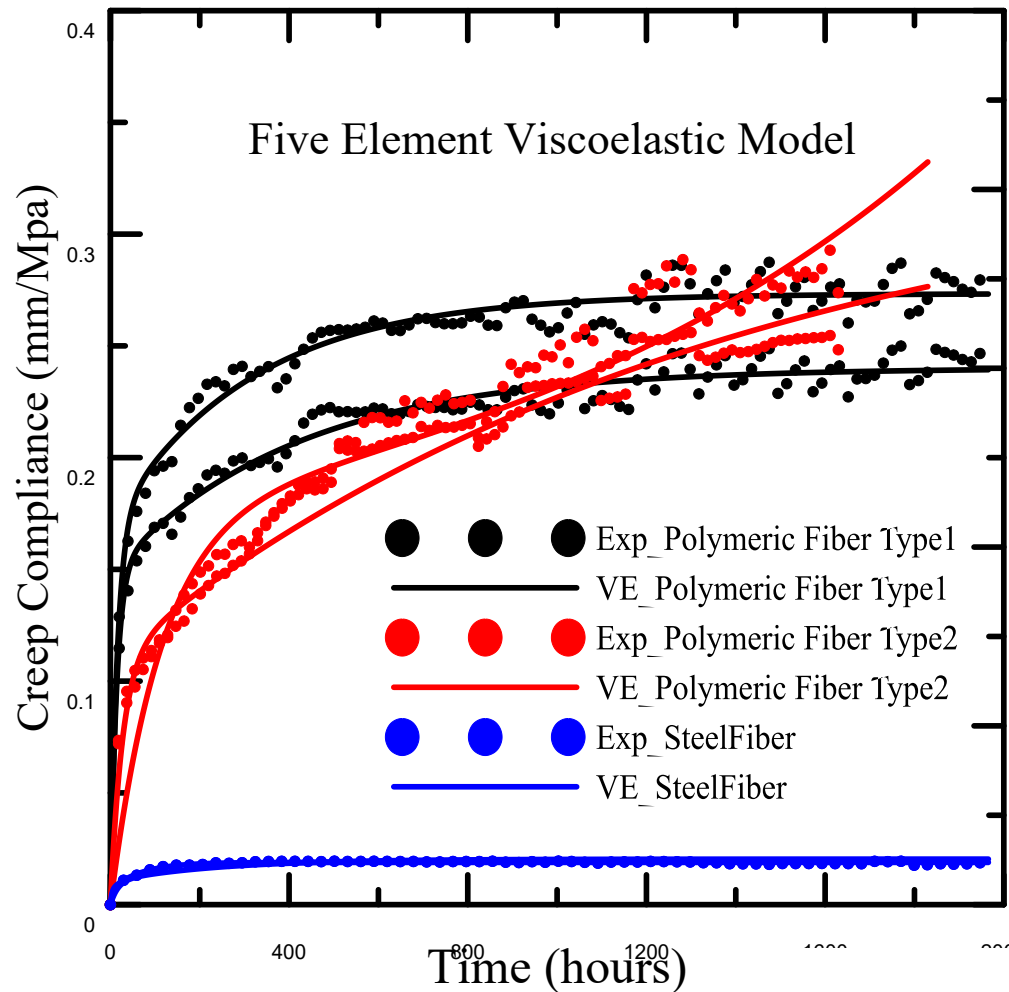
$$J(t) = \frac{1}{E_1}(1 - e^{-t/\tau_1}) + \frac{1}{E_2}(1 - e^{-t/\tau_2})$$



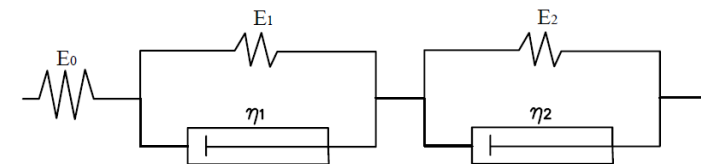
Four Element Hybrid Model



Five Element Hybrid Model and Compliance Viscoelastic Fit



$$J(t) = \frac{1}{E_0} + \frac{1}{E_1} (1 - e^{-t/\tau_1}) + \frac{1}{E_2} (1 - e^{-t/\tau_2})$$



Conclusions

- Main parameters affecting the shrinkage cracking of slabs and role of fibers in their prevention are presented
- Primary parameters affecting the drying shrinkage induced restrained cracking are lumped into several parameters
- Mechanical restrictions by sub-grade are modelled and correlated to the subgrade reaction modulus
- Empirical and numerical models are proposed to simulate the cracking behavior in joint free slab
- The two models agree well with each other by addressing equivalent parameters and one can be calibrated with respect to the other
- A new approach to estimate the curling deflection of slab is proposed. This approach needs to be calibrated against measured data.

Conclusions

- The presented test method is capable of characterizing evaporation parameters and simulating sequential formation of shrinkage cracks in two-dimensional samples.
- Drying of concrete can be segmented into two distinct stages: constant drying rate period (stage I) and falling drying rate period (stage II).
- A dual-stage drying model based on two different boundary conditions at the top surface and a diffusional moisture transfer within the pores can be used to simulate experimental drying results.
- Moisture diffusivity is higher in stage I by more than one order of magnitude than its values in stage II.